Elementary Analysis

Chapter 1: Introduction

All six sections, all 56 exercises, and every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 1. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, letters in Section 3 denote real numbers. The empty sum is 0, and $0! = 1$.

Coverage. Sections 1–6, including the enrichment section on Dedekind cuts. Exercise counts by section are 12, 8, 8, 16, 7, and 5.

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1 The set of natural numbers

Throughout this section, $\mathbb{N} = \{1, 2, 3, \ldots\}$. An induction proof has two separate obligations: establish an initial statement, and establish the implication from an arbitrary statement to the next one. In the second obligation, the induction hypothesis is an assumption used inside that implication; it is not a substitute for the initial case. When induction begins at an integer m instead of at 1, its conclusion concerns precisely the integers $n \geq m$.

We write $\sum_{k=r}^{s} u_k$ for the sum of the terms indexed by $r, r+1, \ldots, s$; if $r > s$, the sum is empty and equals 0. For integers $d \neq 0$ and M, saying that d divides M means that $M = dq$ for some integer q. The proofs of divisibility below identify this integer explicitly.

1.1 Exercise 1.1: Sum of the first squares

What we must show. For every $n \in \mathbb{N}$, establish

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Plan. Increasing the upper limit from n to $n+1$ adds exactly the new term $(n+1)^2$. We substitute the proposed formula for the old sum, then factor the resulting polynomial into the expression required at $n+1$.

1. Specify the proposition and check the initial case.

Let P_n be the displayed identity. At $n = 1$, its left side is $1^2 = 1$, and its right side is

$$\frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = 1.$$

Thus P_1 is true.

2. State the hypothesis and the next target.

Fix an arbitrary $n \in \mathbb{N}$ and assume P_n, namely

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

We must prove

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}.$$

3. Add the new square and carry out the factorization.

The induction hypothesis gives

$$\sum_{k=1}^{n+1} k^2 = \left(\sum_{k=1}^{n} k^2\right) + (n+1)^2$$

$$= \frac{n(n+1)(2n+1)}{6} + (n+1)^2$$
$$= \frac{n(n+1)(2n+1) + 6(n+1)^2}{6}$$
$$= \frac{(n+1)\big[n(2n+1) + 6(n+1)\big]}{6}.$$

To identify the bracket, expand its two products:

$$n(2n+1) + 6(n+1) = 2n^2 + n + 6n + 6 = 2n^2 + 7n + 6.$$

The needed factorization is verified by multiplying it back out:

$$(n+2)(2n+3) = 2n^2 + 3n + 4n + 6 = 2n^2 + 7n + 6.$$

Consequently,

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+2)(2n+3)}{6},$$

which is exactly P_{n+1}.

Conclusion. The initial case and the implication $P_n \Rightarrow P_{n+1}$ have both been proved. Mathematical induction therefore establishes the formula for every positive integer n.

1.2 Exercise 1.2: An arithmetic progression

What we must show. For every $n \in \mathbb{N}$, prove

$$\sum_{k=1}^{n} (8k - 5) = 4n^2 - n.$$

The summands at $k = 1$ and $k = 2$ are $8 - 5 = 3$ and $16 - 5 = 11$, so this notation describes the progression in the exercise.

Plan. Append its next term, $8(n+1) - 5$, and compare the result with the proposed expression evaluated at $n + 1$.

1. Initial case.

For $n = 1$, the left side is 3, and the right side is $4 \cdot 1^2 - 1 = 4 - 1 = 3$. Thus the statement holds at 1.

2. Induction hypothesis and target.

Let $n \geq 1$ and assume

$$\sum_{k=1}^{n} (8k - 5) = 4n^2 - n.$$

The required statement at the next index is

$$\sum_{k=1}^{n+1} (8k - 5) = 4(n+1)^2 - (n+1).$$

Before manipulating the sum, expand this target:

$$4(n+1)^2 - (n+1) = 4(n^2 + 2n + 1) - n - 1$$

$$= 4n^2 + 8n + 4 - n - 1$$
$$= 4n^2 + 7n + 3.$$

3. Compute the new sum.

The appended term is

$$8(n+1) - 5 = 8n + 8 - 5 = 8n + 3.$$

Using the induction hypothesis,

$$\begin{aligned}\sum_{k=1}^{n+1}(8k-5) &= \sum_{k=1}^{n}(8k-5) + \big(8(n+1)-5\big)\\ &= (4n^2 - n) + (8n + 3)\\ &= 4n^2 + (-n + 8n) + 3\\ &= 4n^2 + 7n + 3\\ &= 4(n+1)^2 - (n+1).\end{aligned}$$

This proves the next statement from the assumed one.

Conclusion. By induction, $3 + 11 + \cdots + (8n-5) = 4n^2 - n$ for all $n \in \mathbb{N}$.

1.3 Exercise 1.3: Sum of cubes and the square of a sum

What we must show. Prove, for every $n \in \mathbb{N}$, that

$$\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k\right)^2.$$

Plan. First record the formula for the sum inside the square. Then the asserted identity becomes a formula whose right side can be factored after the next cube is added. All square expansions used here follow from ordinary multiplication; no binomial theorem is being assumed.

1. Establish the needed formula for the first-power sum.

Write $T_n = 1 + 2 + \cdots + n$. We claim

$$T_n = \frac{n(n+1)}{2} \qquad (n \in \mathbb{N}).$$

For $n = 1$, both sides are 1. If $T_n = n(n+1)/2$ for some $n \geq 1$, then

$$\begin{aligned}T_{n+1} &= T_n + (n+1)\\ &= \frac{n(n+1)}{2} + \frac{2(n+1)}{2}\\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

This is the same formula with n replaced by $n+1$. Induction therefore proves the claim for every positive integer n.

It follows that the target identity can be written equivalently as

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2 = \frac{n^2(n+1)^2}{4}.$$

2. Check the initial case for the cube identity.

For $n = 1$, the sum of cubes is $1^3 = 1$, while

$$\frac{1^2(1+1)^2}{4} = \frac{1 \cdot 4}{4} = 1.$$

Thus the proposed formula holds at 1.

3. State the induction hypothesis and next target.

Fix $n \geq 1$ and suppose

$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}.$$

We must show

$$\sum_{k=1}^{n+1} k^3 = \frac{(n+1)^2(n+2)^2}{4}.$$

4. Append the next cube and factor.

Using the hypothesis and a common denominator,

$$\begin{aligned}
\sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^{n} k^3 + (n+1)^3 \\
&= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\
&= \frac{n^2(n+1)^2 + 4(n+1)^3}{4} \\
&= \frac{(n+1)^2\left[n^2 + 4(n+1)\right]}{4} \\
&= \frac{(n+1)^2(n^2+4n+4)}{4}.
\end{aligned}$$

Multiplication gives

$$(n+2)^2 = (n+2)(n+2) = n^2 + 2n + 2n + 4 = n^2 + 4n + 4.$$

Substituting this identity yields

$$\sum_{k=1}^{n+1} k^3 = \frac{(n+1)^2(n+2)^2}{4} = \left(\frac{(n+1)(n+2)}{2}\right)^2 = T_{n+1}^2.$$

This is precisely the assertion required at $n + 1$.

Conclusion. Mathematical induction proves $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$ for every positive integer n.

1.4 Exercise 1.4: The sum of consecutive odd numbers

What we must show. In part (a), use the first four sums to propose a formula. In part (b), prove the proposed formula for every positive integer n.

(a) Compute the requested examples and formulate the claim.

Let

$$O_n = 1 + 3 + \cdots + (2n-1) = \sum_{k=1}^{n} (2k-1).$$

The index k runs from 1 through n, so there are n summands. The requested calculations are

n	sum	O_n	n^2
1	1	1	1
2	$1+3$	4	4
3	$1+3+5$	9	9
4	$1+3+5+7$	16	16

These examples suggest $O_n = n^2$. Checking four values gives a candidate formula, but it does not establish what happens at every larger integer.

(b) Check the initial case.

At $n = 1$, $O_1 = 1 = 1^2$, so the formula starts correctly.

(b) State the induction hypothesis and next target.

Fix $n \geq 1$ and assume $O_n = n^2$. The target is

$$O_{n+1} = (n+1)^2.$$

The new last summand is the $(n+1)$st odd number:

$$2(n+1) - 1 = 2n + 2 - 1 = 2n + 1.$$

(b) Add the new odd number.

The hypothesis gives

$$\begin{aligned} O_{n+1} &= O_n + \big(2(n+1) - 1\big) \\ &= n^2 + (2n+1) \\ &= n^2 + 2n + 1 \\ &= (n+1)^2. \end{aligned}$$

For the last equality, expanding $(n+1)(n+1)$ gives $n^2 + n + n + 1 = n^2 + 2n + 1$.

Conclusion. The formula guessed in (a) is $1 + 3 + \cdots + (2n-1) = n^2$. Part (b) proves it for all $n \in \mathbb{N}$ by induction.

1.5 Exercise 1.5: A finite geometric sum

What we must show. For every positive integer n, prove

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}.$$

There are $n+1$ terms on the left, since the leading 1 is 2^0. Equivalently, the sum is $G_n = \sum_{k=0}^{n} 2^{-k}$.

Plan. The next sum adds $1/2^{n+1}$. Expressing the old final fraction with denominator 2^{n+1} leaves precisely the new final fraction.

1. Initial case.

For $n = 1$, the left side is $1 + 1/2 = 3/2$, and the right side is $2 - 1/2 = 4/2 - 1/2 = 3/2$. Thus the required statement holds at 1.

2. Hypothesis and target.

Let $n \geq 1$ and assume $G_n = 2 - 1/2^n$. We must prove

$$G_{n+1} = 2 - \frac{1}{2^{n+1}}.$$

3. Add the next fraction with a common denominator.

Because $2^{n+1} = 2 \cdot 2^n$, we have $1/2^n = 2/2^{n+1}$. Hence

$$\begin{aligned} G_{n+1} &= G_n + \frac{1}{2^{n+1}} \\ &= 2 - \frac{1}{2^n} + \frac{1}{2^{n+1}} \\ &= 2 - \frac{2}{2^{n+1}} + \frac{1}{2^{n+1}} \\ &= 2 + \frac{-2+1}{2^{n+1}} \\ &= 2 - \frac{1}{2^{n+1}}. \end{aligned}$$

All denominators are nonzero because powers of 2 are positive.

Conclusion. The identity holds for every $n \in \mathbb{N}$ by mathematical induction.

1.6 Exercise 1.6: A difference of powers divisible by seven

What we must show. Show that, for every $n \in \mathbb{N}$, there is an integer q such that

$$11^n - 4^n = 7q.$$

Here both 11 and 4 are raised to the power n.

Plan. Express the next difference of powers as 11 times the old difference plus a term with a visible factor of 7.

1. Initial case and its integer witness.

For $n = 1$,

$$11^1 - 4^1 = 11 - 4 = 7 = 7 \cdot 1.$$

The witness is the integer $q = 1$, so 7 divides the difference at 1.

2. Induction hypothesis and next target.

Fix an arbitrary $n \geq 1$ and assume 7 divides $11^n - 4^n$. By the definition of divisibility, there is some $q \in \mathbb{Z}$ satisfying

$$11^n - 4^n = 7q.$$

We must produce an integer q' such that $11^{n+1} - 4^{n+1} = 7q'$.

3. Add and subtract a useful intermediate term.

Exponent rules give $11^{n+1} = 11 \cdot 11^n$ and $4^{n+1} = 4 \cdot 4^n$. Therefore

$$\begin{aligned} 11^{n+1} - 4^{n+1} &= 11 \cdot 11^n - 4 \cdot 4^n \\ &= 11 \cdot 11^n - 11 \cdot 4^n + 11 \cdot 4^n - 4 \cdot 4^n \\ &= 11(11^n - 4^n) + (11 - 4)4^n \\ &= 11(7q) + 7 \cdot 4^n \\ &= 7(11q + 4^n). \end{aligned}$$

The second line adds $-11 \cdot 4^n + 11 \cdot 4^n = 0$, so it leaves the quantity unchanged. Define $q' = 11q + 4^n$. Since q is an integer and 4^n is an integer, q' is an integer. The last displayed equality is the required divisibility witness at $n + 1$.

Conclusion. Induction proves that 7 divides $11^n - 4^n$ for every positive integer n.

1.7 Exercise 1.7: An exponential expression divisible by thirty-six

What we must show. Prove that 36 divides $7^n - 6n - 1$ for every $n \in \mathbb{N}$.

Plan. Multiplying the old expression by 7 nearly produces the next one. The remaining difference will be $36n$.

1. Initial case.

At $n = 1$,

$$7^1 - 6 \cdot 1 - 1 = 7 - 6 - 1 = 0 = 36 \cdot 0.$$

Thus the divisibility statement holds, with integer witness 0.

2. Hypothesis and next target.

Let $n \geq 1$ and assume the current statement. Then some $q \in \mathbb{Z}$ satisfies

$$7^n - 6n - 1 = 36q.$$

Our task is to express $7^{n+1} - 6(n + 1) - 1$ as 36 times an integer.

3. Relate the next expression to the old one.

First expand the expression at $n+1$:

$$7^{n+1} - 6(n+1) - 1 = 7 \cdot 7^n - 6n - 6 - 1 = 7 \cdot 7^n - 6n - 7.$$

On the other hand,

$$\begin{aligned} 7(7^n - 6n - 1) + 36n &= 7 \cdot 7^n - 42n - 7 + 36n \\ &= 7 \cdot 7^n + (-42 + 36)n - 7 \\ &= 7 \cdot 7^n - 6n - 7. \end{aligned}$$

These two computations establish the exact identity

$$7^{n+1} - 6(n+1) - 1 = 7(7^n - 6n - 1) + 36n.$$

Now insert the induction hypothesis:

$$\begin{aligned} 7^{n+1} - 6(n+1) - 1 &= 7(36q) + 36n \\ &= 36(7q + n). \end{aligned}$$

The number $7q + n$ is an integer because $q, n \in \mathbb{Z}$. It is therefore the needed witness for divisibility at $n+1$.

Conclusion. The base case and the induction step show that $36 \mid (7^n - 6n - 1)$ for every positive integer n.

1.8 Exercise 1.8: Induction with a later starting index

What we must show. Prove the inequality in (a) for all integers $n \geq 2$, and the inequality in (b) for all integers $n \geq 4$.

Plan. Use induction beginning at the first integer in each stated domain. In part (b), the inequality from (a) will supply the comparison needed after multiplying the induction hypothesis by $n+1$.

(a) Initial case for $n^2 > n + 1$.

At $n = 2$, the inequality is

$$2^2 = 4 > 3 = 2 + 1.$$

This gives the required base at the starting index 2.

(a) Induction hypothesis and next target.

Let n be an arbitrary integer with $n \geq 2$ and assume $n^2 > n + 1$. We must show

$$(n+1)^2 > (n+1) + 1 = n + 2.$$

Expand and then use the assumed strict inequality:

$$\begin{aligned} (n+1)^2 &= n^2 + 2n + 1 \\ &> (n+1) + 2n + 1 \\ &= 3n + 2. \end{aligned}$$

The inequality in the second line comes from adding the same number $2n + 1$ to both sides of $n^2 > n + 1$. Also,

$$(3n + 2) - (n + 2) = 2n \geq 4 > 0,$$

so $3n + 2 > n + 2$. Combining the two strict inequalities yields $(n + 1)^2 > n + 2$, as required.

Conclusion. Induction starting at 2 proves $n^2 > n + 1$ for every integer $n \geq 2$.

(b) Initial case for $n! > n^2$.

Recall that $n! = 1 \cdot 2 \cdots n$. At $n = 4$,

$$4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24 > 16 = 4^2.$$

Thus the required base is true at 4.

(b) Induction hypothesis and next target.

Let $n \geq 4$ be an integer and assume $n! > n^2$. The next target is

$$(n + 1)! > (n + 1)^2.$$

The factorial definition gives $(n + 1)! = (n + 1)n!$. Since $n + 1 > 0$, multiplying the induction hypothesis by $n + 1$ preserves its direction:

$$(n + 1)! = (n + 1)n! > (n + 1)n^2.$$

Because $n \geq 4$ also implies $n \geq 2$, part (a) applies and gives $n^2 > n + 1$. Multiplying this inequality by the positive number $n + 1$ gives the second comparison

$$(n + 1)n^2 > (n + 1)(n + 1) = (n + 1)^2.$$

Putting the comparisons together,

$$(n + 1)! > (n + 1)n^2 > (n + 1)^2.$$

This is the needed induction step.

Conclusion. Induction beginning at 4 proves $n! > n^2$ for every integer $n \geq 4$. The different starting indices are essential: the conclusions of the two induction arguments are restricted to their stated domains.

1.9 Exercise 1.9: Exactly when an exponential exceeds a square

What we must show. Determine the integers n for which $2^n > n^2$, then justify the infinite portion of the answer by induction.

Plan. The strict inequality fails at some small positive integers, so an induction argument cannot begin at 1 and continue through every subsequent index. We check the exceptional small cases separately, begin the induction at 5, and also address zero and negative integers because the question says "integers."

(a) Exclude every negative integer.

Suppose $n \leq -1$ is an integer, and put $m = -n$. Then m is a positive integer and $n^2 = m^2 \geq 1$. By the definition of negative powers,

$$2^n = 2^{-m} = \frac{1}{2^m}.$$

The denominator 2^m is a product of $m \geq 1$ factors equal to 2, so $2^m \geq 2$. Since the denominator is positive, taking reciprocals gives $0 < 1/2^m \leq 1/2$. Hence

$$0 < 2^n \leq \frac{1}{2} < 1 \leq n^2.$$

In particular, $2^n > n^2$ is false for every negative integer n.

(a) Compute zero and the first five positive cases.

n	2^n	n^2	$2^n > n^2$?
0	1	0	yes
1	2	1	yes
2	4	4	no: equality
3	8	9	no: $8 < 9$
4	16	16	no: equality
5	32	25	yes

These calculations suggest that all integers at least 5 work. They also show that 0 and 1 are additional solutions, while $2, 3, 4$ are not.

(b) Set up the induction beginning at 5.

Let P_n denote $2^n > n^2$, now with the domain $n \geq 5$. The initial case is the checked inequality

$$2^5 = 32 > 25 = 5^2.$$

Fix an arbitrary integer $n \geq 5$ and assume $2^n > n^2$. We must prove

$$2^{n+1} > (n+1)^2.$$

Multiplication of the hypothesis by the positive number 2 gives

$$2^{n+1} = 2 \cdot 2^n > 2n^2.$$

To complete the step, we compare $2n^2$ with the next square.

(b) Verify the remaining polynomial inequality.

Expand the difference:

$$\begin{aligned} 2n^2 - (n+1)^2 &= 2n^2 - (n^2 + 2n + 1) \\ &= n^2 - 2n - 1 \\ &= n(n-2) - 1. \end{aligned}$$

Since $n \geq 5$, we have $n - 2 \geq 3 > 0$. Multiplication by $n - 2$ therefore preserves the inequality $n \geq 5$, and multiplication by 5 preserves $n - 2 \geq 3$. Thus

$$n(n-2) \geq 5(n-2) \geq 5 \cdot 3 = 15.$$

It follows that

$$2n^2 - (n+1)^2 = n(n-2) - 1 \geq 15 - 1 = 14 > 0.$$

Consequently $2n^2 > (n+1)^2$, and hence

$$2^{n+1} > 2n^2 > (n+1)^2.$$

This proves the implication $P_n \Rightarrow P_{n+1}$ for every integer $n \geq 5$. Induction now proves P_n for every such integer.

Conclusion. Taken literally over $\mathbb{Z}$, the complete solution set is

$$\{0, 1\} \ \cup \ \{n \in \mathbb{Z} : n \geq 5\}.$$

If "integers" is restricted to the positive integers in the surrounding discussion, the answer is $n = 1$ or $n \geq 5$. Zero is a solution whenever zero is included in the domain. The negative integers and $2, 3, 4$ have all been excluded, so no cases remain undecided.

1.10 Exercise 1.10: A sum whose lower and upper limits both change

What we must show. Show that for every positive integer n,

$$(2n+1) + (2n+3) + \cdots + (4n-1) = 3n^2.$$

Plan. Write the sum using an index before attempting induction. When n increases, the first old term disappears and two new terms enter. Accounting for all three changes is the central step.

1. Identify the summation limits and check the initial case.

Define

$$S_n = \sum_{k=n}^{2n-1} (2k+1).$$

At the lower limit $k = n$, the term is $2n+1$; at the upper limit $k = 2n-1$, it is $2(2n-1)+1 = 4n-1$. Consecutive values of k increase the term by 2. There are $(2n-1) - n + 1 = n$ terms, so this is exactly the sum in the question.

For $n = 1$, both limits equal 1. Thus

$$S_1 = 2 \cdot 1 + 1 = 3 = 3 \cdot 1^2.$$

The formula is true at its first index.

2. State the hypothesis and identify the next sum.

Fix $n \geq 1$ and assume $S_n = 3n^2$. We must prove $S_{n+1} = 3(n+1)^2$. Replacing n by $n+1$ in *both* limits gives

$$S_{n+1} = \sum_{k=n+1}^{2(n+1)-1} (2k+1) = \sum_{k=n+1}^{2n+1} (2k+1).$$

The old sum used the indices $n, n+1, \ldots, 2n-1$. The new sum uses $n+1, \ldots, 2n-1, 2n, 2n+1$. Therefore it removes the old index n and adds the indices $2n$ and $2n+1$.

3. Record the common portion, including the empty case.

Let

$$R_n = \sum_{k=n+1}^{2n-1} (2k+1).$$

For $n = 1$, this is the empty sum $\sum_{k=2}^{1}(2k+1) = 0$; the following decomposition remains valid in that case. For every $n \geq 1$,

$$\begin{aligned} S_n &= (2n+1) + R_n, \\ S_{n+1} &= R_n + \big(2(2n)+1\big) + \big(2(2n+1)+1\big) \\ &= R_n + (4n+1) + (4n+3). \end{aligned}$$

Solving the first equation for R_n and substituting it into the second establishes

$$S_{n+1} = S_n - (2n+1) + (4n+1) + (4n+3).$$

In particular, the change cannot be described merely by appending one term to S_n, since its lower limit also changes.

4. Substitute the hypothesis and simplify the new value.

$$\begin{aligned} S_{n+1} &= 3n^2 - (2n+1) + (4n+1) + (4n+3) \\ &= 3n^2 - 2n - 1 + 4n + 1 + 4n + 3 \\ &= 3n^2 + (-2+4+4)n + (-1+1+3) \\ &= 3n^2 + 6n + 3 \\ &= 3(n^2 + 2n + 1) \\ &= 3(n+1)^2. \end{aligned}$$

This is exactly the next assertion.

Conclusion. Induction proves the stated sum equals $3n^2$ for all $n \in \mathbb{N}$.

1.11 Exercise 1.11: A valid induction implication without a true base

What we must show. For P_n defined by "$n^2 + 5n + 1$ is an even integer," prove the implication requested in (a), determine its actual truth values in (b), and explain what this teaches about induction.

Plan. The difference between two consecutive expressions is even. That fact proves the requested implication but does not supply a single true initial statement. A separate parity calculation determines whether any initial statement could exist.

(a) Assume P_n and state what must follow.

Fix any $n \in \mathbb{N}$ and suppose P_n is true. An integer is even when it has the form $2q$ with $q \in \mathbb{Z}$, so the assumption says that

$$n^2 + 5n + 1 = 2q$$

for some integer q. We must prove $(n+1)^2 + 5(n+1) + 1$ is also twice an integer.

(a) Expand the next expression and give its witness.

Distributing the products gives

$$\begin{aligned}(n+1)^2 + 5(n+1) + 1 &= (n^2 + 2n + 1) + (5n + 5) + 1 \\ &= n^2 + 7n + 7 \\ &= (n^2 + 5n + 1) + (2n + 6) \\ &= (n^2 + 5n + 1) + 2(n + 3).\end{aligned}$$

Substitute the assumed representation:

$$(n+1)^2 + 5(n+1) + 1 = 2q + 2(n+3) = 2(q + n + 3).$$

Because $q + n + 3 \in \mathbb{Z}$, this expression is even. We have proved $P_n \Rightarrow P_{n+1}$ for every $n \in \mathbb{N}$.

(b) Determine the parity for an arbitrary n.

Rewrite the expression as

$$n^2 + 5n + 1 = n(n+5) + 1.$$

Every integer is either even or odd. Consider each case explicitly.

If n is even, write $n = 2r$ for some $r \in \mathbb{Z}$. Then

$$n(n+5) + 1 = 2r(n+5) + 1 = 2\big[r(n+5)\big] + 1.$$

The bracket is an integer, so this has the form $2\ell + 1$ and is odd.

If n is odd, write $n = 2r + 1$ for some $r \in \mathbb{Z}$. Now

$$n + 5 = 2r + 1 + 5 = 2r + 6 = 2(r + 3),$$

and hence

$$n(n+5) + 1 = n \cdot 2(r+3) + 1 = 2\big[n(r+3)\big] + 1.$$

Again the bracket is an integer, so the expression is odd.

Both cases show that $n^2 + 5n + 1$ is odd for every $n \in \mathbb{N}$. An odd integer cannot also be even: equality $2\ell + 1 = 2q$ would imply $1 = 2(q - \ell)$, whereas no integer multiplied by 2 equals 1. Thus P_n is false for every positive integer n.

(b) Explain the missing logical requirement.

The proposed initial statement P_1 is false because

$$1^2 + 5 \cdot 1 + 1 = 1 + 5 + 1 = 7,$$

and 7 is odd. Starting induction at a larger positive integer would not help, since every one of the proposed starting statements is false.

Part (a) is nevertheless a valid implication. It says that *if* one of these expressions were even, then the next would be even too. Its antecedent P_n is never true, so the implication has no true antecedent at which to start a chain of conclusions. In logical terms, the implication is *vacuously true*; it does not assert that either P_n or P_{n+1} is actually true.

Conclusion. There is no $n \in \mathbb{N}$ for which P_n is true, despite the validity of $P_n \Rightarrow P_{n+1}$ at every index. The moral is that induction requires both a true base case and the induction implication. Proving only the implication proves none of the assertions unconditionally.

1.12 Exercise 1.12: The binomial theorem from factorials and induction

What we must show. With

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \qquad 0 \le k \le n,$$

verify the first three cases, prove the identity in (b), and then prove

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k \qquad (n \in \mathbb{N},\ a, b \in \mathbb{R}).$$

Plan. Multiplication by $a+b$ produces two sums. After shifting the index in the sum multiplied by b, the terms with the same powers have adjacent coefficients. Part (b) combines those coefficients into the ones required for exponent $n+1$.

Conventions and endpoint coefficients.

For a positive integer r, $r! = 1 \cdot 2 \cdots r$, and $0! = 1$. All factorials used in denominators are therefore positive. In particular, for each positive integer n,

$$\binom{n}{0} = \frac{n!}{0!n!} = \frac{n!}{1 \cdot n!} = 1, \qquad \binom{n}{n} = \frac{n!}{n!0!} = \frac{n!}{n! \cdot 1} = 1.$$

The endpoint coefficients will be treated separately from the interior ones, so no factorial of a negative integer will be needed.

For the polynomial expressions here, a power with exponent 0 is an empty product and denotes the constant 1. Thus the expression x^0 in a monomial has value 1 even when the value substituted for x is 0. This convention identifies the endpoint monomials as a^n and b^n. No step below divides by a or by b, so zero values of either variable are permitted.

(a) Verify the theorem at $n = 1$.

Here $1! = 1$ and $0! = 1$, so

$$\binom{1}{0} = \frac{1!}{0!1!} = 1, \qquad \binom{1}{1} = \frac{1!}{1!0!} = 1.$$

The proposed right side is

$$\binom{1}{0} a^1 b^0 + \binom{1}{1} a^0 b^1 = 1 \cdot a \cdot 1 + 1 \cdot 1 \cdot b = a + b,$$

which equals $(a+b)^1$. The identity therefore holds for every pair of real numbers a, b at $n = 1$.

(a) Verify the theorem at $n = 2$ by multiplication.

Since $2! = 2$, the coefficients are

$$\binom{2}{0} = \frac{2}{1 \cdot 2} = 1, \qquad \binom{2}{1} = \frac{2}{1 \cdot 1} = 2, \qquad \binom{2}{2} = \frac{2}{2 \cdot 1} = 1.$$

The distributive law and commutativity of real multiplication give

$$(a+b)^2 = (a+b)(a+b)$$

$$
\begin{aligned}
&= a(a+b) + b(a+b) \\
&= a^2 + ab + ba + b^2 \\
&= a^2 + ab + ab + b^2 \\
&= a^2 + 2ab + b^2.
\end{aligned}
$$

Using the coefficients just computed, this is

$$\binom{2}{0}a^2b^0 + \binom{2}{1}a^1b^1 + \binom{2}{2}a^0b^2,$$

as required. This calculation does not appeal to a binomial expansion; it establishes this particular expansion directly.

(a) Verify the theorem at $n = 3$ by another multiplication.

The relevant factorials are $3! = 6$, $2! = 2$, $1! = 1$, and $0! = 1$. Consequently

$$\binom{3}{0} = \frac{6}{1 \cdot 6} = 1, \quad \binom{3}{1} = \frac{6}{1 \cdot 2} = 3, \quad \binom{3}{2} = \frac{6}{2 \cdot 1} = 3, \quad \binom{3}{3} = \frac{6}{6 \cdot 1} = 1.$$

Multiply the identity already established for the square by $a + b$:

$$
\begin{aligned}
(a+b)^3 &= (a+b)(a^2 + 2ab + b^2) \\
&= a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2) \\
&= a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3 \\
&= a^3 + (2+1)a^2b + (1+2)ab^2 + b^3 \\
&= a^3 + 3a^2b + 3ab^2 + b^3.
\end{aligned}
$$

The coefficients are exactly $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$, so this is the asserted formula at $n = 3$, for all real a, b.

(b) Set the range of the indices and obtain a common denominator.

Fix $n \in \mathbb{N}$ and an integer k with $1 \le k \le n$. We must prove

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}.$$

The bounds on k give $k - 1 \ge 0$ and $n - k \ge 0$, so every factorial in the two coefficients is defined. From the factorial definition,

$$(n-k+1)! = (n-k+1)(n-k)!, \qquad k! = k(k-1)!.$$

The numbers k and $n-k+1$ are both positive. Therefore the following rewritings use nonzero factors and are valid:

$$
\begin{aligned}
\binom{n}{k} &= \frac{n!}{k!(n-k)!} = \frac{n!(n-k+1)}{k!(n-k+1)!}, \\
\binom{n}{k-1} &= \frac{n!}{(k-1)!(n-(k-1))!} = \frac{n!}{(k-1)!(n-k+1)!} = \frac{n!k}{k!(n-k+1)!}.
\end{aligned}
$$

(b) Add the numerators and identify the next coefficient.

Both fractions now have denominator $k!(n-k+1)!$, and hence

$$\begin{aligned}\binom{n}{k}+\binom{n}{k-1} &= \frac{n!(n-k+1)+n!k}{k!(n-k+1)!}\\ &= \frac{n!\big[(n-k+1)+k\big]}{k!(n-k+1)!}\\ &= \frac{n!(n+1)}{k!(n-k+1)!}\\ &= \frac{(n+1)!}{k!((n+1)-k)!}\\ &= \binom{n+1}{k}.\end{aligned}$$

The penultimate line uses $(n+1)! = (n+1)n!$ and $n-k+1 = (n+1)-k$. This proves the claimed identity for every integer k in the stipulated range $1,\dots,n$. It is often called Pascal's identity. This derivation uses only the definition of the coefficients, not the binomial theorem being proved.

(c) State the induction proposition and verify its base.

Let P_n be the statement that

$$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k \quad \text{for every } a,b\in\mathbb{R}.$$

Part (a) established P_1: the two terms in that sum are a and b. Thus this proposition has a valid starting case. The checks at 2 and 3 in (a) are consistent with the result but are not needed as additional bases for induction.

(c) State the induction hypothesis and the complete next target.

Let $n\geq 1$ be arbitrary and assume P_n. Fix arbitrary real numbers a and b. The hypothesis gives

$$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$$

for these particular numbers. We must prove

$$(a+b)^{n+1} = \sum_{k=0}^{n+1}\binom{n+1}{k}a^{n+1-k}b^k.$$

Because a and b were chosen arbitrarily, obtaining this equality will prove the assertion for every pair of real numbers, as P_{n+1} requires.

(c) Multiply the assumed identity and distribute both sums.

Multiply the induction hypothesis by $a+b$. Distributivity of finite sums gives

$$\begin{aligned}(a+b)^{n+1} &= (a+b)(a+b)^n\\ &= (a+b)\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k\end{aligned}$$

$$= a\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k + b\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$$
$$= \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^{k+1}.$$

In the first sum, multiplying a^{n-k} by a raises its exponent to $n-k+1 = n+1-k$. In the second sum, multiplying b^k by b raises its exponent to $k+1$. All of these are products of nonnegative integer powers, so these rules are valid at $a = 0$ and $b = 0$ as well.

(c) Reindex the second sum with its new bounds.

In that second sum, set $j = k+1$. Then $k = j-1$, and the bounds change as follows:

$$k = 0 \implies j = 1, \qquad k = n \implies j = n+1.$$

Also the power of a becomes

$$n - k = n - (j-1) = n + 1 - j.$$

Every old index corresponds to exactly one new index and conversely. Therefore

$$\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^{k+1} = \sum_{j=1}^{n+1}\binom{n}{j-1}a^{n+1-j}b^j.$$

Substituting the reindexed sum into the preceding expression gives

$$(a+b)^{n+1} = \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{j=1}^{n+1}\binom{n}{j-1}a^{n+1-j}b^j.$$

(c) Separate the two endpoints before combining interior terms.

The first sum contains an endpoint at $k = 0$, equal to

$$\binom{n}{0}a^{n+1}b^0 = 1 \cdot a^{n+1} \cdot 1 = a^{n+1}.$$

The second sum contains an endpoint at $j = n+1$, equal to

$$\binom{n}{n}a^0b^{n+1} = 1 \cdot 1 \cdot b^{n+1} = b^{n+1}.$$

Removing these endpoints leaves the common index range $1, \ldots, n$. Rename the dummy index j in the second interior sum as k; this changes only the name of the summation variable. We obtain

$$(a+b)^{n+1} = a^{n+1} + \sum_{k=1}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=1}^{n}\binom{n}{k-1}a^{n+1-k}b^k + b^{n+1}$$
$$= a^{n+1} + \sum_{k=1}^{n}\left(\binom{n}{k} + \binom{n}{k-1}\right)a^{n+1-k}b^k + b^{n+1}.$$

For each interior index k, both terms contain the identical monomial $a^{n+1-k}b^k$; the distributive law combines their coefficients.

(c) Apply part (b) and restore the full target sum.

Every index in the remaining sum satisfies $1 \le k \le n$, so part (b) applies to every coefficient. Thus

$$(a+b)^{n+1} = a^{n+1} + \sum_{k=1}^{n} \binom{n+1}{k} a^{n+1-k} b^k + b^{n+1}.$$

The endpoint computations at the beginning of the exercise, applied with $n+1$ in place of n, give

$$\binom{n+1}{0} = 1, \qquad \binom{n+1}{n+1} = 1.$$

Consequently the first and last terms can be written as

$$a^{n+1} = \binom{n+1}{0} a^{n+1} b^0, \qquad b^{n+1} = \binom{n+1}{n+1} a^0 b^{n+1}.$$

Including these two endpoint terms with the interior sum produces

$$(a+b)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k,$$

which is exactly the required next statement for the arbitrary real numbers a, b. Therefore P_n implies P_{n+1}.

Conclusion. Part (a) verifies the first three cases, part (b) establishes Pascal's identity directly from factorials, and part (c) proves the binomial theorem for every positive integer n and every $a, b \in \mathbb{R}$ by mathematical induction. The proof includes $a = 0$, $b = 0$, and $a = b = 0$, since it uses no division by either variable and interprets zero exponents as constant monomials.

2 The set of rational numbers

The exercises in this section ask whether particular numbers belong to $\mathbb{Q}$. Recall that a number is rational when it can be written as m/n, where $m, n \in \mathbb{Z}$ and $n \neq 0$. A number is irrational when it is a real number that is not rational. The following elementary tools explain the steps used in the solutions.

Tool 1: Why rational arithmetic stays rational

Suppose $r = m/n$ and $s = p/q$, where $m, n, p, q \in \mathbb{Z}$ and $n, q \neq 0$. Putting fractions over a common denominator gives

$$r + s = \frac{mq + pn}{nq}, \qquad r - s = \frac{mq - pn}{nq}, \qquad rs = \frac{mp}{nq}.$$

Each numerator is an integer, and nq is a nonzero integer. Each displayed quantity is therefore rational by the definition of $\mathbb{Q}$. If $s \neq 0$, then $p \neq 0$, and

$$\frac{r}{s} = \frac{mq}{np}$$

also has an integer numerator and a nonzero integer denominator. Thus division by a nonzero rational number preserves rationality. Integers are rational because $j = j/1$. In particular, multiplication by an integer and subtraction of an integer preserve rationality.

For a positive integer k, repeated multiplication gives

$$r^k = \underbrace{r \cdot r \cdots r}_{k \text{ factors}} = \frac{m^k}{n^k} \in \mathbb{Q},$$

since $m^k, n^k \in \mathbb{Z}$ and $n^k \neq 0$. These formulas justify every use of "rational closure" below.

Tool 2: The integer cancellation fact behind the rational zeros theorem

Two integers are relatively prime when they have no common positive divisor other than 1. We use this elementary fact: if $u, v \in \mathbb{Z}$ are relatively prime, $v \neq 0$, and v divides uw, then v divides w. Here "v divides z" means that $z = vk$ for some $k \in \mathbb{Z}$.

For completeness, the cancellation fact follows from a short integer argument. The positive integers of the form $Au + Bv$, with $A, B \in \mathbb{Z}$, form a nonempty set: one of v and $-v$ is positive. Let d be the least such positive integer, and write $d = A_0u + B_0v$. The division algorithm gives $u = hd + t$, with $h \in \mathbb{Z}$ and $0 \leq t < d$. Then

$$t = u - hd = (1 - hA_0)u - hB_0v.$$

If $t > 0$, this would be a smaller positive integer of the same form, contradicting the choice of d. Hence $t = 0$, so d divides u. Applying the same argument to division of v by d shows that d divides v. Relative primality forces $d = 1$. Consequently there are integers A_0, B_0 with

$$A_0u + B_0v = 1.$$

Multiplying this identity by w gives

$$w = A_0(uw) + B_0vw.$$

If v divides uw, both terms on the right are integer multiples of v, so w is also an integer multiple of v. This proves the cancellation fact. The integer division algorithm and the existence of a least positive integer are the only integer facts used here.

One consequence needed below is worth spelling out. If u, v are relatively prime and v divides u^k, then cancellation in $u^k = u\,u^{k-1}$ shows that v divides u^{k-1}. Repeating this step eventually shows that v divides 1. When $u \neq 0$, the last power in this argument is $u^0 = 1$. When $u = 0$, relative primality already forces $|v| = 1$, so the same conclusion holds directly.

Tool 3: A monic polynomial can have only integer rational zeros

A polynomial is *monic* when the coefficient of its highest power is 1. Suppose

$$P(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0, \qquad a_0, \ldots, a_{d-1} \in \mathbb{Z},$$

where $d \geq 1$. Assume that $r \in \mathbb{Q}$ and $P(r) = 0$. Write $r = m/n$ in lowest terms, with $n > 0$. Thus $m, n \in \mathbb{Z}$ are relatively prime. If $m = 0$, then $r = 0$ is already an integer. For the denominator argument we may therefore assume $m \neq 0$. Substituting and multiplying by n^d gives

$$m^d + \sum_{j=0}^{d-1} a_j m^j n^{d-j} = 0.$$

Every term in the sum contains a factor n. Equivalently,

$$m^d = -n \left(\sum_{j=0}^{d-1} a_j m^j n^{d-j-1} \right).$$

The expression in parentheses is an integer, so n divides m^d. Tool 2, applied repeatedly to the relatively prime integers m, n, shows that n divides 1. As n is positive, $n = 1$. Therefore $r = m \in \mathbb{Z}$.

There is a further restriction if $a_0 \neq 0$. In that case $r \neq 0$, because $P(0) = a_0 \neq 0$. Substituting the integer m into $P(m) = 0$ and collecting all terms other than the constant gives

$$a_0 = -m \left(m^{d-1} + \sum_{j=1}^{d-1} a_j m^{j-1} \right).$$

For $d = 1$ the sum is empty and equals 0. In every case the expression in parentheses is an integer. Hence m divides a_0.

We have proved the monic case of Ross's Rational Zeros Theorem (Theorem 2.2 and Corollary 2.3):

> A rational zero of a monic polynomial with integer coefficients must be an integer. If the constant coefficient is nonzero, that integer must divide the constant coefficient.

In particular, if N is a positive integer and k is a positive integer, then a rational solution of $x^k = N$ must be an integer, because it is a zero of the monic integer polynomial $x^k - N$.

Tool 4: Bounding a positive root by consecutive integers

For $0 \leq u < v$ and a positive integer k,

$$v^k - u^k = (v - u)(v^{k-1} + v^{k-2}u + \cdots + u^{k-1}) > 0.$$

For $k = 1$ the factor in parentheses is 1. For $k \geq 2$, all its terms are nonnegative and its first term v^{k-1} is positive. Thus raising nonnegative numbers to the kth power preserves their strict order. This is an algebraic fact; no limiting argument is needed.

Let x be the nonnegative kth root of a positive integer N. Suppose a nonnegative integer a satisfies

$$a^k < N < (a+1)^k.$$

If $x \leq a$, then $x^k \leq a^k < N$, contradicting $x^k = N$. If $x \geq a+1$, then $x^k \geq (a+1)^k > N$, giving the same contradiction. It follows that

$$a < x < a+1.$$

There is no integer strictly between the consecutive integers a and $a+1$. Tool 3 therefore implies that x is irrational. This gives the criterion used in the first two exercises:

> If a positive integer lies strictly between two consecutive nonnegative integer kth powers, its positive kth root is irrational.

2.1 Exercise 2.1: Five square roots of integers

What we must show. Prove that $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, $\sqrt{24}$, and $\sqrt{31}$ are irrational.

Plan. Each number is a root of a monic integer polynomial $x^2 - N$. If it were rational, Tool 3 would make it an integer. We locate each root strictly between consecutive integers to rule that out.

1. Record the exact integer-square comparisons

The needed computations are

$$1^2 = 1, \quad 2^2 = 4, \quad 3^2 = 9, \quad 4^2 = 16, \quad 5^2 = 25, \quad 6^2 = 36.$$

They give the following bounds by Tool 4:

N	Consecutive square bounds	Consequent root bounds
3	$1^2 = 1 < 3 < 4 = 2^2$	$1 < \sqrt{3} < 2$
5	$2^2 = 4 < 5 < 9 = 3^2$	$2 < \sqrt{5} < 3$
7	$2^2 = 4 < 7 < 9 = 3^2$	$2 < \sqrt{7} < 3$
24	$4^2 = 16 < 24 < 25 = 5^2$	$4 < \sqrt{24} < 5$
31	$5^2 = 25 < 31 < 36 = 6^2$	$5 < \sqrt{31} < 6$

2. Apply the rational-root argument to each number

(a) Suppose $\sqrt{3}$ were rational. Since $(\sqrt{3})^2 - 3 = 0$, Tool 3 applied to $x^2 - 3$ would imply $\sqrt{3} \in \mathbb{Z}$. But $1 < \sqrt{3} < 2$, and no integer satisfies those two inequalities. This contradiction proves $\sqrt{3} \notin \mathbb{Q}$.

(b) Suppose $\sqrt{5}$ were rational. The equation $(\sqrt{5})^2 - 5 = 0$ and Tool 3 imply $\sqrt{5} \in \mathbb{Z}$. The bounds $2 < \sqrt{5} < 3$ exclude every integer, so $\sqrt{5} \notin \mathbb{Q}$.

(c) Suppose $\sqrt{7}$ were rational. The monic integer equation $(\sqrt{7})^2 - 7 = 0$ would make $\sqrt{7}$ an integer. The strict bounds $2 < \sqrt{7} < 3$ make this impossible. Hence $\sqrt{7} \notin \mathbb{Q}$.

(d) Suppose $\sqrt{24}$ were rational. Because it satisfies $x^2 - 24 = 0$, it would be an integer by Tool 3. Its bounds $4 < \sqrt{24} < 5$ contradict that conclusion. Thus $\sqrt{24} \notin \mathbb{Q}$.

(e) Suppose $\sqrt{31}$ were rational. Tool 3 applied to x^2-31 would make it an integer. Since $5 < \sqrt{31} < 6$, there is no such integer. Therefore $\sqrt{31} \notin \mathbb{Q}$.

3. Establish one auxiliary case for later exercises

We will also need that $\sqrt{2}$ is irrational. If it were rational, the equation $(\sqrt{2})^2 - 2 = 0$ would make it an integer. But

$$1^2 = 1 < 2 < 4 = 2^2 \quad \Longrightarrow \quad 1 < \sqrt{2} < 2,$$

which is impossible for an integer. Thus $\sqrt{2} \notin \mathbb{Q}$ as well.

Conclusion. All five requested square roots are irrational. The same argument also proves the auxiliary fact $\sqrt{2} \notin \mathbb{Q}$.

2.2 Exercise 2.2: Three higher roots of integers

What we must show. Prove that $\sqrt[3]{2}$, $\sqrt[7]{5}$, and $\sqrt[4]{13}$ are irrational.

Plan. Tool 3 applies to $x^3 - 2$, $x^7 - 5$, and $x^4 - 13$. We compute suitable consecutive integer powers, then apply Tool 4.

1. The cube root of 2

Let $x = \sqrt[3]{2}$. By the definition of the positive cube root, $x > 0$ and $x^3 = 2$. Assume that $x \in \mathbb{Q}$. Since x is a zero of the monic polynomial $x^3 - 2$, whose coefficients are integers, Tool 3 gives $x \in \mathbb{Z}$.

Now $1^3 = 1$ and $2^3 = 2 \cdot 2 \cdot 2 = 8$. Consequently

$$1^3 = 1 < 2 < 8 = 2^3.$$

Tool 4 gives $1 < x < 2$. No integer lies in that interval, contradicting $x \in \mathbb{Z}$. Hence $\sqrt[3]{2}$ is irrational.

2. The seventh root of 5

Let $y = \sqrt[7]{5}$. Then $y > 0$ and $y^7 = 5$. If y were rational, Tool 3 applied to $x^7 - 5$ would imply $y \in \mathbb{Z}$. The exact powers are

$$1^7 = 1, \qquad 2^7 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 128.$$

Thus

$$1^7 = 1 < 5 < 128 = 2^7,$$

so $1 < y < 2$. This excludes all integers and contradicts the consequence of the rationality assumption. Therefore $\sqrt[7]{5}$ is irrational.

3. The fourth root of 13

Let $z = \sqrt[4]{13}$, where the fourth-root notation denotes the positive real root. Then $z > 0$ and $z^4 = 13$. If $z \in \mathbb{Q}$, the monic integer equation $z^4 - 13 = 0$ gives $z \in \mathbb{Z}$ by Tool 3. However,

$$1^4 = 1 < 13 < 16 = 2 \cdot 2 \cdot 2 \cdot 2 = 2^4.$$

It follows that $1 < z < 2$, contradicting integer membership. Hence $\sqrt[4]{13}$ is irrational.

Conclusion. The numbers $\sqrt[3]{2}$, $\sqrt[7]{5}$, and $\sqrt[4]{13}$ are all irrational.

2.3 Exercise 2.3: A nested square root

What we must show. Prove that $\sqrt{2+\sqrt{2}}$ is irrational.

Plan. Assume the outer square root is rational. Square it and isolate $\sqrt{2}$. Rational closure would then force $\sqrt{2}$ to be rational, contradicting the auxiliary case in Exercise 2.1.

1. Name the number and record its defining equation

Put

$$a = \sqrt{2+\sqrt{2}}.$$

The inner root $\sqrt{2}$ is positive, so $2+\sqrt{2} > 0$. The outer square root therefore denotes a positive real number. Its defining equation is

$$a^2 = 2+\sqrt{2}.$$

2. Assume rationality and rearrange

Suppose, for a contradiction, that $a \in \mathbb{Q}$. Then $a^2 \in \mathbb{Q}$ by rational closure under multiplication. Subtracting the rational integer 2 from both sides gives

$$\sqrt{2} = a^2 - 2 \in \mathbb{Q}.$$

In fraction form, if $a = m/n$ with $m, n \in \mathbb{Z}$ and $n \neq 0$, this is

$$a^2 - 2 = \frac{m^2}{n^2} - \frac{2n^2}{n^2} = \frac{m^2-2n^2}{n^2},$$

an integer divided by a nonzero integer. Thus the rationality conclusion is justified directly by the definition, as well as by Tool 1.

3. Identify the contradiction

Exercise 2.1 established $\sqrt{2} \notin \mathbb{Q}$. The rationality assumption for a has produced its negation, $\sqrt{2} \in \mathbb{Q}$. Therefore the assumption is false.

Conclusion. $\sqrt{2+\sqrt{2}}$ is irrational.

2.4 Exercise 2.4: A nested cube root

What we must show. Prove that $\sqrt[3]{5-\sqrt{3}}$ is irrational.

Plan. Cube the number to remove the outer radical, then isolate $\sqrt{3}$. If the original number were rational, this rearrangement would express $\sqrt{3}$ using rational operations only.

1. Check the sign and write the cube equation

Exercise 2.1 gives $1 < \sqrt{3} < 2$. In particular,

$$5 - \sqrt{3} > 5 - 2 = 3 > 0.$$

Thus

$$a = \sqrt[3]{5-\sqrt{3}}$$

is a positive real number, and the definition of cube root gives

$$a^3 = 5 - \sqrt{3}.$$

2. Derive a rational expression for $\sqrt{3}$ under the assumption

Assume $a \in \mathbb{Q}$. Then $a^3 \in \mathbb{Q}$ by Tool 1. Adding $\sqrt{3}$ to the cube equation and subtracting a^3 gives

$$a^3 + \sqrt{3} = 5, \qquad \text{hence} \qquad \sqrt{3} = 5 - a^3.$$

Both 5 and a^3 are rational, so their difference is rational. Explicitly, if $a = m/n$ with integer m, n and $n \neq 0$, then

$$5 - a^3 = \frac{5n^3 - m^3}{n^3} \in \mathbb{Q}.$$

3. Finish the contradiction

The preceding step gives $\sqrt{3} \in \mathbb{Q}$, contrary to Exercise 2.1. Therefore $a \notin \mathbb{Q}$.

Conclusion. $\sqrt[3]{5 - \sqrt{3}}$ is irrational.

2.5 Exercise 2.5: A positive real fractional power

What we must show. Prove that $(3 + \sqrt{2})^{2/3}$ is irrational.

Plan. First explain the fractional exponent using the positive real cube root. Cubing the proposed number then produces a simple expression $11 + 6\sqrt{2}$, from which $\sqrt{2}$ can be isolated.

1. Interpret the exponent without a sign ambiguity

Since $\sqrt{2} > 0$, the base

$$u = 3 + \sqrt{2}$$

is positive. Let $t = \sqrt[3]{u} > 0$. Thus $t^3 = u$, and the positive real fractional power in the problem is

$$a = u^{2/3} = (\sqrt[3]{u})^2 = t^2.$$

It follows from integer power rules that

$$a^3 = (t^2)^3 = t^6 = (t^3)^2 = u^2 = (3 + \sqrt{2})^2.$$

This explains precisely why cubing removes the exponent $2/3$.

2. Expand the square completely

Using $(v + w)^2 = v^2 + 2vw + w^2$ gives

$$\begin{aligned} a^3 &= (3 + \sqrt{2})^2 \\ &= 3^2 + 2 \cdot 3 \cdot \sqrt{2} + (\sqrt{2})^2 \\ &= 9 + 6\sqrt{2} + 2 \\ &= 11 + 6\sqrt{2}. \end{aligned}$$

Subtracting 11 and dividing by 6 therefore yields

$$a^3 - 11 = 6\sqrt{2}, \qquad \sqrt{2} = \frac{a^3 - 11}{6}.$$

3. Use rational closure to obtain the contradiction

Suppose $a \in \mathbb{Q}$. Then $a^3 \in \mathbb{Q}$, hence $a^3 - 11 \in \mathbb{Q}$. Because 6 is a nonzero rational integer, division by 6 preserves rationality. The last displayed equation would imply $\sqrt{2} \in \mathbb{Q}$. This contradicts the auxiliary irrationality result in Exercise 2.1.

Conclusion. $(3 + \sqrt{2})^{2/3}$, with the positive real interpretation of the fractional power, is irrational.

2.6 Exercise 2.6: A rational quadratic expression

What we must show. Explain why $b \in \mathbb{Q}$ implies $4 - 7b^2 \in \mathbb{Q}$.

Plan. Start with the defining fraction for a rational number and write $4 - 7b^2$ as one fraction. Check its numerator and denominator separately.

1. Write the rationality hypothesis explicitly

Assume $b \in \mathbb{Q}$. By definition, there are integers m, n such that

$$b = \frac{m}{n}, \qquad n \neq 0.$$

The numerator m may be positive, negative, or zero. No restriction on the sign of b is needed for this argument.

2. Perform the arithmetic over one denominator

Squaring and multiplying gives

$$b^2 = \frac{m^2}{n^2}, \qquad 7b^2 = \frac{7m^2}{n^2}.$$

As $n \neq 0$, its square n^2 is nonzero, so we may also write $4 = 4n^2/n^2$. Subtracting now gives

$$4 - 7b^2 = \frac{4n^2}{n^2} - \frac{7m^2}{n^2} = \frac{4n^2 - 7m^2}{n^2}.$$

The integer operations of squaring, multiplication, and subtraction show that $4n^2 - 7m^2 \in \mathbb{Z}$. The denominator n^2 is a nonzero integer. The displayed fraction is therefore rational by definition.

3. Explain the connection with Example 6

The same reasoning can be summarized using Tool 1: $b \in \mathbb{Q}$ implies $b^2 \in \mathbb{Q}$, then $7b^2 \in \mathbb{Q}$, and then $4 - 7b^2 \in \mathbb{Q}$. This is the rational-closure step used in Example 6. For the particular number considered there, the radical equation also gives

$$7b^2 = 4 - 2\sqrt{3}, \qquad \text{so} \qquad 4 - 7b^2 = 2\sqrt{3}.$$

If that b were rational, the result just proved would make $2\sqrt{3}$ rational. Dividing by the nonzero rational number 2 would make $\sqrt{3}$ rational, contradicting Exercise 2.1. This shows how the closure statement supplies the needed step in an irrationality proof.

Conclusion. For every rational $b = m/n$ with $n \neq 0$, $4 - 7b^2 = (4n^2 - 7m^2)/n^2 \in \mathbb{Q}$.

2.7 Exercise 2.7: Two radicals that simplify to rational numbers

What we must show. Show that

$$\text{(a)} \quad \sqrt{4+2\sqrt{3}}-\sqrt{3}, \qquad \text{(b)} \quad \sqrt{6+4\sqrt{2}}-\sqrt{2}$$

are rational, and find their values. In each expression the final subtracted square root lies outside the first radical.

Plan. Recognize each radicand as the square of a positive sum. Positivity is essential: for a real v, the principal square root satisfies $\sqrt{v^2}=|v|$. To replace that absolute value by v, we check that $v>0$.

1. Part (a): Expand the proposed positive square root

Consider $v=1+\sqrt{3}$. Because $\sqrt{3}>0$, we have $v>0$. Expanding its square gives

$$\begin{aligned}(1+\sqrt{3})^2 &= 1^2+2\cdot 1\cdot\sqrt{3}+(\sqrt{3})^2\\ &= 1+2\sqrt{3}+3\\ &= 4+2\sqrt{3}.\end{aligned}$$

The radicand is positive, and $1+\sqrt{3}$ is a positive number whose square equals that radicand. It is therefore the principal square root:

$$\sqrt{4+2\sqrt{3}}=1+\sqrt{3}.$$

One can also justify the choice of sign from uniqueness. If two nonnegative numbers v,w have the same positive square, then $(v-w)(v+w)=v^2-w^2=0$. Since $v+w>0$, it follows that $v-w=0$, so $v=w$. There is thus only one nonnegative square root here.

Substituting the identity into the expression in part (a) gives

$$\sqrt{4+2\sqrt{3}}-\sqrt{3}=(1+\sqrt{3})-\sqrt{3}=1+(\sqrt{3}-\sqrt{3})=1+0=1.$$

As $1=1/1$, this value is rational.

2. Part (b): Expand the second proposed positive square root

Now consider $w=2+\sqrt{2}$. It is positive because $\sqrt{2}>0$. Its square expands as

$$\begin{aligned}(2+\sqrt{2})^2 &= 2^2+2\cdot 2\cdot\sqrt{2}+(\sqrt{2})^2\\ &= 4+4\sqrt{2}+2\\ &= 6+4\sqrt{2}.\end{aligned}$$

Thus $2+\sqrt{2}$ is the positive square root of $6+4\sqrt{2}$. The sign check permits the identity

$$\sqrt{6+4\sqrt{2}}=2+\sqrt{2}.$$

It follows that

$$\sqrt{6+4\sqrt{2}}-\sqrt{2}=(2+\sqrt{2})-\sqrt{2}=2+(\sqrt{2}-\sqrt{2})=2+0=2.$$

Since $2=2/1$, this value is rational as well.

Conclusion. Part (a) equals 1 and part (b) equals 2; both expressions are rational numbers.

2.8 Exercise 2.8: All rational zeros of an eighth-degree polynomial

What we must show. Find every rational solution of

$$x^8 - 4x^5 + 13x^3 - 7x + 1 = 0.$$

Plan. The polynomial is monic, has integer coefficients, and has constant term 1. The monic rational-zero result restricts every rational root to an integer divisor of 1. We must test both candidates and explain why this exhausts the rational possibilities.

1. Obtain the complete candidate list

Let

$$P(x) = x^8 - 4x^5 + 13x^3 - 7x + 1.$$

The coefficient of x^8 is 1, so P is monic. Its missing powers have coefficient 0, which is also an integer. Its constant coefficient is $1 \neq 0$. Thus Tool 3 applies to every rational zero of P. It says that such a zero must be an integer dividing 1. If an integer m divides 1, then $1 = mk$ for some integer k. Taking absolute values gives $1 = |m|\,|k|$, so both nonnegative integer factors must equal 1. In particular $|m| = 1$, and hence $m = 1$ or $m = -1$. The full rational candidate list is therefore

$$\{1, -1\}.$$

Here is the denominator step directly for this particular polynomial. If a rational root is m/n in lowest terms, with $n > 0$, multiplication of $P(m/n) = 0$ by n^8 gives

$$m^8 - 4m^5n^3 + 13m^3n^5 - 7mn^7 + n^8 = 0.$$

Solving for m^8 gives

$$m^8 = n(4m^5n^2 - 13m^3n^4 + 7mn^6 - n^7),$$

so n divides m^8. Since m, n are relatively prime, repeated coprime cancellation gives $n \mid 1$, hence $n = 1$. After this reduction, $P(m) = 0$ can be rearranged as

$$1 = -m^8 + 4m^5 - 13m^3 + 7m = m(-m^7 + 4m^4 - 13m^2 + 7).$$

This proves $m \mid 1$ and recovers precisely the two candidates above. Also $P(0) = 1$, so zero has not been omitted as a possible root.

2. Evaluate the candidate 1

Every positive power of 1 equals 1. Consequently

$$\begin{aligned} P(1) &= 1^8 - 4 \cdot 1^5 + 13 \cdot 1^3 - 7 \cdot 1 + 1 \\ &= 1 - 4 + 13 - 7 + 1 \\ &= -3 + 13 - 7 + 1 \\ &= 10 - 7 + 1 \\ &= 3 + 1 = 4. \end{aligned}$$

Since $4 \neq 0$, the number 1 is not a solution.

3. Evaluate the candidate -1

The eighth power of -1 is 1, while its fifth and third powers are -1. Substituting with the signs retained gives

$$\begin{aligned} P(-1) &= (-1)^8 - 4(-1)^5 + 13(-1)^3 - 7(-1) + 1 \\ &= 1 - 4(-1) + 13(-1) + 7 + 1 \\ &= 1 + 4 - 13 + 7 + 1 \\ &= 5 - 13 + 7 + 1 \\ &= -8 + 7 + 1 \\ &= -1 + 1 = 0. \end{aligned}$$

Thus -1 is a solution, and it is rational because $-1 = (-1)/1$.

4. Explain why the search is complete

Every rational solution must belong to the candidate set $\{1, -1\}$ from Step 1. Step 2 rejects 1, and Step 3 verifies -1. There are no other rational candidates to examine.

Conclusion. The set of all rational solutions is $\{-1\}$.

3 The set of real numbers

This section concerns the rules of arithmetic, inequalities, and absolute values. We work with the usual operations on real numbers. In particular, *an inverse must belong to the set under discussion*: an integer does not have a multiplicative inverse *in the integers* merely because it has one in the real numbers.

A reference sheet for the rules

Ross's labels will be useful when explaining individual steps. For real numbers x, y, z, the field rules are

$$\begin{aligned} &\text{A1}: \quad x+(y+z)=(x+y)+z, && \text{A2}: \quad x+y=y+x,\\ &\text{A3}: \quad x+0=x, && \text{A4}: \quad x+(-x)=0,\\ &\text{M1}: \quad x(yz)=(xy)z, && \text{M2}: \quad xy=yx,\\ &\text{M3}: \quad x\cdot 1=x, && \text{M4}: \quad xx^{-1}=1 \quad (x\neq 0),\\ &\text{DL}: \quad x(y+z)=xy+xz. \end{aligned}$$

The additive identity 0, the multiplicative identity 1, the additive inverse $-x$, and (when $x \neq 0$) the multiplicative inverse x^{-1} must be elements of the field. A field has more than one element.

The order rules say that any two real numbers can be compared (O1), $x \leq y$ and $y \leq x$ together imply $x = y$ (O2), and $x \leq y \leq z$ implies $x \leq z$ (O3). They also give

$$\text{O4}: \quad x \leq y \implies x+z \leq y+z, \qquad \text{O5}: \quad x \leq y,\ 0 \leq z \implies xz \leq yz.$$

By definition, $x < y$ means that $x \leq y$ and $x \neq y$. Subtraction is addition of the additive inverse: $x - y = x + (-y)$. Division by a nonzero number is multiplication by its multiplicative inverse: $x/y = xy^{-1}$.

We may use the results already established in the text before these exercises, including additive cancellation, $x \cdot 0 = 0$, $(-x)y = -(xy)$, nonnegativity of squares, and the triangle inequality. When an exercise asks for a missing part of a theorem, that missing part is proved below rather than assumed.

3.1 Exercise 3.1: Which rules fail for natural numbers and integers?

What we must show. Determine which of A1–A4, M1–M4, DL, and O1–O5 are valid when the operations and order are the usual ones and all required elements must lie in the specified set.

(a), Step 1: fix the convention for natural numbers.

Ross uses $\mathbb{N} = \{1, 2, 3, \ldots\}$. Thus $1 \in \mathbb{N}$ but $0 \notin \mathbb{N}$. This distinction matters for the identity requirement A3.

(a), Step 2: test the additive identity.

A3 requires an additive identity *in the set*. The usual identity is 0, which is not in $\mathbb{N}$. There is no replacement positive integer e: if $e \in \mathbb{N}$, then $e \geq 1$, so

$$1 + e \geq 1 + 1 = 2 > 1.$$

Consequently $1 + e \neq 1$. Hence no natural number can act as an additive identity, and the internal identity requirement A3 fails.

There is a small interpretive issue here. The equality $n + 0 = n$ is true if we calculate in $\mathbb{Z}$ or $\mathbb{R}$. It is not a false numerical equation. It fails as a *field axiom for* $\mathbb{N}$ because its required constant 0 is outside $\mathbb{N}$. We use this internal interpretation in the answer.

(a), Step 3: test additive inverses.

A4 would require an element of $\mathbb{N}$ that can be added to 1 to give 0. But if $m \in \mathbb{N}$, then $m \geq 1$, and therefore $1 + m \geq 2$. In particular, $1 + m \neq 0$. The additive inverse -1 exists in $\mathbb{Z}$ and $\mathbb{R}$, but not in $\mathbb{N}$. Thus A4 fails.

(a), Step 4: test multiplicative inverses.

M4 would require an element $m \in \mathbb{N}$ with $2m = 1$. Every such m satisfies $m \geq 1$, so $2m \geq 2$. Therefore $2m \neq 1$. Equivalently, the required inverse $1/2$ is not a natural number. Thus M4 fails. It is enough to find the single nonzero element 2 for which the requirement fails; M4 must hold for *every* nonzero element.

(a), Step 5: check the remaining rules.

Addition and multiplication of positive integers again give positive integers. Associativity and commutativity remain true, so A1, A2, M1, and M2 hold. Since $1 \in \mathbb{N}$ and $n \cdot 1 = n$, M3 holds. The distributive identity holds for positive integers, so DL holds. The usual comparisons of natural numbers satisfy O1–O4. Finally, multiplying an inequality between natural numbers by a natural number, which is positive, preserves its direction; this verifies the numerical inequality in O5. The occurrence of 0 in the condition of O5 is understood in the ambient usual order. No additional order inequality becomes false upon restricting its variables to $\mathbb{N}$.

Conclusion. For $\mathbb{N}$ the failed internal field requirements are $\boxed{\text{A3, A4, M4}}$. The remaining listed arithmetic identities and order implications are valid with the usual arithmetic and order.

(b), Step 1: identify what changes for integers.

Now the set is $\mathbb{Z}$, so $0 \in \mathbb{Z}$ and $-n \in \mathbb{Z}$ whenever $n \in \mathbb{Z}$. Thus $n + 0 = n$ takes place inside the set, and $n + (-n) = 0$ also takes place inside the set. A3 and A4 both hold.

(b), Step 2: exhibit the remaining failure.

The element 2 is a nonzero integer. If it had an inverse $m \in \mathbb{Z}$, we would have $2m = 1$. The left side is an even integer, since it is twice an integer; the right side is odd. Hence no such integer m exists. This again says that $1/2 \notin \mathbb{Z}$, so M4 fails.

(b), Step 3: check every other requirement.

The sum and product of integers are integers. The associative, commutative, and distributive identities remain valid. Both identities, 0 and 1, are integers. Additive inverses are integers. The usual order of integers satisfies O1–O4, and multiplication by a nonnegative integer preserves a weak inequality, as required by O5.

Conclusion. For $\mathbb{Z}$ the only failed listed requirement is $\boxed{\text{M4}}$. Some integers, namely 1 and -1, do have integer multiplicative inverses, but that does not meet a requirement imposed on every nonzero integer.

3.2 Exercise 3.2: Locate the hidden uses of additive commutativity

What we must show. Find the uses of A2 in the proofs of Theorem 3.1(ii) and (iii).

Plan. The cancellation rule proved in Theorem 3.1(i) is written with the common term on the *right*:

$$x + c = y + c \quad \Longrightarrow \quad x = y.$$

Before using this particular form, both equalities must have that shape.

(a), Step 1: write what distributivity actually gives.

Set $t = a \cdot 0$. Since $0 + 0 = 0$ by A3, distributivity gives

$$t = a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0 = t + t.$$

To use right cancellation with the common term t, the proof rewrites the leftmost t as $0 + t$.

(a), Step 2: justify that rewriting.

A3 gives $t + 0 = t$, with the zero on the *right*. A2 is needed to move it to the left:

$$0 + t = t + 0 = t.$$

Thus $t = t + t$ becomes $0 + t = t + t$. Right cancellation of t now gives $0 = t = a \cdot 0$.

Conclusion. The hidden use of A2 is the passage from $a \cdot 0$ to $0 + a \cdot 0$: first A3 permits $a \cdot 0 + 0$, then A2 switches the two summands.

(b), Step 1: identify the equality just before cancellation.

The proof obtains

$$ab + (-a)b = ab + \big(-(ab)\big).$$

Here the common summand ab is on the *left*, whereas the previously proved cancellation statement has the common summand on the right.

(b), Step 2: apply A2 to both sums.

The two relevant switches are

$$ab + (-a)b = (-a)b + ab, \qquad ab + \big(-(ab)\big) = \big(-(ab)\big) + ab.$$

The equality therefore becomes

$$(-a)b + ab = \big(-(ab)\big) + ab.$$

Now right cancellation with common term ab proves $(-a)b = -(ab)$.

(b), Step 3: distinguish addition from multiplication.

There are also commutative switches in the preceding factorization, but those use M2, not A2. For example,

$$\begin{aligned} ab + (-a)b &= ba + b(-a) && \text{by M2 in the two products,} \\ &= b(a + (-a)) && \text{by DL, read in reverse,} \\ &= b \cdot 0 && \text{by A4,} \end{aligned}$$

$$= 0 \qquad \text{by Theorem 3.1(ii).}$$

No exchange of *summands* occurs in this displayed chain.

Conclusion. The A2 needed for the final use of Theorem 3.1(i) moves the common additive term ab from the left to the right on both sides. Product switches elsewhere in the argument use M2.

3.3 Exercise 3.3: A product of two negatives and multiplicative cancellation

What we must show. Prove Theorem 3.1(iv), $(-a)(-b) = ab$, and Theorem 3.1(v), $ac = bc$ with $c \neq 0$ implies $a = b$.

Part (iv), Step 1: prove that taking the negative twice returns the number.

For any real u, A4 gives

$$u + (-u) = 0.$$

Apply A4 again, now to the number $-u$:

$$(-u) + \big(-(-u)\big) = 0.$$

By A2 the second left side is $-(-u) + (-u)$. Hence

$$u + (-u) = -(-u) + (-u).$$

The common right summand is $-u$. Additive cancellation gives

$$u = -(-u).$$

This proves the double-negative rule without using a product of two negative numbers.

Part (iv), Step 2: reduce the product to the rule already proved.

Theorem 3.1(iii) states $(-x)y = -(xy)$ for all real x, y. First apply it with $x = a$ and $y = -b$:

$$(-a)(-b) = -\big(a(-b)\big).$$

Next rewrite the product inside the parentheses:

$$\begin{aligned} a(-b) &= (-b)a && \text{by M2,} \\ &= -(ba) && \text{by Theorem 3.1(iii) with } x = b,\ y = a, \\ &= -(ab) && \text{by M2 inside the negative.} \end{aligned}$$

Substitute this expression into the previous equality:

$$(-a)(-b) = -\big(-(ab)\big) = ab.$$

The last equality is Step 1 with $u = ab$.

Conclusion. For all real a, b, $\boxed{(-a)(-b) = ab}$.

Part (v), Step 1: use the nonzero assumption.

Suppose $ac = bc$ and $c \neq 0$. By M4 the number c^{-1} exists and satisfies $cc^{-1} = 1$. Multiply both sides of the given equality on the right by this *same* real number:

$$(ac)c^{-1} = (bc)c^{-1}.$$

Part (v), Step 2: simplify each side, naming each rule.

For the left side,

$$(ac)c^{-1} = a(cc^{-1}) = a \cdot 1 = a.$$

These equalities use M1, M4, and M3, respectively. The same rules give

$$(bc)c^{-1} = b(cc^{-1}) = b \cdot 1 = b.$$

Replacing the two sides of $(ac)c^{-1} = (bc)c^{-1}$ by these simplified forms gives $a = b$.

Conclusion. If $ac = bc$ and $c \neq 0$, then $\boxed{a = b}$. The condition $c \neq 0$ is necessary: $1 \cdot 0 = 2 \cdot 0$, but $1 \neq 2$.

3.4 Exercise 3.4: Positivity of one and reversal of reciprocals

What we must show. Prove Theorem 3.2(v), $0 < 1$, and Theorem 3.2(vii):

$$0 < a < b \quad \Longrightarrow \quad 0 < b^{-1} < a^{-1}.$$

Part (v), Step 1: show that 1 is nonnegative.

Theorem 3.2(iv), already proved in the text, says $0 \leq x^2$ for every real x. Take $x = 1$. By M3, $1^2 = 1 \cdot 1 = 1$. Therefore

$$0 \leq 1.$$

Part (v), Step 2: rule out equality with zero.

Suppose, for a contradiction, that $1 = 0$. Let x be any element of the field. Then

$$x = x \cdot 1 = x \cdot 0 = 0.$$

The first equality uses M3; the second substitutes $1 = 0$; the third uses Theorem 3.1(ii). Thus *every* field element would equal 0, leaving only one element. This contradicts the requirement that a field have more than one element. Consequently $1 \neq 0$.

Conclusion. Since $0 \leq 1$ and $0 \neq 1$, the definition of strict inequality gives $\boxed{0 < 1}$.

Part (vii), Step 1: record why the inverses are positive.

If $t > 0$, then $t \neq 0$, so M4 supplies t^{-1} with $tt^{-1} = 1$. Here is the proof of its positivity, as in Theorem 3.2(vi). First t^{-1} cannot be zero, because then $1 = tt^{-1} = t \cdot 0 = 0$, contradicting Part (v). If $t^{-1} < 0$, order reversal under negation gives $0 < -t^{-1}$. The product of nonnegative numbers is nonnegative (Theorem 3.2(iii)), so

$$0 \leq t(-t^{-1}) = -(tt^{-1}) = -1.$$

The product identity in the middle follows from Theorem 3.1(iii) and M2. Negating $0 \leq -1$ reverses its direction and gives $1 \leq 0$. Together with $0 \leq 1$, O2 would give $1 = 0$, again contradicting Part (v). Thus t^{-1} is neither zero nor negative, so $t^{-1} > 0$.

In the hypothesis $0 < a < b$, we have $a > 0$ and $b > 0$. Applying the preceding argument to $t = a$ and $t = b$ gives

$$a^{-1} > 0, \qquad b^{-1} > 0.$$

Part (vii), Step 2: justify multiplication of strict inequalities.

If $x < y$ and $c > 0$, then O5 first gives the weak inequality $xc \leq yc$. Equality $xc = yc$ would imply $x = y$ by Exercise 3.3(v), since $c \neq 0$. But $x < y$ includes $x \neq y$. Therefore equality is impossible, and

$$x < y, \quad c > 0 \quad \Longrightarrow \quad xc < yc.$$

This explains why strictness is retained in the following two steps.

Part (vii), Step 3: multiply first by a^{-1}.

Start from $a < b$ and use the positive multiplier a^{-1}:

$$aa^{-1} < ba^{-1}.$$

M4 turns the left side into 1, so

$$1 < ba^{-1}.$$

Part (vii), Step 4: multiply next by b^{-1}.

The multiplier b^{-1} is also positive. Thus

$$1 \cdot b^{-1} < (ba^{-1})b^{-1}.$$

The left side equals b^{-1} by M2 and M3. On the right,

$$(ba^{-1})b^{-1} = (a^{-1}b)b^{-1} = a^{-1}(bb^{-1}) = a^{-1} \cdot 1 = a^{-1}.$$

These steps use M2, M1, M4, and M3, in that order. Hence $b^{-1} < a^{-1}$.

Conclusion. Combining this strict inequality with $b^{-1} > 0$ gives $\boxed{0 < b^{-1} < a^{-1}}$. Positivity of a and b is essential to the choice of positive multipliers.

3.5 Exercise 3.5: Absolute-value bounds and the reverse triangle inequality

What we must show. Prove (a) $|b| \leq a$ if and only if $-a \leq b \leq a$, and (b) $||a| - |b|| \leq |a - b|$.

Part (a), Step 1: unpack the definition of absolute value.

For a real number x,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

In both cases $|x| \geq 0$. We also have

$$-|x| \leq x \leq |x|.$$

To verify the second statement rather than assume it, consider the two cases. If $x \geq 0$, then $|x| = x$, the upper inequality is equality, and the lower inequality is $-x \leq x$, since $-x \leq 0 \leq x$. If $x < 0$, then $|x| = -x$, the lower inequality is equality $-|x| = x$, and the upper inequality follows from $x < 0 < -x$.

Part (a), Step 2: prove the forward implication.

Assume $|b| \leq a$. The bounds from Step 1 give $b \leq |b|$. Transitivity then gives

$$b \leq |b| \leq a,$$

so $b \leq a$. Negating the assumed inequality reverses its direction:

$$-a \leq -|b|.$$

Since $-|b| \leq b$ by Step 1, transitivity gives $-a \leq b$. The two conclusions together are $-a \leq b \leq a$.

Part (a), Step 3: prove the reverse implication.

Assume both $-a \leq b$ and $b \leq a$. If $b \geq 0$, then $|b| = b$, so the second assumption gives $|b| \leq a$. If $b < 0$, negate the first assumption $-a \leq b$ to obtain $-b \leq a$. In this case $|b| = -b$, so again $|b| \leq a$. These two cases exhaust all real b by O1.

Part (a), Step 4: check zero and negative bounds.

If $a = 0$, the left condition says $|b| \leq 0$, which forces $|b| = 0$ and hence $b = 0$ by the definition. The right condition is $0 \leq b \leq 0$, which forces $b = 0$ by O2. If $a < 0$, the left condition is impossible because $0 \leq |b|$, whereas the right condition is impossible because $-a > 0 > a$: no real b can simultaneously be at least $-a$ and at most a. Thus the equivalence includes these bounds as well.

Conclusion. For all real a, b, including $a \leq 0$, $\boxed{|b| \leq a \iff -a \leq b \leq a}$.

Part (b), Step 1: obtain a one-sided bound from the triangle inequality.

The equality $a = (a - b) + b$ follows by associating $a + ((-b) + b)$ and using $(-b) + b = 0$. Apply the triangle inequality to the sum $(a - b) + b$:

$$|a| = |(a - b) + b| \leq |a - b| + |b|.$$

Add $-|b|$ to both sides using O4. After associating and cancelling $|b| + (-|b|)$, the result is

$$|a| - |b| \leq |a - b|. \tag{1}$$

Part (b), Step 2: bound the negative of the same expression.

Interchange the roles of a and b in the preceding argument:

$$|b| - |a| \leq |b - a|. \tag{2}$$

We need to identify the right side. First, $b - a = -(a - b)$, as follows by distributing the negative over $a + (-b)$ and using the double-negative rule. Also $|-x| = |x|$ for any real x: when $x > 0$, the two absolute values are $-(-x) = x$ and x; when $x < 0$, both are $-x$; when $x = 0$, both are 0. Therefore

$$|b - a| = |-(a - b)| = |a - b|.$$

Substitute this into (2):

$$|b| - |a| \leq |a - b| .$$

Negating both sides reverses the inequality, so

$$-|a - b| \leq |a| - |b| . \tag{3}$$

Part (b), Step 3: combine and apply part (a).

Together (1) and (3) give the two-sided bound

$$-|a - b| \leq |a| - |b| \leq |a - b| .$$

In part (a), take its bounded number to be $|a| - |b|$ and its bound to be $|a - b|$. That equivalence turns this two-sided bound into

$$||a| - |b|| \leq |a - b| .$$

Conclusion. The reverse triangle inequality is $\boxed{||a| - |b|| \leq |a - b|}$ for every real a, b.

3.6 Exercise 3.6: The triangle inequality for three and for finitely many terms

What we must show. Prove (a) $|a + b + c| \leq |a| + |b| + |c|$ and (b) $|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$ for every positive integer n and every choice of real summands.

Part (a), Step 1: group the sum into two terms.

By A1, $a + b + c = (a + b) + c$. The two-term triangle inequality, applied to the real numbers $a + b$ and c, gives

$$|a + b + c| \leq |a + b| + |c| . \tag{4}$$

Part (a), Step 2: apply the two-term inequality again.

Apply it to a and b:

$$|a + b| \leq |a| + |b| .$$

Add $|c|$ to both sides by O4:

$$|a + b| + |c| \leq |a| + |b| + |c| . \tag{5}$$

By transitivity, combining (4) with (5) gives

$$|a + b + c| \leq |a| + |b| + |c| .$$

Conclusion. The three-term inequality follows from two applications of the two-term inequality; no enumeration of the signs of a, b, c is needed.

Part (b), Step 1: state exactly what is being proved by induction.

Let P_n be the assertion that

$$|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$$

for *every* list of n real numbers $a_1, \ldots, a_n$. This quantifier is important: in the induction step we may choose any list of $n + 1$ numbers and apply P_n to its first n entries.

Part (b), Step 2: verify the base case.

For $n = 1$, the claimed inequality is simply $|a_1| \le |a_1|$. The two sides are equal, so P_1 holds for every real a_1.

Part (b), Step 3: state the induction hypothesis and the next target.

Fix $n \ge 1$ and assume P_n. Let $a_1, \dots, a_n, a_{n+1}$ be arbitrary real numbers. The target is

$$|a_1 + \cdots + a_n + a_{n+1}| \le |a_1| + \cdots + |a_n| + |a_{n+1}| \,.$$

Set $s = a_1 + \cdots + a_n$. The induction hypothesis gives $|s| \le \sum_{j=1}^{n} |a_j|$.

Part (b), Step 4: prove the induction step.

The two-term triangle inequality and then the induction hypothesis give

$$\begin{aligned} |a_1 + \cdots + a_n + a_{n+1}| &= |s + a_{n+1}| && \text{by the definition of } s \text{ and A1,} \\ &\le |s| + |a_{n+1}| && \text{by the triangle inequality,} \\ &\le \sum_{j=1}^{n} |a_j| + |a_{n+1}| && \text{by the induction hypothesis and O4,} \\ &= \sum_{j=1}^{n+1} |a_j| && \text{by the definition of the sum.} \end{aligned}$$

The list of $n + 1$ real numbers was arbitrary, so P_{n+1} follows.

Conclusion. Since P_1 holds and P_n implies P_{n+1} for every $n \ge 1$, mathematical induction proves the finite-sum triangle inequality for every $n \in \mathbb{N}$.

3.7 Exercise 3.7: Strict absolute-value bounds and intervals centered at a point

What we must show. Establish all three equivalences:

$$\begin{array}{lll} \text{(a)} & |b| < a & \iff \quad -a < b < a, \\ \text{(b)} & |a - b| < c & \iff \quad b - c < a < b + c, \\ \text{(c)} & |a - b| \le c & \iff \quad b - c \le a \le b + c. \end{array}$$

Part (a), Step 1: prove the forward implication.

Suppose $|b| < a$. From Exercise 3.5(a), Step 1, we know $b \le |b|$ and $-|b| \le b$. Thus

$$b \le |b| < a \quad \Longrightarrow \quad b < a.$$

Negating $|b| < a$ gives $-a < -|b|$. This strict reversal is justified by Theorem 3.2(i) for the weak inequality; equality of the negatives would imply equality of the originals by taking their negatives again, so equality is excluded. Combining it with $-|b| \le b$ gives $-a < b$. Together these conclusions are $-a < b < a$.

Part (a), Step 2: prove the reverse implication.

Assume $-a < b < a$. If $b \geq 0$, then $|b| = b < a$ by the right-hand assumed inequality. If $b < 0$, negate the left-hand assumed inequality $-a < b$ to get $-b < a$; since $|b| = -b$ in this case, we again obtain $|b| < a$. The case $b = 0$ belongs to the first case.

Part (a), Step 3: check nonpositive values of the bound.

If $a \leq 0$, the left condition cannot hold: it would give $0 \leq |b| < a \leq 0$. The right condition cannot hold either. Indeed $a \leq 0$ implies $0 \leq -a$, hence $a \leq -a$, while $-a < b < a$ would imply $-a < a$. Those inequalities contradict each other. Thus the equivalence remains true when both conditions are false.

Conclusion. For all real a, b, $\boxed{|b| < a \iff -a < b < a}$. Either condition, when true, forces $a > 0$.

Part (b), Step 1: prove the forward implication.

Assume $|a - b| < c$. Apply part (a) to the real number $a - b$ with bound c. It gives

$$-c < a - b < c.$$

Add b to the three expressions. Addition preserves strict inequality: O4 preserves the weak inequality, and equality after addition would contradict additive cancellation. Thus

$$-c + b < (a - b) + b < c + b.$$

A1 and A4 give $(a - b) + b = a$; A2 gives $-c + b = b - c$ and $c + b = b + c$. Therefore $b - c < a < b + c$.

Part (b), Step 2: prove the reverse implication.

Assume $b - c < a < b + c$. Add $-b$ to every expression:

$$(b - c) - b < a - b < (b + c) - b.$$

Associativity and commutativity let b and $-b$ become adjacent; A4 and A3 then give $(b - c) - b = -c$ and $(b + c) - b = c$. We obtain $-c < a - b < c$, and part (a) gives $|a - b| < c$.

Conclusion. For all real a, b, c, $\boxed{|a - b| < c \iff b - c < a < b + c}$. If $c \leq 0$, both conditions are false, as follows from part (a); thus no unstated assumption $c > 0$ is needed for the equivalence.

Part (c), Step 1: prove the forward implication.

Assume $|a - b| \leq c$. Exercise 3.5(a), with bounded number $a - b$ and bound c, gives

$$-c \leq a - b \leq c.$$

Use O4 to add b throughout:

$$-c + b \leq (a - b) + b \leq c + b.$$

The same algebra as in part (b) simplifies the three expressions, giving $b - c \leq a \leq b + c$.

Part (c), Step 2: prove the reverse implication.

Assume $b - c \leq a \leq b + c$. Add $-b$ throughout by O4 and simplify:

$$(b - c) - b \leq a - b \leq (b + c) - b \quad \Longrightarrow \quad -c \leq a - b \leq c.$$

Exercise 3.5(a) now gives $|a - b| \leq c$.

Part (c), Step 3: check the boundary cases.

If $c < 0$, an absolute value cannot be at most c, so the left condition is false. Also $b - c > b + c$, since subtracting gives $(b - c) - (b + c) = -2c > 0$; hence no a can lie between those endpoints in the specified order. The right condition is false as well. If $c = 0$, the left condition forces $a - b = 0$, or $a = b$; the right condition becomes $b \le a \le b$, which also forces $a = b$ by O2.

Conclusion. For all real a, b, c, $\boxed{|a - b| \le c \iff b - c \le a \le b + c}$.

3.8 Exercise 3.8: Comparison with every number above a fixed bound

What we must show. Suppose that $a, b \in \mathbb{R}$ and

$$a \le b_1 \qquad \text{for every real number } b_1 \text{ satisfying } b_1 > b. \tag{6}$$

Prove $a \le b$.

Plan. If a were above b, the arithmetic midpoint between a and b would be above b but still below a. That specific choice would contradict (6). We will calculate both comparisons explicitly.

Step 1: negate the desired conclusion.

Suppose, for a contradiction, that $a \le b$ is false. By O1, either $a \le b$ or $b \le a$. Since the first alternative is excluded, $b \le a$. Equality $a = b$ would imply $a \le b$, so equality is also excluded. Consequently $b < a$, or equivalently $a > b$. Adding $-b$ to this strict inequality gives

$$d := a - b > 0.$$

Step 2: justify division by two and its sign.

Exercise 3.4 gives $1 > 0$. Adding 1 to $0 < 1$ gives $1 < 2$, and transitivity with $0 < 1$ gives $2 > 0$. Thus $2 \ne 0$, so its inverse $1/2$ exists and is positive by Theorem 3.2(vi). Multiplying $0 < d$ by the positive number $1/2$ preserves strict inequality, by the argument in Exercise 3.4(vii), Step 2. Hence

$$\frac{d}{2} > 0.$$

Step 3: construct a number to which the hypothesis applies.

Define

$$m = \frac{a + b}{2}.$$

We compute its difference from b using a common denominator:

$$\begin{aligned}
m - b &= \frac{a + b}{2} - \frac{2b}{2} && \text{because } 2 \ne 0 \text{ and } (2b)/2 = b, \\
&= \frac{a + b - 2b}{2} && \text{by DL and the definition of division,} \\
&= \frac{a - b}{2} && \text{since } b - 2b = -b, \\
&= \frac{d}{2} > 0 && \text{by the definition of } d \text{ and Step 2.}
\end{aligned}$$

Adding b to $m - b > 0$ gives $m > b$. Thus $b_1 = m$ satisfies the restriction in (6).

Step 4: show that this same number is below a.

Again use a common denominator:

$$\begin{aligned} a - m &= \frac{2a}{2} - \frac{a+b}{2}, \\ &= \frac{2a - a - b}{2}, \\ &= \frac{a-b}{2}, \\ &= \frac{d}{2} > 0. \end{aligned}$$

Adding m throughout $a - m > 0$ gives $a > m$, or $m < a$.

Step 5: apply the universal hypothesis and exhibit the contradiction.

Since $m > b$, (6) applies with $b_1 = m$ and gives $a \leq m$. Step 4 gave $m < a$, which includes $m \leq a$ and $m \neq a$. From $a \leq m$ and $m \leq a$, O2 gives $a = m$, contrary to $m \neq a$. The assumption $a > b$ leads to a contradiction. Therefore $a \leq b$.

Conclusion. The hypothesis forces $\boxed{a \leq b}$. Only ordered-field arithmetic was used: the point m was constructed directly, without a completeness axiom or an appeal to a limit.

4 The completeness axiom

4.1 Exercise 4.1: Three upper bounds and a common analysis of the sets

What we must show. For each of the twenty-three sets in parts (a)–(w), give three distinct real upper bounds when such bounds exist. Otherwise, give the requested answer "NBA," meaning "not bounded above."

Plan. Exercises 4.2, 4.3, and 4.4 ask about the very same sets. We first analyze each set once, including the sharpness of its endpoint bounds. Four answer lists then state the answer to every part of every exercise and link it to its proof. This keeps the proofs of the intersections, the prime-number assertion, and the trigonometric values available without repeating them four times.

Definitions and tools for all four exercises

An *upper bound* $U \in \mathbb{R}$ for a set S must satisfy $s \le U$ for every $s \in S$. A *lower bound* $L \in \mathbb{R}$ must satisfy $L \le s$ for every $s \in S$. Thus, once U is known to be an upper bound, every number at least U is another upper bound: if $s \le U$ and $U \le V$, then $s \le V$. Similarly, every number at most a known lower bound is another lower bound.

For a proposed supremum u, we prove two things:

1. $s \le u$ for every member s of the set;

2. for each real $U < u$, there is a member $s > U$.

The second condition means that no number smaller than u is an upper bound, so u is the *least* upper bound. For a proposed infimum ℓ, the corresponding conditions are that $\ell \le s$ for every member, and that every real $L > \ell$ has some member $s < L$. If a bound belongs to the set, that member itself proves sharpness: any smaller upper bound, or larger lower bound, fails on that member. Thus a maximum is a supremum, and a minimum is an infimum.

All suprema and infima in these four exercises are *real* numbers. An unbounded-above set has "NO sup," and an unbounded-below set has "NO inf," in the conventions of this section. These answers are not assignments of extended-real values.

Tool 1: The Archimedean property and small reciprocals.

The Archimedean property implies that, for every real B, there is $n \in \mathbb{N}$ with $n > B$. If $B \le 0$, $n = 1$ works; if $B > 0$, apply the Archimedean property with $a = 1$ and $b = B$. Consequently, for every $\delta > 0$, there is an $n \in \mathbb{N}$ such that

$$n > \frac{1}{\delta} \quad \text{and therefore} \quad 0 < \frac{1}{n} < \delta.$$

To justify the last strict inequality, multiply $n > 1/\delta$ by the positive number δ, obtaining $n\delta > 1$, and then divide by the positive integer n.

Tool 2: Rational points in an open interval make its endpoints sharp.

Let $c, d \in \mathbb{R}$ with $c < d$, and consider

$$T = \{r \in \mathbb{Q} : c < r < d\}.$$

The density of $\mathbb{Q}$ shows that T is nonempty. Its defining inequalities give the upper bound d and the lower bound c. If $U < d$, put $v = \max\{U, c\}$. Both U and c are less than d, so $v < d$. Density provides a rational r with

$$\max\{U, c\} < r < d.$$

Then $c < r < d$, so $r \in T$, and $r > U$, so U is not an upper bound. This proves $\sup T = d$.

If $L > c$, put $w = \min\{L, d\}$. Both L and d are greater than c, so $c < w$. Density provides a rational r with

$$c < r < \min\{L, d\}.$$

This r belongs to T and satisfies $r < L$, proving that L is not a lower bound. Hence $\inf T = c$. Neither endpoint belongs to T, since membership requires strict inequalities. The same witnesses also prove that the real interval (c, d) has supremum d and infimum c, because those rational witnesses are real members of it.

Tool 3: Converting a quadratic inequality into an interval.

For $c > 0$ and $x \in \mathbb{R}$,

$$x^2 < c^2 \quad \Longleftrightarrow \quad -c < x < c.$$

For the reverse direction, if $-c < x < c$, then $c + x > 0$ and $c - x > 0$. Their product is positive, so

$$0 < (c - x)(c + x) = c^2 - x^2,$$

which says $x^2 < c^2$. For the forward direction, if $x \geq c > 0$, then $x^2 \geq c^2$, contradicting the strict inequality. If $x \leq -c$, then $-x \geq c > 0$ and $x^2 = (-x)^2 \geq c^2$, again a contradiction. Thus neither $x \geq c$ nor $x \leq -c$ is possible, proving $-c < x < c$.

Tool 4: Named numerical bounds used for the familiar constants.

We use the familiar estimates

$$3 < \pi < \frac{22}{7}, \qquad 2 < e < 3.$$

In particular $2 < e < 3 < \pi < 4$, because $22/7 < 4$ follows from $22 < 28$. These estimates suffice to order the finite sets involving π and e; decimal approximations are unnecessary.

A proof for every set (a)–(w)

Analysis (a): The closed interval $[0, 1]$.

Every member s satisfies $0 \leq s \leq 1$, by the definition of this interval. Both 0 and 1 belong to it. Thus 0 is its minimum and infimum, and 1 is its maximum and supremum. In particular $1, 2, 3$ are upper bounds and $0, -1, -2$ are lower bounds.

Analysis (b): The open interval $(0, 1)$.

Every member satisfies $0 < s < 1$, so 0 is a lower bound and 1 is an upper bound. Tool 2 with $c = 0$, $d = 1$ proves that no upper bound smaller than 1, and no lower bound larger than 0, can work. Therefore $\sup(0, 1) = 1$ and $\inf(0, 1) = 0$. Neither endpoint is a member, so neither extremal bound is attained. Upper bounds include $1, 2, 3$; lower bounds include $0, -1, -2$.

Analysis (c): The finite set $\{2, 7\}$.

Its two members satisfy $2 \leq 2 \leq 7$ and $2 \leq 7 \leq 7$. Consequently 2 bounds every member from below and 7 bounds every member from above. Each is itself in the set. Hence the minimum and infimum are 2, while the maximum and supremum are 7. Examples of upper bounds are $7, 8, 9$; examples of lower bounds are $2, 1, 0$.

Analysis (d): The finite set $\{\pi, e\}$.

Tool 4 gives $2 < e < 3 < \pi < 4$, and therefore $e < \pi$. Thus every member of this two-element set is at least e and at most π. Both endpoints are members, so its infimum is e and its supremum is π, both attained. Since every member is less than 4, the three distinct numbers $4, 5, 6$ are upper bounds. Since every member is greater than 2, the distinct numbers $2, 1, 0$ are lower bounds.

Analysis (e): The reciprocal set $\{1/n : n \in \mathbb{N}\}$.

For $n \geq 1$, positivity and $n \geq 1$ give

$$0 < \frac{1}{n} \leq 1.$$

At $n = 1$, the value is 1, so 1 is the maximum and supremum. The number 0 is a lower bound, but it is not a member since a positive reciprocal cannot equal 0. If $L > 0$, Tool 1 supplies n with $1/n < L$. This member defeats L as a lower bound, so 0 is the greatest lower bound. Thus the infimum is 0 and is not attained. Upper bounds include $1, 2, 3$; lower bounds include $0, -1, -2$.

Analysis (f): The singleton $\{0\}$.

Its only member is 0. Therefore 0 is both a lower and an upper bound, and is both the minimum and the maximum. It follows that the supremum and infimum both equal 0. The numbers $0, 1, 2$ are three distinct upper bounds, and $0, -1, -2$ are three distinct lower bounds.

Analysis (g): The union $[0, 1] \cup [2, 3]$.

A member s belongs to at least one of the two intervals. If $s \in [0, 1]$, then $0 \leq s \leq 1 \leq 3$. If $s \in [2, 3]$, then $0 \leq 2 \leq s \leq 3$. Thus every member satisfies $0 \leq s \leq 3$. The member 0 comes from the first interval, and the member 3 comes from the second. Hence the infimum is the attained minimum 0, and the supremum is the attained maximum 3. The gap between the two intervals changes neither endpoint bound. Upper bounds include $3, 4, 5$ and lower bounds include $0, -1, -2$.

Analysis (h): The infinite union $\bigcup_{n=1}^{\infty}[2n, 2n + 1]$.

Membership in the union means membership in *at least one* interval. If s belongs to it, there is an $n \in \mathbb{N}$ with $2n \leq s \leq 2n + 1$. Because $n \geq 1$, we obtain $2 \leq 2n \leq s$. The number 2 belongs to the interval at $n = 1$, namely $[2, 3]$. Thus 2 is the minimum and infimum, and $2, 1, 0$ are lower bounds.

There is no upper bound. To see this for an arbitrary proposed $M \in \mathbb{R}$, use Tool 1 to choose

$$n > \max\{1, M/2\}.$$

The member $2n \in [2n, 2n + 1]$ satisfies $2n > M$. Thus every proposed real upper bound fails. This set is NBA and has NO sup.

Analysis (i): The intersection $\bigcap_{n=1}^{\infty}[-1/n, 1 + 1/n]$.

Call this set I. We prove the exact identity $I = [0, 1]$ by both inclusions. If $0 \leq x \leq 1$, then for every $n \in \mathbb{N}$,

$$-\frac{1}{n} < 0 \leq x \leq 1 < 1 + \frac{1}{n}.$$

Thus x belongs to every displayed closed interval, proving $[0,1] \subseteq I$.

For the other inclusion, suppose $x \notin [0,1]$. If $x < 0$, let $\delta = -x > 0$. Tool 1 gives an n with $1/n < \delta = -x$. Multiplying this inequality by -1 reverses its direction:

$$-\frac{1}{n} > x.$$

Therefore x fails the lower-endpoint requirement in that particular interval, so $x \notin I$. If $x > 1$, let $\delta = x - 1 > 0$ and choose n with $1/n < x - 1$. Then $1 + 1/n < x$, so x fails the upper-endpoint requirement and again is not in I. These two cases exclude every real number outside $[0,1]$, proving $I \subseteq [0,1]$.

It follows that $I = [0,1]$. Its infimum is the member 0 and its supremum is the member 1. Upper bounds are $1, 2, 3$, for example, and lower bounds are $0, -1, -2$.

Analysis (j): The set $\{1 - 1/3^n : n \in \mathbb{N}\}$.

The denominator is 3^n, so the expression is $1 - 3^{-n}$. For $n \geq 1$, $3^n \geq 3$, and hence

$$0 < \frac{1}{3^n} \leq \frac{1}{3}, \qquad \frac{2}{3} = 1 - \frac{1}{3} \leq 1 - \frac{1}{3^n} < 1.$$

At $n = 1$, the lower value $2/3$ occurs. Thus $2/3$ is the minimum and infimum. The number 1 is an upper bound but is not a member.

To prove that this upper bound is sharp without assuming convergence, we first verify $3^n \geq n$ for $n \in \mathbb{N}$. At $n = 1$, $3 \geq 1$. If $3^n \geq n$, multiplication by $3 > 0$ gives $3^{n+1} \geq 3n$. Also $3n - (n+1) = 2n - 1 \geq 1 > 0$, so $3n \geq n + 1$. Induction proves the auxiliary inequality.

Now fix a proposed smaller bound $U < 1$ and set $\delta = 1 - U > 0$. Choose n by Tool 1 so that $1/n < \delta$. Since $3^n \geq n > 0$,

$$\frac{1}{3^n} \leq \frac{1}{n} < 1 - U, \qquad 1 - \frac{1}{3^n} > U.$$

This member exceeds U, so U is not an upper bound. Therefore the supremum is 1, not attained. Upper bounds include $1, 2, 3$; lower bounds include $2/3, 0, -1$.

Analysis (k): The set $\{n + (-1)^n/n : n \in \mathbb{N}\}$.

The numerator $(-1)^n$ equals either 1 or -1. Therefore, for each $n \in \mathbb{N}$,

$$n + \frac{(-1)^n}{n} \geq n - \frac{1}{n} \geq n - 1 \geq 0.$$

Here the first inequality uses $(-1)^n \geq -1$ and division by $n > 0$; the second uses $1/n \leq 1$; the third uses $n \geq 1$. At $n = 1$, the expression is $1 + (-1)/1 = 0$. Thus 0 is the minimum and infimum, and $0, -1, -2$ are lower bounds.

For any proposed real upper bound M, choose $n \in \mathbb{N}$ with $n > M + 1$, using Tool 1. The same estimate gives

$$n + \frac{(-1)^n}{n} \geq n - 1 > M.$$

Thus the set is NBA and has NO sup. The oscillating correction $(-1)^n/n$ does not keep the growing term n bounded.

Analysis (l): The rational half-line $\{r \in \mathbb{Q} : r < 2\}$.

All members are less than 2, so 2 is an upper bound. If $U < 2$, density provides a rational r with $U < r < 2$. That r is a member and is larger than U, so no $U < 2$ is an upper bound. Hence the supremum is 2. It is not attained because $r = 2$ fails the strict membership inequality. Upper bounds include $2, 3, 4$.

There is no lower bound: for a proposed $L \in \mathbb{R}$, choose $n \in \mathbb{N}$ with $n > \max\{1, -L\}$. Then $-n$ is an integer, hence rational, and $-n < 0 < 2$, so $-n$ is in the set. Also $n > -L$ gives $-n < L$. Thus every proposed lower bound fails. The answers for lower boundedness and infimum are NBB and NO inf.

Analysis (m): The rational set $\{r \in \mathbb{Q} : r^2 < 4\}$.

Tool 3 with $c = 2$ gives

$$\{r \in \mathbb{Q} : r^2 < 4\} = \{r \in \mathbb{Q} : -2 < r < 2\}.$$

Thus -2 is a lower bound and 2 is an upper bound. To see sharpness explicitly, for $U < 2$ density supplies a rational $\max\{U, -2\} < r < 2$, a member exceeding U. For $L > -2$ it supplies a rational $-2 < r < \min\{L, 2\}$, a member below L. These are exactly the two endpoint arguments of Tool 2. Thus the supremum is 2 and the infimum is -2. Neither is a member: $(2)^2 = (-2)^2 = 4$, whereas membership requires a square strictly less than 4. Three upper bounds are $2, 3, 4$; three lower bounds are $-2, -3, -4$.

Analysis (n): The rational set $\{r \in \mathbb{Q} : r^2 < 2\}$.

Let $c = \sqrt{2}$, the positive real number with $c^2 = 2$. Tool 3 gives

$$\{r \in \mathbb{Q} : r^2 < 2\} = \{r \in \mathbb{Q} : -\sqrt{2} < r < \sqrt{2}\}.$$

For $U < \sqrt{2}$, density gives a rational $\max\{U, -\sqrt{2}\} < r < \sqrt{2}$; it is a member above U. For $L > -\sqrt{2}$, density gives a rational $-\sqrt{2} < r < \min\{L, \sqrt{2}\}$; it is a member below L. Together with the bounding inequalities, this proves

$$\sup S = \sqrt{2}, \qquad \inf S = -\sqrt{2}.$$

Neither endpoint belongs to the set, since its square equals 2 instead of being less than 2. This exclusion already follows from the strict inequality; it does not require a separate irrationality argument.

Since $0 < \sqrt{2} < 2$, every member lies strictly between -2 and 2. For the last comparison, $\sqrt{2} \geq 2$ would imply $2 = (\sqrt{2})^2 \geq 4$, a contradiction. Therefore $2, 3, 4$ are upper bounds and $-2, -3, -4$ are lower bounds.

Analysis (o): The negative half-line $\{x \in \mathbb{R} : x < 0\}$.

The number 0 is an upper bound. If $U < 0$, then

$$U < \frac{U}{2} < 0,$$

because $U/2 - U = -U/2 > 0$ and $U/2$ is still negative. Thus $U/2$ is a member above the proposed bound U, proving that the supremum is 0. The value 0 is not a member. Upper bounds include $0, 1, 2$.

For an arbitrary proposed lower bound $L \in \mathbb{R}$, take $x = \min\{L, 0\} - 1$. Then

$$x < \min\{L, 0\} \leq L, \qquad x < \min\{L, 0\} \leq 0.$$

Hence x is a negative real number below L. This proves NBB and therefore NO inf.

Analysis (p): The finite set $\{1, \pi/3, \pi^2, 10\}$.

The third entry is π^2, not a cube. We verify the ordering using Tool 4. Dividing $3 < \pi < 22/7$ by the positive number 3 gives

$$1 < \frac{\pi}{3} < \frac{22}{21} < 2,$$

where $22/21 < 2$ follows from $22 < 42$. Squaring the positive inequalities $3 < \pi < 22/7$ gives

$$9 < \pi^2 < \frac{484}{49} < 10.$$

The final strict inequality holds since $484 < 490 = 10 \cdot 49$. Combining these comparisons with $2 < 9$, we obtain

$$1 < \frac{\pi}{3} < 2 < 9 < \pi^2 < 10.$$

Thus 1 is the member of smallest value, and 10 is the member of largest value. The infimum is the attained minimum 1, and the supremum is the attained maximum 10. Upper bounds include $10, 11, 12$; lower bounds include $1, 0, -1$.

Analysis (q): The finite set $\{0, 1, 2, 4, 8, 16\}$.

The displayed members satisfy

$$0 < 1 < 2 < 4 < 8 < 16.$$

Every member is at least 0 and at most 16, and both 0 and 16 appear in the set. Hence the minimum and infimum equal 0, and the maximum and supremum equal 16. Upper bounds include $16, 17, 18$, while lower bounds include $0, -1, -2$.

Analysis (r): The intersection $\bigcap_{n=1}^{\infty}(1 - 1/n, 1 + 1/n)$.

Let J denote this intersection. For each $n \in \mathbb{N}$, $1/n > 0$, so $1 - 1/n < 1 < 1 + 1/n$. Therefore 1 belongs to every one of the open intervals and hence to J.

Now suppose $x \neq 1$. If $x < 1$, take $\delta = 1 - x > 0$ and use Tool 1 to choose n with $1/n < 1 - x$. Rearranging gives $x < 1 - 1/n$. Thus x lies below the lower endpoint of that particular open interval and cannot belong to J. If $x > 1$, choose n with $1/n < x - 1$. Then $1 + 1/n < x$, and x lies above the upper endpoint of that interval, so again $x \notin J$. These two cases exclude every $x \neq 1$.

It follows that $J = \{1\}$. Thus both its infimum and its supremum are 1, and both are attained. The fact that the intersected intervals are open does not prevent their intersection from being a singleton. Three upper bounds are $1, 2, 3$; three lower bounds are $1, 0, -1$.

Analysis (s): Reciprocals of positive primes.

The set is

$$P = \{1/p : p \in \mathbb{N} \text{ and } p \text{ is prime}\}.$$

Every positive prime p satisfies $p \geq 2$, so $0 < 1/p \leq 1/2$. The integer 2 is prime: its only positive integer factors are 1 and 2. Consequently $1/2$ belongs to P, and is its maximum and supremum. The number 0 is a lower bound and is not a member. To show it is the *greatest* lower bound, we need primes as large as we wish.

The needed prime-number fact, with proof. First, each integer $m \geq 2$ has a prime divisor. The finite nonempty set

$$D = \{d \in \{2, 3, \ldots, m\} : d \text{ divides } m\}$$

contains m, so it has a smallest member p. If p were not prime, it would have a positive divisor a other than 1 and p, with $2 \leq a < p$. Writing $p = ab$ and $m = pc$ with integers b, c gives $m = a(bc)$, so a would divide m and belong to D. That contradicts the minimality of p. Hence p is prime.

Now let $B \in \mathbb{R}$ be arbitrary. By Tool 1 choose an integer $N > \max\{B, 2\}$. The integer $m = N! + 1$ is at least 2, so it has a prime divisor p. If $p \leq N$, then p is one of the factors in the product $N! = 1 \cdot 2 \cdots N$. Thus both $N!$ and $N! + 1$ would be divisible by p. Subtracting their integer-multiple representations would make

$$(N! + 1) - N! = 1$$

divisible by p. This is impossible for an integer $p \geq 2$: if $1 = pq$ for an integer q, then $q \neq 0$ and $1 = p\,|q| \geq p \geq 2$ after taking absolute values, a contradiction. Therefore $p > N > B$. This proves that the primes are unbounded above.

Finally let $L > 0$ be an arbitrary proposed lower bound for P. Apply the fact just proved with $B = 1/L$ to obtain a prime $p > 1/L$. Multiplication by $L > 0$ gives $pL > 1$, and division by $p > 0$ then gives $1/p < L$. This member of P defeats the proposed lower bound L. Therefore

$$\inf P = 0, \qquad \sup P = \frac{1}{2}.$$

Only the supremum is attained. Three upper bounds are $1/2, 1, 2$; three lower bounds are $0, -1, -2$.

Analysis (t): The real set $\{x \in \mathbb{R} : x^3 < 8\}$.

We first identify the inequality's exact solution set. Multiplication shows

$$\begin{aligned}(x - 2)(x^2 + 2x + 4) &= x^3 + 2x^2 + 4x - 2x^2 - 4x - 8 \\ &= x^3 - 8.\end{aligned}$$

The second factor is always positive, because

$$x^2 + 2x + 4 = (x + 1)^2 + 3 \geq 3 > 0.$$

The sign of $x^3 - 8$ is therefore the sign of $x - 2$. It follows that $x^3 < 8$ if and only if $x < 2$.

Thus 2 is an upper bound. If $U < 2$, the midpoint $x = (U + 2)/2$ satisfies

$$x - U = \frac{2 - U}{2} > 0, \qquad 2 - x = \frac{2 - U}{2} > 0.$$

Hence $U < x < 2$, so x belongs to the set and is above U. This proves that the supremum is 2. It is not attained since $2^3 = 8$ fails the strict inequality. Upper bounds include $2, 3, 4$.

For any $L \in \mathbb{R}$, let $x = \min\{L, 2\} - 1$. Then $x < L$ and $x < 2$, so x is a member below the proposed lower bound L. Thus the set is NBB and has NO inf.

Analysis (u): The set of real squares $\{x^2 : x \in \mathbb{R}\}$.

Every square is nonnegative, so 0 is a lower bound. Taking $x = 0$ gives the member $0^2 = 0$. Therefore 0 is the minimum and infimum, and $0, -1, -2$ are lower bounds.

To defeat a proposed upper bound $M \in \mathbb{R}$, use Tool 1 to choose $n \in \mathbb{N}$ with $n > \max\{M, 1\}$. Since $n \geq 1$, $n^2 \geq n > M$. Taking $x = n$ produces the member $x^2 = n^2$ above M. Thus this set is NBA and has NO sup.

Analysis (v): The values of $\cos(n\pi/3)$.

We record a period table, which also supplies the proof for (w). For a positive integer n, division of $n - 1$ by 6 gives $n = 6q + r$ with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6\}$. Therefore

$$\frac{n\pi}{3} = 2q\pi + \frac{r\pi}{3}.$$

The period 2π of sine and cosine shows that their values depend only on r. The familiar unit-circle values are

r	$r\pi/3$	$\cos(r\pi/3)$	$\sin(r\pi/3)$
1	$\pi/3$	$1/2$	$\sqrt{3}/2$
2	$2\pi/3$	$-1/2$	$\sqrt{3}/2$
3	π	-1	0
4	$4\pi/3$	$-1/2$	$-\sqrt{3}/2$
5	$5\pi/3$	$1/2$	$-\sqrt{3}/2$
6	2π	1	0

Every positive integer has one of these residues, so no other values occur. Every row occurs at $n = r$, so none of the listed values is missing from the corresponding set.

In particular the cosine set is exactly

$$\{-1, -1/2, 1/2, 1\}.$$

Its minimum and infimum are -1, attained at $n = 3$, and its maximum and supremum are 1, attained at $n = 6$. Upper bounds include $1, 2, 3$; lower bounds include $-1, -2, -3$.

Analysis (w): The values of $\sin(n\pi/3)$.

The period argument and all six sine values were displayed in Analysis (v). They show that the sine set is exactly

$$\{-\sqrt{3}/2, 0, \sqrt{3}/2\}.$$

Because $\sqrt{3} > 0$, these three numbers have the displayed increasing order. Thus its infimum is the member $-\sqrt{3}/2$ at $n = 4$, and its supremum is the member $\sqrt{3}/2$ at $n = 1$. Also $\sqrt{3} < 2$: the contrary inequality would imply $3 = (\sqrt{3})^2 \geq 4$. Hence $0 < \sqrt{3}/2 < 1$, so all members are between -1 and 1. Three upper bounds are $1, 2, 3$; three lower bounds are $-1, -2, -3$.

Exercise 4.1: Explicit answers for all twenty-three parts

Each row gives three *distinct* upper bounds, or NBA. A number equal to the supremum is permitted as an upper bound whether or not it belongs to the set. In a bounded row, the reason proves that the smallest listed number is an upper bound; the larger two then follow by transitivity of $\leq$. The linked analysis supplies the full membership or unboundedness argument.

Part	Three upper bounds / NBA	Why these are valid
(a)	$1, 2, 3$	Every member is at most 1; Analysis (a).
(b)	$1, 2, 3$	Every member is less than 1; Analysis (b).
(c)	$7, 8, 9$	Both members are at most 7; Analysis (c).
(d)	$4, 5, 6$	$e < 3 < \pi < 4$; Analysis (d).
(e)	$1, 2, 3$	$1/n \leq 1$ for every $n \geq 1$; Analysis (e).
(f)	$0, 1, 2$	The only member is 0; Analysis (f).
(g)	$3, 4, 5$	Each interval lies at or below 3; Analysis (g).
(h)	NBA	The members $2n$ exceed any proposed bound; Analysis (h).
(i)	$1, 2, 3$	The intersection equals $[0, 1]$; Analysis (i).
(j)	$1, 2, 3$	$1 - 3^{-n} < 1$; Analysis (j).
(k)	NBA	Members are at least $n - 1$, unbounded above; Analysis (k).
(l)	$2, 3, 4$	Membership requires $r < 2$; Analysis (l).

Part	Three upper bounds / NBA	Why these are valid
(m)	$2, 3, 4$	$r^2 < 4$ implies $r < 2$; Analysis (m).
(n)	$2, 3, 4$	$r < \sqrt{2} < 2$; Analysis (n).
(o)	$0, 1, 2$	Membership requires $x < 0$; Analysis (o).
(p)	$10, 11, 12$	10 is the largest of the four members; Analysis (p).
(q)	$16, 17, 18$	16 is the largest listed member; Analysis (q).
(r)	$1, 2, 3$	The intersection is $\{1\}$; Analysis (r).
(s)	$1/2, 1, 2$	Every prime is at least 2; Analysis (s).
(t)	$2, 3, 4$	$x^3 < 8$ is equivalent to $x < 2$; Analysis (t).
(u)	NBA	Squares n^2 exceed any real bound; Analysis (u).
(v)	$1, 2, 3$	The four cosine values are at most 1; Analysis (v).
(w)	$1, 2, 3$	The largest sine value is $\sqrt{3}/2 < 1$; Analysis (w).

Conclusion. The only sets in Exercise 4.1 that are not bounded above are (h), (k), and (u). Every other part has the three distinct upper bounds explicitly listed and justified above.

4.2 Exercise 4.2: Three lower bounds for each of the same sets

What we must show. For each set (a)–(w) from Exercise 4.1, list three distinct real lower bounds or give NBB, meaning "not bounded below."

Plan. Use the lower-endpoint conclusions of the shared analyses. If ℓ is a lower bound and $L \leq \ell$, then $L \leq \ell \leq s$ for each member s, so L is also a lower bound. Consequently it suffices in each bounded row to justify the largest of the three listed candidates; the other two are smaller. For an NBB row, the linked proof constructs, for every $L \in \mathbb{R}$, a member strictly below L.

Part	Three lower bounds / NBB	Why these are valid
(a)	$0, -1, -2$	Every member is at least 0; Analysis (a).
(b)	$0, -1, -2$	Every member is positive; Analysis (b).
(c)	$2, 1, 0$	Both members are at least 2; Analysis (c).
(d)	$2, 1, 0$	$2 < e < \pi$; Analysis (d).
(e)	$0, -1, -2$	Every $1/n$ is positive; Analysis (e).
(f)	$0, -1, -2$	The only member is 0; Analysis (f).
(g)	$0, -1, -2$	Both intervals lie at or above 0; Analysis (g).
(h)	$2, 1, 0$	Each interval starts at $2n \geq 2$; Analysis (h).
(i)	$0, -1, -2$	The intersection is $[0, 1]$; Analysis (i).
(j)	$2/3, 0, -1$	$1 - 3^{-n} \geq 2/3$; Analysis (j).
(k)	$0, -1, -2$	$n + (-1)^n/n \geq 0$; Analysis (k).
(l)	NBB	Negative integers in the set go below any L; Analysis (l).

Part	Three lower bounds / NBB	Why these are valid
(m)	$-2, -3, -4$	$r^2 < 4$ implies $r > -2$; Analysis (m).
(n)	$-2, -3, -4$	$r > -\sqrt{2} > -2$; Analysis (n).
(o)	NBB	$\min\{L, 0\} - 1$ is a member below L; Analysis (o).
(p)	$1, 0, -1$	1 is the smallest of the four members; Analysis (p).
(q)	$0, -1, -2$	0 is the smallest listed member; Analysis (q).
(r)	$1, 0, -1$	The only member of the intersection is 1; Analysis (r).
(s)	$0, -1, -2$	All prime reciprocals are positive; Analysis (s).
(t)	NBB	$\min\{L, 2\} - 1$ satisfies $x^3 < 8$ and $x < L$; Analysis (t).
(u)	$0, -1, -2$	Every real square is nonnegative; Analysis (u).
(v)	$-1, -2, -3$	The cosine values are at least -1; Analysis (v).
(w)	$-1, -2, -3$	The least sine value is $-\sqrt{3}/2 > -1$; Analysis (w).

Conclusion. The only sets in Exercise 4.2 that are not bounded below are (l), (o), and (t). Each remaining part has three distinct lower bounds in its row.

4.3 Exercise 4.3: The real supremum of each set

What we must show. For each of the twenty-three sets, give its supremum when that real least upper bound exists; otherwise write NO sup.

Plan. A mere upper bound is insufficient: we also need leastness. The shared analyses prove leastness either by exhibiting the proposed bound as a member or by constructing members above every smaller candidate. The word "attained" in the table means the supremum belongs to the set, in which case it is also the maximum. Failure of attainment does not prevent a supremum from existing.

Part	Supremum	Bound and leastness proof
(a)	1	An upper bound and a member; attained. Analysis (a).
(b)	1	Upper endpoint is sharp by Tool 2; not attained. Analysis (b).
(c)	7	The larger member is an upper bound; attained. Analysis (c).
(d)	π	$e < \pi$ and π is a member; attained. Analysis (d).
(e)	1	$1/n \leq 1$, with equality at $n = 1$; attained. Analysis (e).
(f)	0	The sole member is its upper bound; attained. Analysis (f).
(g)	3	All members are at most 3 and 3 is a member; attained. Analysis (g).
(h)	NO sup	There is no real upper bound at all. Analysis (h).
(i)	1	The intersection equals $[0, 1]$; attained. Analysis (i).
(j)	1	Members exceed any $U < 1$ but never equal 1; not attained. Analysis (j).
(k)	NO sup	The members exceed every real upper-bound candidate. Analysis (k).
(l)	2	Density gives a rational member above any $U < 2$; not attained. Analysis (l).

Part	Supremum	Bound and leastness proof
(m)	2	The rational interval has sharp upper endpoint 2; not attained. Analysis (m).
(n)	$\sqrt{2}$	Density proves the upper endpoint is sharp; not attained. Analysis (n).
(o)	0	For $U < 0$, the member $U/2$ is above U; not attained. Analysis (o).
(p)	10	All four values are at most the member 10; attained. Analysis (p).
(q)	16	The largest listed member is 16; attained. Analysis (q).
(r)	1	The intersection is the singleton $\{1\}$; attained. Analysis (r).
(s)	1/2	Every prime is at least 2, and 2 is prime; attained. Analysis (s).
(t)	2	The set is $\{x : x < 2\}$, with members above every $U < 2$; not attained. Analysis (t).
(u)	NO sup	Real squares exceed every proposed upper bound. Analysis (u).
(v)	1	The period table has maximum 1, occurring at $n = 6$; attained. Analysis (v).
(w)	$\sqrt{3}/2$	The period table has this largest value at $n = 1$; attained. Analysis (w).

Conclusion. Parts (h), (k), and (u) have NO sup in $\mathbb{R}$. Every other supremum is the real number stated in its row, with its upper-bound and leastness requirements proved in the linked analysis.

4.4 Exercise 4.4: The real infimum of each set

What we must show. Give the real greatest lower bound of each set (a)–(w), or write NO inf if the set has no real infimum.

Plan. Reverse the roles of upper and lower bounds in Exercise 4.3. For an attained lower bound, the member itself defeats every larger candidate. For a nonattained infimum, the linked analysis constructs members below every larger candidate. "NO inf" is used exactly where there is no real lower bound.

Part	Infimum	Bound and greatestness proof
(a)	0	A lower bound and a member; attained. Analysis (a).
(b)	0	The open interval has sharp lower endpoint 0; not attained. Analysis (b).
(c)	2	The smaller member is a lower bound; attained. Analysis (c).
(d)	e	$e < \pi$ and e is a member; attained. Analysis (d).
(e)	0	Positive reciprocals lie below any $L > 0$; not attained. Analysis (e).
(f)	0	The sole member is its lower bound; attained. Analysis (f).
(g)	0	Every member is at least the member 0; attained. Analysis (g).
(h)	2	Every interval starts at least at 2, and 2 occurs; attained. Analysis (h).
(i)	0	The intersection is $[0, 1]$; attained. Analysis (i).
(j)	2/3	Every value is at least 2/3, with equality at $n = 1$; attained. Analysis (j).
(k)	0	Every value is nonnegative, with equality at $n = 1$; attained. Analysis (k).
(l)	NO inf	Negative integer members go below any proposed lower bound. Analysis (l).

Part	Infimum	Bound and greatestness proof
(m)	-2	Density makes the strict lower endpoint sharp; not attained. Analysis (m).
(n)	$-\sqrt{2}$	Density makes the strict lower endpoint sharp; not attained. Analysis (n).
(o)	NO inf	Negative real members go below every real L. Analysis (o).
(p)	1	The other three members are greater than the member 1; attained. Analysis (p).
(q)	0	The least of the listed members is 0; attained. Analysis (q).
(r)	1	The intersection equals $\{1\}$; attained. Analysis (r).
(s)	0	Unbounded primes give a reciprocal below any $L > 0$; not attained. Analysis (s).
(t)	NO inf	The half-line $x < 2$ has members below every real L. Analysis (t).
(u)	0	Every square is nonnegative and $0^2 = 0$; attained. Analysis (u).
(v)	-1	The smallest period-table value occurs at $n = 3$; attained. Analysis (v).
(w)	$-\sqrt{3}/2$	The smallest sine value occurs at $n = 4$; attained. Analysis (w).

Conclusion. Parts (l), (o), and (t) have NO inf in $\mathbb{R}$. Every other part has the real infimum in its row. In particular, (e) and (s) have infimum 0 without containing 0, while the open-interval intersection in (r) has both supremum and infimum equal to its actual member 1.

4.5 Exercise 4.5: When a supremum is a maximum

What we must show. Let S be a nonempty subset of $\mathbb{R}$ that is bounded above. Prove that if $\sup S \in S$, then $\sup S = \max S$.

Step 1: recall the two requirements for a maximum.

A number M is the maximum of S exactly when

(i) M is an element of S;

(ii) every $s \in S$ satisfies $s \leq M$.

The first requirement concerns membership; the second concerns order.

Step 2: check those requirements for the supremum.

Since S is nonempty and bounded above, completeness supplies the real number $M = \sup S$. The hypothesis of the exercise is precisely $M \in S$, so requirement (i) holds. A supremum is an upper bound, so $s \leq M$ for every $s \in S$; requirement (ii) holds as well.

Conclusion. The number $\sup S$ belongs to S and is at least every member of S. By the definition of maximum, $\boxed{\sup S = \max S}$. The additional membership hypothesis is essential: an upper bound need not be an element of its set.

4.6 Exercise 4.6: Comparing the infimum and supremum

What we must show. For a nonempty bounded set $S \subseteq \mathbb{R}$, prove (a) $\inf S \leq \sup S$, and (b) describe S when these two numbers are equal.

Part (a), Step 1: identify the real bounds and choose an element.

Because S is nonempty and bounded, the completeness axiom and its lower-bound version give real numbers

$$m = \inf S, \qquad M = \sup S.$$

Nonemptiness permits us to choose a specific $s_0 \in S$.

Part (a), Step 2: apply the meanings of lower and upper bound.

Since m is a lower bound, $m \leq s_0$. Since M is an upper bound, $s_0 \leq M$. Transitivity of $\leq$ gives

$$m \leq s_0 \leq M \quad \Longrightarrow \quad m \leq M.$$

Conclusion. For every nonempty bounded $S \subseteq \mathbb{R}$, $\boxed{\inf S \leq \sup S}$. Choosing s_0 is why nonemptiness matters in this argument.

Part (b), Step 1: let the common value be c.

Assume $\inf S = \sup S = c$. For an arbitrary $s \in S$, the lower-bound and upper-bound properties give

$$c \leq s \leq c.$$

By antisymmetry of the order, $s = c$. Since s was arbitrary, every element of S equals c, and hence $S \subseteq \{c\}$.

Part (b), Step 2: use nonemptiness to get the reverse inclusion.

There is an $s_0 \in S$. The preceding step gives $s_0 = c$, so $c \in S$. Thus $\{c\} \subseteq S$, and the two inclusions give $S = \{c\}$. Conversely, for the singleton $\{c\}$, the element c is both its maximum and its minimum. Every upper bound must be at least c, and every lower bound must be at most c. Therefore both its supremum and its infimum equal c.

Conclusion. Equality of the infimum and supremum occurs exactly when S is a singleton: $\boxed{\inf S = \sup S = c \iff S = \{c\}}$.

4.7 Exercise 4.7: Bounds of a subset and the supremum of a union

What we must show. Let S, T be nonempty bounded subsets of $\mathbb{R}$. Prove (a) that $S \subseteq T$ implies $\inf T \leq \inf S \leq \sup S \leq \sup T$, and (b) that $\sup(S \cup T) = \max\{\sup S, \sup T\}$ without assuming either set contains the other.

Part (a), Step 1: prove the comparison between the infima.

Assume $S \subseteq T$. Since $\inf T$ is a lower bound of T, it satisfies $\inf T \leq t$ for every $t \in T$. Every $s \in S$ is also an element of T, so

$$\inf T \leq s \qquad \text{for every } s \in S.$$

Consequently $\inf T$ is a lower bound of S. The number $\inf S$ is the *greatest* lower bound of S, so every lower bound of S is at most $\inf S$. In particular,

$$\inf T \leq \inf S.$$

Part (a), Step 2: prove the middle comparison.

Exercise 4.6(a), applied to the nonempty bounded set S, gives

$$\inf S \leq \sup S.$$

Part (a), Step 3: prove the comparison between the suprema.

Because $\sup T$ is an upper bound of T, every $s \in S \subseteq T$ satisfies $s \leq \sup T$. Thus $\sup T$ is an upper bound of S. Since $\sup S$ is the *least* upper bound of S,

$$\sup S \leq \sup T.$$

Conclusion. Combining the three inequalities gives $\boxed{\inf T \leq \inf S \leq \sup S \leq \sup T}$ whenever $S \subseteq T$.

Part (b), Step 1: verify that the union has real bounds.

Write

$$U = S \cup T, \qquad L = \min\{\inf S, \inf T\}, \qquad M = \max\{\sup S, \sup T\}.$$

The numbers L, M are real. Since S is nonempty, the union U is nonempty. If $u \in U$, then $u \in S$ or $u \in T$. In the first case,

$$L \leq \inf S \leq u \leq \sup S \leq M;$$

in the second case,

$$L \leq \inf T \leq u \leq \sup T \leq M.$$

Therefore U is bounded below by L and above by M. In particular, $\sup U$ exists as a real number.

Part (b), Step 2: prove that $\sup U \leq M$.

Step 1 showed $u \leq M$ for every $u \in U$, so M is an upper bound of U. The least-upper-bound property gives $\sup U \leq M$.

Part (b), Step 3: prove the inequality in the other direction.

Since $S \subseteq U$, every member of S is at most $\sup U$. Hence $\sup U$ is an upper bound of S, and $\sup S \leq \sup U$. The same reasoning with $T \subseteq U$ gives $\sup T \leq \sup U$. The maximum M is one of these two numbers; both are at most $\sup U$. Thus $M \leq \sup U$.

Conclusion. The two comparisons $\sup U \leq M$ and $M \leq \sup U$ give

$$\boxed{\sup(S \cup T) = \max\{\sup S, \sup T\}}.$$

We used the containments $S \subseteq S \cup T$ and $T \subseteq S \cup T$, not a containment between S and T.

4.8 Exercise 4.8: Two sets ordered entirely below and above each other

What we must show. Let $S, T \subseteq \mathbb{R}$ be nonempty and suppose

$$s \leq t \qquad \text{for every } s \in S \text{ and every } t \in T.$$

Show (a) that S is bounded above and T is bounded below, (b) that $\sup S \leq \inf T$, and supply the examples requested in (c) and (d).

Part (a), Step 1: choose one member from each set.

Because both sets are nonempty, there are $s_0 \in S$ and $t_0 \in T$. Keep these two particular numbers fixed for this step. The hypothesis, with $t = t_0$, says

$$s \leq t_0 \qquad \text{for every } s \in S.$$

Thus the real number t_0 is an upper bound of S. The hypothesis, with $s = s_0$, says

$$s_0 \leq t \qquad \text{for every } t \in T.$$

Thus the real number s_0 is a lower bound of T.

Conclusion. S is bounded above, and T is bounded below. No lower bound for S, or upper bound for T, is required by this conclusion.

Part (b), Step 1: establish existence of the two numbers.

The set S is nonempty and bounded above, so the completeness axiom gives $\sup S \in \mathbb{R}$. The set T is nonempty and bounded below, so Corollary 4.5 gives $\inf T \in \mathbb{R}$.

Part (b), Step 2: turn the supremum into a lower bound of T.

Take an arbitrary $t \in T$. The hypothesis says that $s \leq t$ for every $s \in S$, so t is an upper bound of S. Since $\sup S$ is the least upper bound of S, we have $\sup S \leq t$. This is true for every $t \in T$. Therefore $\sup S$ is a lower bound of T. Its greatest lower bound is $\inf T$, so

$$\sup S \leq \inf T.$$

Conclusion. The separation of all pairs of set elements implies $\boxed{\sup S \leq \inf T}$. The proof does not assume that either extremal bound belongs to its set.

Part (c), Step 1: give sets that share an element.

Take

$$S = [0, 1], \qquad T = [1, 2].$$

Both sets are nonempty because 1 belongs to each. If $s \in S$ and $t \in T$, the interval definitions give $s \leq 1$ and $1 \leq t$; transitivity gives $s \leq t$, as required.

Part (c), Step 2: identify their intersection and bounds.

If $x \in S \cap T$, membership in S gives $x \leq 1$, and membership in T gives $1 \leq x$. Antisymmetry forces $x = 1$. Conversely, 1 belongs to both intervals, so

$$S \cap T = \{1\}.$$

The number 1 is the maximum of S and the minimum of T. Hence $\sup S = 1 = \inf T$.

Conclusion. The example $\boxed{S = [0,1],\ T = [1,2]}$ meets the pairwise order condition and has nonempty intersection $\{1\}$.

Part (d), Step 1: give disjoint sets with a common extremal bound.

Take

$$S = (0,1), \qquad T = (1,2).$$

The first contains $1/2$, and the second contains $3/2$, so both are nonempty. For $s \in S$ and $t \in T$, we have $s < 1 < t$, and hence $s \le t$. If x belonged to their intersection, we would have both $x < 1$ and $x > 1$, which is impossible. Thus $S \cap T = \varnothing$.

Part (d), Step 2: prove directly that $\sup S = 1$.

Every $s \in (0,1)$ satisfies $s < 1$, so 1 is an upper bound of S. To prove it is the least, take any real $u < 1$ and put

$$v = \max\{0, u\}, \qquad s = \frac{v+1}{2}.$$

Both 0 and u are less than 1, so $v < 1$. Since $v \ge 0$, the inequalities $v < s < 1$ imply $0 < s < 1$. For completeness,

$$s - v = \frac{1-v}{2} > 0, \qquad 1 - s = \frac{1-v}{2} > 0.$$

Thus $s \in S$ and $s > v \ge u$. Consequently u cannot be an upper bound of S. Every number less than 1 fails to be an upper bound, so $\sup S = 1$.

Part (d), Step 3: prove directly that $\inf T = 1$.

Every $t \in (1,2)$ satisfies $t > 1$, so 1 is a lower bound of T. For an arbitrary real $\ell > 1$, put

$$w = \min\{2, \ell\}, \qquad t = \frac{1+w}{2}.$$

Both 2 and ℓ exceed 1, so $w > 1$. We have

$$t - 1 = \frac{w-1}{2} > 0, \qquad w - t = \frac{w-1}{2} > 0.$$

Therefore $1 < t < w \le 2$, which gives $t \in T$. Also $t < w \le \ell$, so $t < \ell$. Thus ℓ cannot be a lower bound of T. Every number greater than 1 fails to be a lower bound, so $\inf T = 1$.

Conclusion. The example $\boxed{S = (0,1),\ T = (1,2)}$ has $\boxed{\sup S = \inf T = 1}$ and $\boxed{S \cap T = \varnothing}$. Equality of the bounds does not force either set to contain the common bound.

4.9 Exercise 4.9: Obtain an infimum by negating a supremum

What we must show. Complete the proof of Corollary 4.5. For a nonempty set $S \subseteq \mathbb{R}$ that is bounded below, put

$$-S = \{-s : s \in S\}, \qquad s_0 = \sup(-S).$$

Prove the two assertions left open in the text:

$$(1) \quad -s_0 \le s \qquad \text{for every } s \in S,$$

$$(2) \quad t \le s \text{ for every } s \in S \qquad \Longrightarrow \quad t \le -s_0.$$

Step 1: justify that s_0 exists as a real number.

Since S is nonempty, choose $s_* \in S$. Then $-s_* \in -S$, so $-S$ is also nonempty. Since S is bounded below, there is a real number m with $m \le s$ for every $s \in S$. Negating this inequality reverses it, so $-s \le -m$ for every $s \in S$. Every element of $-S$ has the form $-s$ for such an s. Therefore $-m$ is an upper bound of $-S$. The completeness axiom applies and gives the real number $s_0 = \sup(-S)$.

Step 2: prove assertion (1), the lower-bound property.

Let $s \in S$ be arbitrary. By the definition of $-S$, its negative $-s$ belongs to $-S$. Because s_0 is an upper bound of that set,

$$-s \le s_0.$$

Negate both sides. The direction reverses, and $-(-s) = s$, giving

$$-s_0 \le s.$$

This holds for every $s \in S$, so $-s_0$ is a lower bound of S.

Step 3: prove assertion (2), the greatest-lower-bound property.

Let t be any real lower bound of S. This means $t \le s$ for every $s \in S$. Negating each of those inequalities gives

$$-s \le -t \qquad \text{for every } s \in S.$$

Hence $-t$ is an upper bound of $-S$. The least-upper-bound property of $s_0 = \sup(-S)$ now gives

$$s_0 \le -t.$$

Negating once more yields $t \le -s_0$. Thus every lower bound t of S is at most $-s_0$.

Conclusion. Assertion (1) says $-s_0$ is a lower bound, and assertion (2) says it is greater than or equal to every lower bound. Together they prove $\boxed{\inf S = -s_0 = -\sup(-S)}$. This establishes the lower-bound form of completeness from its upper-bound form.

4.10 Exercise 4.10: One natural number meeting two inequalities

What we must show. Given a real number $a > 0$, find one $n \in \mathbb{N}$ such that $1/n < a < n$.

Plan. It is not enough to find one integer for $1/n < a$ and a different integer for $a < n$. We choose a single integer larger than both a and $1/a$.

Step 1: form one positive real target.

Since $a > 0$, its inverse $1/a$ exists and is positive. Define

$$B = \max\{a, 1/a\}.$$

The number B is positive because both entries in this maximum are positive.

Step 2: apply the Archimedean property.

Use Theorem 4.6 with the positive increment 1 and the positive target B. It gives a natural number n satisfying

$$n \cdot 1 > B.$$

Thus $n > B \geq a$, so $a < n$, and $n > B \geq 1/a$, so $n > 1/a$.

Step 3: convert $n > 1/a$ into the reciprocal inequality.

Multiply $n > 1/a$ by the positive number a to get $na > 1$. Because $n \in \mathbb{N}$, we have $n > 0$, so division by n preserves strict inequality:

$$na > 1 \quad \Longrightarrow \quad a > \frac{1}{n}.$$

Conclusion. The same natural number n satisfies both comparisons: $\boxed{\frac{1}{n} < a < n}$. Both positivity checks are needed when multiplying and dividing inequalities.

4.11 Exercise 4.11: Every nonempty open interval contains infinitely many rationals

What we must show. If $a, b \in \mathbb{R}$ and $a < b$, use the denseness of $\mathbb{Q}$ to prove that there are infinitely many rational numbers r with $a < r < b$.

Step 1: define the set whose size is at issue.

Let

$$E = \{r \in \mathbb{Q} : a < r < b\}.$$

Theorem 4.7, applied to $a < b$, gives at least one rational in this interval. Thus E is nonempty. This first application alone does not prove infinitude; we must rule out the possibility that the set is finite.

Step 2: suppose there were only finitely many.

Assume, for a contradiction, that

$$E = \{r_1, r_2, \ldots, r_k\}$$

for some positive integer k. A nonempty finite set of real numbers has a maximum: compare its first two entries, keep the larger, and continue through the finitely many entries. Let

$$r_* = \max\{r_1, \ldots, r_k\}.$$

Since $r_* \in E$, it satisfies $a < r_* < b$.

Step 3: use density in a smaller interval.

The strict inequality $r_* < b$ permits another application of Theorem 4.7. There is a rational q with

$$r_* < q < b.$$

Because $a < r_* < q$, transitivity gives $a < q$. Together with $q < b$ and $q \in \mathbb{Q}$, this shows $q \in E$. But $q > r_*$, contradicting the choice of r_* as the largest member of E.

Conclusion. The set E is nonempty and cannot be finite. Therefore there are **infinitely many rational numbers between a and b**. The same density theorem can always be applied above the largest member of any proposed finite list.

4.12 Exercise 4.12: Irrational numbers are dense as well

What we must show. Let $I = \mathbb{R} \setminus \mathbb{Q}$. Prove that for any real numbers $a < b$ there is an $x \in I$ such that $a < x < b$.

Step 1: prove that a rational translate of $\sqrt{2}$ is irrational.

The irrationality of $\sqrt{2}$ was established in Section 2. Let $r \in \mathbb{Q}$ be arbitrary and suppose, for a contradiction, that $r + \sqrt{2}$ were rational. Write

$$r = \frac{m}{n}, \qquad r + \sqrt{2} = \frac{h}{k},$$

where $m, n, h, k \in \mathbb{Z}$, $n \neq 0$, and $k \neq 0$. Subtracting gives

$$\sqrt{2} = \frac{h}{k} - \frac{m}{n} = \frac{hn - mk}{kn}.$$

The numerator $hn - mk$ is an integer, the denominator kn is an integer, and $kn \neq 0$ because both k and n are nonzero. The last quotient would make $\sqrt{2}$ rational, a contradiction. Thus $r + \sqrt{2}$ is irrational for every $r \in \mathbb{Q}$, proving

$$\{r + \sqrt{2} : r \in \mathbb{Q}\} \subseteq I.$$

Step 2: shift the desired interval before choosing a rational.

Adding $-\sqrt{2}$ to $a < b$ preserves strict inequality:

$$a - \sqrt{2} < b - \sqrt{2}.$$

By Theorem 4.7, there is a rational number r satisfying

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

The endpoints here are real numbers; they need not themselves be rational for the density theorem to apply.

Step 3: shift that rational back.

Define $x = r + \sqrt{2}$. Step 1 gives $x \in I$. Add $\sqrt{2}$ to all three expressions in the preceding two inequalities:

$$(a - \sqrt{2}) + \sqrt{2} < r + \sqrt{2} < (b - \sqrt{2}) + \sqrt{2}.$$

Associativity and additive cancellation simplify the endpoints, giving $a < x < b$.

Conclusion. The irrational number $\boxed{x = r + \sqrt{2}}$, where $a - \sqrt{2} < r < b - \sqrt{2}$ and $r \in \mathbb{Q}$, lies strictly between a and b. The initial shift is what ensures that adding $\sqrt{2}$ does not move our chosen point outside the desired interval.

4.13 Exercise 4.13: Three descriptions of an open interval

What we must show. For real a, b, c, prove the equivalence of

$$\text{(i)} \quad |a - b| < c,$$

$$\text{(ii)} \quad b - c < a < b + c,$$
$$\text{(iii)} \quad a \in (b - c, b + c).$$

We interpret (u, v) as the set $\{x \in \mathbb{R} : u < x < v\}$; this set is empty when $u \geq v$.

Step 1: prove that (i) implies (ii).

Assume $|a - b| < c$. Exercise 3.7(a), with bounded number $a - b$ and bound c, gives

$$-c < a - b < c.$$

Add b throughout. Strict inequalities are preserved by addition, so

$$-c + b < (a - b) + b < c + b.$$

Associativity and the additive inverse rule simplify the middle expression to a. Commutativity turns the endpoints into $b - c$ and $b + c$. We have obtained $b - c < a < b + c$, which is (ii).

Step 2: prove that (ii) implies (i).

Assume $b - c < a < b + c$. Add $-b$ throughout and simplify to obtain

$$-c < a - b < c.$$

Exercise 3.7(a) now gives $|a - b| < c$. This is (i).

Step 3: prove the equivalence of (ii) and (iii).

By the definition of the open interval,

$$(b - c, b + c) = \{x \in \mathbb{R} : b - c < x < b + c\}.$$

Membership of the particular number a in this set means exactly that $b - c < a < b + c$. Thus (ii) implies (iii), and (iii) implies (ii).

Step 4: verify explicitly what happens when $c \leq 0$.

If $c \leq 0$, statement (i) is false because every absolute value is nonnegative: it cannot satisfy $0 \leq |a - b| < c \leq 0$. Also $c \leq 0$ implies $c \leq -c$, so adding b gives $b + c \leq b - c$. Statement (ii) would imply $b - c < b + c$, contrary to that inequality. Thus (ii) is false. The interval in (iii) has its left endpoint at least as large as its right endpoint, so it is empty; statement (iii) is false too. This includes $c = 0$, when both endpoints equal b.

Conclusion. For $c > 0$, the steps above show that the three statements are equivalent; for $c \leq 0$, all three are false. Consequently, for every real a, b, c, $\boxed{\text{(i)} \iff \text{(ii)} \iff \text{(iii)}}$.

4.14 Exercise 4.14: Supremum and infimum of the sum of two sets

What we must show. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Define

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

Prove

$$\text{(a)} \quad \sup(A + B) = \sup A + \sup B, \qquad \text{(b)} \quad \inf(A + B) = \inf A + \inf B.$$

The notation $A + B$ means the set of all such sums, not a union.

Preliminary step: check nonemptiness, boundedness, and existence.

Since A and B are nonempty and bounded, their extrema as bounds exist as real numbers. Write

$$m_A = \inf A, \quad M_A = \sup A, \qquad m_B = \inf B, \quad M_B = \sup B.$$

Choose $a_0 \in A$ and $b_0 \in B$. Then $a_0 + b_0 \in A + B$, so the sum set is nonempty. For arbitrary $a \in A$ and $b \in B$, we have

$$m_A \le a \le M_A, \qquad m_B \le b \le M_B.$$

Adding the lower comparisons and then the upper comparisons gives

$$m_A + m_B \le a + m_B \le a + b \le M_A + b \le M_A + M_B.$$

Thus $A+B$ is bounded below by m_A+m_B and above by M_A+M_B. Completeness and its lower-bound corollary supply real numbers

$$m = \inf(A + B), \qquad M = \sup(A + B).$$

No assertion that an infimum or supremum belongs to A or B has been made.

Part (a), Step 1: prove the first inequality.

We just showed that every $a + b \in A + B$ satisfies $a + b \le M_A + M_B$. Hence $M_A + M_B$ is an upper bound of $A + B$. Its least upper bound is M, so

$$M \le M_A + M_B.$$

Part (a), Step 2: explain how to approach each separate supremum.

Let $\varepsilon > 0$. Because $\varepsilon/2 > 0$, $M_A - \varepsilon/2 < M_A$. The number $M_A - \varepsilon/2$ cannot be an upper bound of A: such an upper bound would be strictly smaller than the least upper bound M_A. Therefore there is an $a_\varepsilon \in A$ with

$$M_A - \frac{\varepsilon}{2} < a_\varepsilon \le M_A.$$

The same argument for B gives a $b_\varepsilon \in B$ with

$$M_B - \frac{\varepsilon}{2} < b_\varepsilon \le M_B.$$

These are approximation statements, not claims that $M_A \in A$ or $M_B \in B$.

Part (a), Step 3: prove the reverse inequality using a gap.

Suppose, for a contradiction, that $M < M_A + M_B$. Set

$$\varepsilon = M_A + M_B - M > 0.$$

Choose $a_\varepsilon, b_\varepsilon$ as in Step 2. Adding their two strict lower bounds gives

$$\begin{aligned} a_\varepsilon + b_\varepsilon &> \left(M_A - \frac{\varepsilon}{2}\right) + \left(M_B - \frac{\varepsilon}{2}\right), \\ &= M_A + M_B - \varepsilon, \end{aligned}$$

$$= M.$$

But $a_\varepsilon + b_\varepsilon \in A + B$, and M is an upper bound of this set, so the same sum must be at most M. This is a contradiction. Therefore $M_A + M_B \leq M$.

Conclusion. Both $M \leq M_A + M_B$ and $M_A + M_B \leq M$ hold. By antisymmetry,

$$\boxed{\sup(A + B) = \sup A + \sup B}.$$

Splitting the positive gap into two halves leaves enough room to choose one near-maximal element from each set.

Part (b), Step 1: prove the first inequality.

The preliminary step showed $m_A + m_B \leq a + b$ for every $a + b \in A + B$. Thus $m_A + m_B$ is a lower bound of the sum set. Since m is the greatest lower bound,

$$m_A + m_B \leq m.$$

Part (b), Step 2: explain how to approach the separate infima.

Let $\varepsilon > 0$. The number $m_A + \varepsilon/2$ is strictly larger than m_A. It cannot be a lower bound of A, since m_A is its greatest lower bound. Therefore there is an $a_\varepsilon \in A$ with

$$m_A \leq a_\varepsilon < m_A + \frac{\varepsilon}{2}.$$

Likewise there is a $b_\varepsilon \in B$ with

$$m_B \leq b_\varepsilon < m_B + \frac{\varepsilon}{2}.$$

Again no attainment of either infimum is being assumed.

Part (b), Step 3: prove the reverse inequality using a gap.

Suppose, for a contradiction, that $m > m_A + m_B$. Put

$$\varepsilon = m - m_A - m_B > 0.$$

Choose $a_\varepsilon, b_\varepsilon$ from Step 2. Adding their two strict upper bounds gives

$$\begin{aligned} a_\varepsilon + b_\varepsilon &< \left(m_A + \frac{\varepsilon}{2}\right) + \left(m_B + \frac{\varepsilon}{2}\right), \\ &= m_A + m_B + \varepsilon, \\ &= m. \end{aligned}$$

This sum belongs to $A + B$. But m is a lower bound, so every member of that set must be at least m. The contradiction rules out $m > m_A + m_B$, giving $m \leq m_A + m_B$.

Conclusion. Combining the two comparisons gives $\boxed{\inf(A + B) = \inf A + \inf B}$. The argument applies whether or not the extremal bounds of the original sets are attained.

4.15 Exercise 4.15: Comparison using reciprocal natural-number errors

What we must show. Suppose $a, b \in \mathbb{R}$ and

$$a \leq b + \frac{1}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Prove $a \leq b$.

Plan. If $a > b$, the positive gap $a - b$ can be made larger than one of the numbers $1/n$. That particular n contradicts the hypothesis.

Step 1: assume the opposite of the desired conclusion.

Suppose $a \leq b$ is false. Total ordering then gives $a > b$, so

$$d = a - b > 0.$$

Step 2: obtain a strict reciprocal estimate from the Archimedean property.

Apply Theorem 4.6 to the positive numbers d and 1. There is an $n \in \mathbb{N}$ with $nd > 1$. Since $n > 0$, division by n gives

$$d > \frac{1}{n}.$$

The strict inequality is needed: an estimate $1/n \leq d$ alone would not exclude equality at the next step.

Step 3: compare with the hypothesis for that particular n.

Substitute $d = a - b$ and add b to get

$$\frac{1}{n} < a - b \quad \Longrightarrow \quad b + \frac{1}{n} < a.$$

The hypothesis applies to every natural number, so it applies to this one and gives $a \leq b + 1/n$. These inequalities cannot both hold: they would give $a \leq b + 1/n < a$.

Conclusion. The contradiction proves $\boxed{a \leq b}$. Exercise 3.8 tested a against every real number above b. Here only the particular numbers $b + 1/n$ are tested; the Archimedean property guarantees that at least one of them detects any hypothetical positive gap $a - b$.

4.16 Exercise 4.16: A real number as the supremum of the rationals below it

What we must show. For each $a \in \mathbb{R}$, prove

$$\sup\{r \in \mathbb{Q} : r < a\} = a.$$

Step 1: define the set and prove that it is nonempty.

Fix an arbitrary real a and let

$$S_a = \{r \in \mathbb{Q} : r < a\}.$$

The inequality $a - 1 < a$ is strict because $1 > 0$. Apply the denseness of $\mathbb{Q}$ to the interval $(a - 1, a)$. It gives a rational r_0 with

$$a - 1 < r_0 < a.$$

Thus $r_0 \in S_a$, proving nonemptiness. This construction works even for $a \leq 0$; one must not assume that 0 belongs to S_a.

Step 2: prove that a is an upper bound.

By definition, every $r \in S_a$ satisfies $r < a$, and therefore $r \leq a$. Hence a is an upper bound of S_a. In particular, S_a is nonempty and bounded above, so its real supremum exists by completeness.

Step 3: prove that every smaller candidate fails to be an upper bound.

Take any real $u < a$. Denseness of $\mathbb{Q}$ gives a rational r with

$$u < r < a.$$

Since $r \in \mathbb{Q}$ and $r < a$, this r belongs to S_a. The additional inequality $u < r$ shows that u is not an upper bound of S_a: it is exceeded by the specific element $r \in S_a$.

Step 4: express the least-upper-bound conclusion.

Step 2 showed that a is an upper bound. Step 3 showed that every number strictly less than a fails to be an upper bound. Equivalently, if U is any upper bound, $U < a$ is impossible, and the real order therefore forces $a \leq U$. Thus a is at most every upper bound while being one itself.

Conclusion. The two defining obligations for a supremum are satisfied, so $\boxed{\sup\{r \in \mathbb{Q} : r < a\} = a}$ for every real a. The number a need not be rational, and it is excluded from the set even when it is rational; the supremum need not be attained.

5 The symbols of positive and negative infinity

The symbols $+\infty$ and $-\infty$ record a failure of a real bound. They are not real numbers and are not elements of any of the subsets of $\mathbb{R}$ considered here. We use the extended ordering

$$-\infty < r < +\infty \qquad (r \in \mathbb{R}).$$

For a nonempty set $E \subseteq \mathbb{R}$, the conventions in this section are

$$\begin{array}{ll} \sup E \in \mathbb{R}, & \text{if } E \text{ is bounded above,} \\ \sup E = +\infty, & \text{if } E \text{ is not bounded above,} \\ \inf E \in \mathbb{R}, & \text{if } E \text{ is bounded below,} \\ \inf E = -\infty, & \text{if } E \text{ is not bounded below.} \end{array}$$

The finite supremum exists by completeness, and the finite infimum exists by its lower-bound version, Corollary 4.5. A set can have a finite infimum while being unbounded above, or a finite supremum while being unbounded below. A conclusion about one side does not require a bound on the other side.

In explicit terms, not being bounded above means that for every real M there is an $e \in E$ with $e > M$. Not being bounded below means that for every real L there is an $e \in E$ with $e < L$. These statements are the negations of the definitions of an upper bound and a lower bound.

For negation we use $-(+\infty) = -\infty$ and $-(-\infty) = +\infty$. The additions needed in Exercise 5.7 are only

$$c + (-\infty) = (-\infty) + c = -\infty \quad (c \in \mathbb{R}), \qquad (-\infty) + (-\infty) = -\infty.$$

These are extended-bound conventions. They do not make infinity a real number, and they do not define the indeterminate expression $(+\infty) + (-\infty)$.

Two elementary algebra facts used for the interval descriptions

First, if $0 \le u < v$, then

$$v^2 - u^2 = (v - u)(v + u) > 0.$$

Both factors on the right are positive. Thus squaring preserves strict order for nonnegative numbers. The corresponding weak inequality follows by allowing $u = v$.

Second, cubing preserves strict order on all of $\mathbb{R}$. To check the possible negative signs, suppose $u < v$. Then

$$v^3 - u^3 = (v - u)(v^2 + vu + u^2),$$

and

$$v^2 + vu + u^2 = \left(v + \frac{u}{2}\right)^2 + \frac{3u^2}{4}.$$

If $u \neq 0$, the second term is positive and the first is nonnegative. If $u = 0$, then $v > 0$ and the expression is $v^2 > 0$. Hence in both cases $v^3 - u^3 > 0$, because $v - u > 0$. It follows, in particular, that

$$x^3 \le 8 \iff x \le 2, \qquad x^3 < 8 \iff x < 2.$$

For example, $x \le 2$ implies $x^3 \le 2^3 = 8$; conversely, $x^3 \le 8$ and $x > 2$ would imply $x^3 > 8$, a contradiction. For the strict equivalence, $x < 2$ implies $x^3 < 8$, while $x \ge 2$ would imply $x^3 \ge 8$.

5.1 Exercise 5.1: Four descriptions in interval notation

What we must show. Write each of the following sets as an interval:

$$\text{(a) } \{x \in \mathbb{R} : x < 0\}, \quad \text{(b) } \{x \in \mathbb{R} : x^3 \leq 8\},$$
$$\text{(c) } \{x^2 : x \in \mathbb{R}\}, \quad \text{(d) } \{x \in \mathbb{R} : x^2 < 8\}.$$

Plan. Translate the defining inequalities, and prove both inclusions when the description involves squares. An infinite endpoint always has a parenthesis, because it is not a real element that could be included.

1. Part (a): All negative real numbers

The notation $(-\infty, 0)$ means precisely the real numbers strictly less than 0. If x belongs to the given set, it satisfies $x < 0$ and therefore belongs to $(-\infty, 0)$. Conversely, if $x \in (-\infty, 0)$, then $x \in \mathbb{R}$ and $x < 0$, so it belongs to the given set. Thus

$$\{x \in \mathbb{R} : x < 0\} = (-\infty, 0).$$

The endpoint 0 is excluded because the defining inequality is strict.

2. Part (b): A non-strict cubic inequality

The cube-order calculation above gives both directions of $x^3 \leq 8 \iff x \leq 2$. Consequently

$$\{x \in \mathbb{R} : x^3 \leq 8\} = \{x \in \mathbb{R} : x \leq 2\} = (-\infty, 2].$$

In particular, 2 is included: $2^3 = 8$ satisfies the non-strict inequality. This is why the finite endpoint has a square bracket.

3. A completeness justification for the square roots needed below

To identify the set of all squares, we need that every nonnegative real number has a nonnegative real square root. Here is a proof using the completeness axiom from Section 4.

The number 0 has square root 0. Fix $y > 0$ and define

$$C = \{t \in \mathbb{R} : t \geq 0 \text{ and } t^2 \leq y\}.$$

The set is nonempty because $0 \in C$. It is bounded above by $y + 1$: if $t > y + 1$, then $t > 1$ and $t^2 > t > y$, so such a t cannot belong to C. Completeness supplies the real number $c = \sup C$, and $c \geq 0$ since $0 \in C$.

We claim $c^2 = y$. If $c^2 < y$, put $d = y - c^2 > 0$ and choose

$$h = \min\left\{1, \frac{d}{2(2c+1)}\right\} > 0.$$

We have $h \leq 1$ and $(2c + 1)h \leq d/2$. Expanding and using $h^2 \leq h$ gives

$$(c+h)^2 = c^2 + 2ch + h^2 \leq c^2 + (2c+1)h \leq c^2 + \frac{d}{2} < c^2 + d = y.$$

Thus $c + h \in C$, but $c + h > c$, contradicting that c is an upper bound for C.

If $c^2 > y$, then $c > 0$. Put $d = c^2 - y > 0$ and choose

$$h = \min\left\{\frac{c}{2}, \frac{d}{4c}\right\} > 0.$$

Now $c - h \geq c/2 > 0$, and $2ch \leq d/2$. Therefore

$$(c-h)^2 = c^2 - 2ch + h^2 \geq c^2 - 2ch \geq c^2 - \frac{d}{2} > c^2 - d = y.$$

If $t \in C$ and $t > c - h$, both numbers are nonnegative, so the square-order fact gives $t^2 > (c-h)^2 > y$, contradicting $t \in C$. Thus $t \leq c - h$ for every $t \in C$. The smaller number $c - h < c$ would be an upper bound for C, contradicting the least-upper-bound property of c.

Both alternatives are impossible, so $c^2 = y$. This proves existence of a nonnegative square root. It is unique: if $a, b \geq 0$ and $a^2 = b^2$, then $(a-b)(a+b) = 0$. If $a + b = 0$, nonnegativity forces $a = b = 0$; otherwise $a - b = 0$, so again $a = b$. This justifies the notation $\sqrt{y}$ without using any later result about continuity.

4. Part (c): The set of all squares

Let $P = \{x^2 : x \in \mathbb{R}\}$. If $z \in P$, then $z = x^2$ for some $x \in \mathbb{R}$. Every real square is nonnegative, so $z \geq 0$ and $z \in [0, +\infty)$. This proves $P \subseteq [0, +\infty)$.

Conversely, let $z \in [0, +\infty)$. Then $z \geq 0$, and Step 3 gives a real number $x = \sqrt{z}$ with $x^2 = z$. Hence $z \in P$. We have proved the reverse inclusion and therefore

$$\{x^2 : x \in \mathbb{R}\} = [0, +\infty).$$

The finite endpoint is included because $0 = 0^2$.

5. Part (d): A strict square inequality

Put $a = \sqrt{8}$. Step 3 gives $a \geq 0$ and $a^2 = 8$; in fact $a > 0$ because $8 > 0$. Let $D = \{x \in \mathbb{R} : x^2 < 8\}$.

First suppose $x \in D$. If $x \geq a$, nonnegative square order would give $x^2 \geq a^2 = 8$, a contradiction. Hence $x < a$. If $x \leq -a$, then $-x \geq a > 0$, so $x^2 = (-x)^2 \geq a^2 = 8$, again a contradiction. Thus $x > -a$. We have shown $-a < x < a$, so $D \subseteq (-a, a)$.

Conversely, suppose $-a < x < a$. If $x \geq 0$, then $0 \leq x < a$ and $x^2 < a^2 = 8$. If $x < 0$, the inequality $x > -a$ becomes $-x < a$ after multiplying by -1, while $-x > 0$. Therefore $x^2 = (-x)^2 < a^2 = 8$ in this case too. Thus $x \in D$, proving the reverse inclusion. We conclude

$$\{x \in \mathbb{R} : x^2 < 8\} = (-\sqrt{8}, \sqrt{8}).$$

Neither endpoint belongs to the set, since $(\sqrt{8})^2 = (-\sqrt{8})^2 = 8$.

Conclusion. The intervals are (a) $(-\infty, 0)$, (b) $(-\infty, 2]$, (c) $[0, +\infty)$, and (d) $(-\sqrt{8}, \sqrt{8})$.

5.2 Exercise 5.2: Extended bounds of the four intervals

What we must show. Find the infimum and supremum, allowing the infinite symbols, of each set in Exercise 5.1.

Plan. A finite included endpoint is a minimum or maximum. For an open endpoint, verify that elements approach it from the allowed side. For an infinite endpoint, show explicitly that no real bound exists on that side.

1. Part (a): The interval $(-\infty, 0)$

The number 0 is an upper bound, since every member is negative. If $U < 0$, let $x = U/2$. Then $x < 0$, and $x - U = -U/2 > 0$, so $x > U$. Thus no number less than 0 is an upper bound. This proves $\sup(-\infty, 0) = 0$.

For any proposed real lower bound L, put $x = \min\{L, 0\} - 1$. Then $x < 0$ and $x < L$, so x belongs to the interval and disproves that L is a lower bound. The interval is not bounded below, and its extended infimum is $-\infty$.

2. Part (b): The interval $(-\infty, 2]$

Every element is at most 2, and 2 itself belongs to the interval. Hence 2 is its maximum. Every upper bound must be at least this member 2, so the supremum is 2.

Given any real L, the number $x = \min\{L, 2\} - 1$ satisfies $x < 2$ and $x < L$. It belongs to $(-\infty, 2]$ and lies below the proposed lower bound. Thus the interval is not bounded below and has infimum $-\infty$.

3. Part (c): The interval $[0, +\infty)$

Every element is at least 0, and 0 belongs to the interval. Thus 0 is a minimum. Any lower bound is at most the member 0, so the greatest lower bound is 0.

For any real proposed upper bound M, choose $z = \max\{M, 0\} + 1$. Then $z \geq 0$, so z belongs to the interval, and $z > M$. There is no real upper bound. Therefore its supremum is $+\infty$.

4. Part (d): The interval $(-\sqrt{8}, \sqrt{8})$

Write $a = \sqrt{8} > 0$. Every $x \in (-a, a)$ satisfies $-a < x < a$. Thus $-a$ is a lower bound and a is an upper bound. We verify the endpoint approximation required for them to be the greatest lower bound and least upper bound.

Fix any $\varepsilon > 0$ and set

$$\delta = \min\left\{\frac{\varepsilon}{2}, a\right\} > 0.$$

Because $\delta \leq a$, the numbers

$$x_- = -a + \delta, \qquad x_+ = a - \delta$$

both belong to $(-a, a)$: we have $-a < x_- \leq 0 < a$ and $-a < 0 \leq x_+ < a$. Moreover, $\delta \leq \varepsilon/2 < \varepsilon$, so

$$x_- < -a + \varepsilon, \qquad x_+ > a - \varepsilon.$$

These inequalities show that no number greater than $-a$ can be a lower bound, and no number less than a can be an upper bound. For example, if $L > -a$, choose $\varepsilon = L + a > 0$ to obtain $x_- < L$; if $U < a$, choose $\varepsilon = a - U > 0$ to obtain $x_+ > U$. Therefore the infimum is $-a$ and the supremum is a.

Conclusion. The results, in the order of Exercise 5.1, are

Part	inf	sup
(a)	$-\infty$	0
(b)	$-\infty$	2
(c)	0	$+\infty$
(d)	$-\sqrt{8}$	$\sqrt{8}$

The finite endpoints in part (d) are not attained.

5.3 Exercise 5.3: The unbounded sets from Exercise 4.1

What we must show. Find the extended infimum and supremum of every unbounded set in Exercise 4.1.

Plan. The unbounded cases are precisely (h), (k), (l), (o), (t), and (u). For each, determine its finite endpoint when one exists and exhibit elements beyond every proposed bound on the unbounded side.

1. Identify all the relevant cases

We retain the letters of Exercise 4.1:

$$H = \bigcup_{n=1}^{\infty} [2n, 2n+1], \qquad K = \left\{ n + \frac{(-1)^n}{n} : n \in \mathbb{N} \right\},$$
$$L = \{r \in \mathbb{Q} : r < 2\}, \qquad O = \{x \in \mathbb{R} : x < 0\},$$
$$T = \{x \in \mathbb{R} : x^3 < 8\}, \qquad P = \{x^2 : x \in \mathbb{R}\}.$$

Here P denotes the set in part (u). In particular, the inequality in part (t) is strict; it differs from the non-strict cubic inequality in Exercise 5.1(b).

To check that no unbounded case was omitted, the remaining sets can be enclosed in finite intervals as follows. The notation in the left column refers to the letters of Exercise 4.1:

Parts	A finite containing interval
(a), (b), (e), (j), (s)	$[0, 1]$
(c)	$[2, 7]$
(f)	$[0, 0]$
(g)	$[0, 3]$
(i)	$[-1, 2]$
(m), (n)	$[-2, 2]$
(q)	$[0, 16]$
(r)	$[0, 2]$
(v), (w)	$[-1, 1]$

For the reciprocal terms, $n \geq 1$ gives $0 < 1/n \leq 1$, and $0 < 1/3^n \leq 1$, which justify the bounds for (e), (j), (s). Each intersection in (i) and (r) is a subset of its $n = 1$ interval, giving the displayed bounds. For (m) and (n), a real number outside $[-2, 2]$ has square greater than 4 and therefore cannot have square less than 4 or less than 2. The usual ranges of sine and cosine give the bounds for (v) and (w). The other displayed finite-interval inclusions follow directly from their listed elements or intervals. Finally, (d) and (p) are finite sets of real numbers. For a finite list $a_1, \ldots, a_m$, the number $C = 1 + |a_1| + \cdots + |a_m|$ satisfies $-C < a_j < C$ for every listed element, so each such set is bounded. This accounts for all the other parts.

2. Part (h): An infinite union of intervals

If $x \in H$, some $n \in \mathbb{N}$ satisfies $x \in [2n, 2n+1]$. Hence $x \geq 2n \geq 2$. Also $2 \in [2, 3] \subseteq H$, so 2 is a minimum and $\inf H = 2$.

Given any real M, the Archimedean property supplies an integer $n > \max\{1, M/2\}$. Then $2n > M$ and $2n \in [2n, 2n+1] \subseteq H$. Thus no real M is an upper bound. Therefore $\sup H = +\infty$.

3. Part (k): A growing sequence with an alternating correction

For every $n \in \mathbb{N}$, the value $(-1)^n$ is either 1 or -1. As $n > 0$, it follows that

$$n + \frac{(-1)^n}{n} \geq n - \frac{1}{n} = \frac{n^2 - 1}{n} \geq 0.$$

The last inequality holds because $n \geq 1$ gives $n^2 - 1 \geq 0$. At $n = 1$, the value is $1 + (-1)/1 = 0$. Thus 0 is the minimum of K, and $\inf K = 0$.

Also $1/n \leq 1$, so

$$n + \frac{(-1)^n}{n} \geq n - \frac{1}{n} \geq n - 1.$$

For a proposed real upper bound M, choose $n > \max\{1, M+1\}$ by the Archimedean property. Then $n - 1 > M$, and the corresponding member of K is greater than M. Hence K is not bounded above and $\sup K = +\infty$.

4. Part (l): The rational numbers less than 2

Every member of L is less than 2, so 2 is an upper bound. If $U < 2$, density of $\mathbb{Q}$ in $\mathbb{R}$ supplies a rational number r with $U < r < 2$. This r belongs to L and exceeds U, so U cannot be an upper bound. Thus $\sup L = 2$, although $2 \notin L$.

Given any real D, choose $n \in \mathbb{N}$ with $n > \max\{1, -D\}$. Then $-n < D$, and the integer $-n$ is rational and satisfies $-n < 0 < 2$. Hence $-n \in L$ lies below D. There is no real lower bound, so $\inf L = -\infty$.

5. Part (o): The negative half-line

This is $O = (-\infty, 0)$, the set already considered in Exercise 5.2(a). For completeness, 0 bounds it above, and if $U < 0$ then $U/2 \in O$ and $U/2 > U$. Thus its supremum is 0. For every real D, the member $\min\{D, 0\} - 1$ lies below D. Thus O is not bounded below and $\inf O = -\infty$.

6. Part (t): A strict cubic inequality

By the cube-order fact proved at the start of the section,

$$T = \{x \in \mathbb{R} : x^3 < 8\} = (-\infty, 2).$$

All its elements are less than 2, so 2 is an upper bound. If $U < 2$, the number $x = (U + 2)/2$ satisfies

$$x - U = \frac{2 - U}{2} > 0, \qquad 2 - x = \frac{2 - U}{2} > 0.$$

Hence $U < x < 2$, so $x \in T$ and U is not an upper bound. Therefore $\sup T = 2$; the supremum is not attained.

For any real D, set $x = \min\{D, 2\} - 1$. Then $x < 2$ and $x < D$, so $x \in T$ lies below D. This proves $\inf T = -\infty$.

7. Part (u): The real squares

Every real square is nonnegative and $0 = 0^2$ belongs to P. Thus $\inf P = 0$. Exercise 5.1(c) established, using completeness, that $P = [0, +\infty)$. For any real M, the number $y = \max\{M, 0\} + 1$ is nonnegative and exceeds M; since $y = (\sqrt{y})^2$, it belongs to P. No real upper bound exists, and $\sup P = +\infty$.

Conclusion. Exactly six sets in Exercise 4.1 are unbounded. Their extended bounds are

Part	inf	sup
(h)	2	$+\infty$
(k)	0	$+\infty$
(l)	$-\infty$	2
(o)	$-\infty$	0
(t)	$-\infty$	2
(u)	0	$+\infty$

5.4 Exercise 5.4: Reflection and extended bounds

What we must show. For every nonempty $S \subseteq \mathbb{R}$, with $-S = \{-s : s \in S\}$, prove

$$\inf S = -\sup(-S).$$

Plan. Negating an inequality changes a lower bound into an upper bound. Treat the case of a finite infimum and the case of no real lower bound separately.

1. Establish how reflection changes bounds

The set $-S$ is nonempty: if $s_0 \in S$, then $-s_0 \in -S$. If a real number d is a lower bound for S, then

$$d \le s \quad (s \in S) \quad \Longrightarrow \quad -s \le -d \quad (s \in S).$$

Thus $-d$ is an upper bound for $-S$. Conversely, if u is a real upper bound for $-S$, then $-s \le u$ for all $s \in S$. Multiplication by -1 reverses the inequality, giving $-u \le s$ for all $s \in S$. Hence $-u$ is a lower bound for S. In particular, S is bounded below if and only if $-S$ is bounded above.

2. The case in which S is bounded below

Let $\alpha = \inf S \in \mathbb{R}$, whose existence follows from Corollary 4.5. This step requires no upper bound on S. Since $\alpha \le s$ for all $s \in S$, we have $-s \le -\alpha$ for all $s \in S$. Therefore $-\alpha$ is an upper bound for $-S$.

Now let u be any real upper bound for $-S$. By Step 1, $-u$ is a lower bound for S. The fact that α is the greatest lower bound gives $-u \le \alpha$. Reversing signs gives $-\alpha \le u$. Thus $-\alpha$ is less than or equal to every upper bound for $-S$, proving

$$\sup(-S) = -\alpha.$$

Negating this real equality yields $-\sup(-S) = \alpha = \inf S$. This is the finite-infimum argument of Exercise 4.9, with the bound correspondence written out explicitly.

3. The case in which S is not bounded below

Here $\inf S = -\infty$ by definition. Let M be any real number. Since S has no real lower bound, there exists $s \in S$ with $s < -M$. Then $-s > M$, and $-s \in -S$. Thus $-S$ has elements greater than every real M, so $\sup(-S) = +\infty$. Using the stated negation convention gives

$$-\sup(-S) = -(+\infty) = -\infty = \inf S.$$

These two cases exhaust a nonempty subset of $\mathbb{R}$, because its infimum is either a real number or $-\infty$.

Conclusion. $\inf S = -\sup(-S)$ holds both for finite infima and for the extended value $-\infty$.

5.5 Exercise 5.5: The infimum cannot exceed the supremum

What we must show. Prove $\inf S \le \sup S$ for every nonempty $S \subseteq \mathbb{R}$, whether or not S is bounded on either side.

Plan. Choose one real member of S. Its position between the extended lower and upper bounds supplies a common comparison point.

1. Compare the infimum to a chosen member

Because S is nonempty, choose $s_0 \in S$. If S is bounded below, its finite infimum is a lower bound and therefore $\inf S \le s_0$. If S is not bounded below, then $\inf S = -\infty < s_0$ by the extended order. Thus in either case

$$\inf S \le s_0.$$

2. Compare the same member to the supremum

If S is bounded above, its finite supremum is an upper bound and therefore $s_0 \le \sup S$. If S is not bounded above, then $\sup S = +\infty > s_0$. Hence in either case

$$s_0 \le \sup S.$$

3. Use transitivity of the extended order

Combining the preceding inequalities gives

$$\inf S \le s_0 \le \sup S.$$

The desired inequality follows. This argument includes the cases in which one or both bounds are infinite, without treating infinity as an element of S.

Conclusion. For every nonempty $S \subseteq \mathbb{R}$, $\inf S \le \sup S$.

5.6 Exercise 5.6: A chain of bounds under set inclusion

What we must show. If S, T are nonempty subsets of $\mathbb{R}$ and $S \subseteq T$, prove

$$\inf T \le \inf S \le \sup S \le \sup T.$$

Plan. A lower bound for the larger set is also a lower bound for the smaller set; the same holds for upper bounds. Check the infinite cases on the relevant side before applying the finite greatest- or least-bound property.

1. Prove the leftmost inequality

If T is not bounded below, then $\inf T = -\infty$. The number $\inf S$ is either real or $-\infty$, so the extended order gives $\inf T \le \inf S$, including equality when both are $-\infty$.

If T is bounded below, put $\alpha = \inf T \in \mathbb{R}$. Every $t \in T$ satisfies $\alpha \le t$. Because every $s \in S$ also belongs to T, we have $\alpha \le s$ for all $s \in S$. Thus S is bounded below as well, and α is one of its real lower bounds. Its greatest lower bound must be at least this lower bound, so

$$\inf T = \alpha \le \inf S.$$

This argument assumes only a lower bound for T, not an upper bound.

2. Prove the middle inequality

Exercise 5.5 applies to the nonempty set S and gives

$$\inf S \le \sup S.$$

In particular, this remains valid if S is unbounded on one or both sides.

3. Prove the rightmost inequality

If T is not bounded above, then $\sup T = +\infty$. Since $\sup S$ is either real or $+\infty$, we have $\sup S \le \sup T$ in the extended order, including equality when both are $+\infty$.

If T is bounded above, put $\beta = \sup T \in \mathbb{R}$. Every element of T is at most β, so every element of the subset S is at most β. Thus S is bounded above, and β is an upper bound for it. The least-upper-bound property gives

$$\sup S \le \beta = \sup T.$$

Here only an upper bound for T was needed.

4. Assemble the comparisons

The three inequalities have been proved in all finite and infinite cases. Placing them in order yields the required chain.

Conclusion. For nonempty $S \subseteq T \subseteq \mathbb{R}$, $\inf T \le \inf S \le \sup S \le \sup T$.

5.7 Exercise 5.7: The infimum of a sum set, including unbounded cases

What we must show. For arbitrary nonempty $A, B \subseteq \mathbb{R}$, define

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

Prove the extended equality

$$\inf(A + B) = \inf A + \inf B.$$

Plan. When both sets are bounded below, add their lower bounds and use the defining approximation property of an infimum with an $\varepsilon/2$ allowance for each summand. When a set is unbounded below, fix one member of the other set and show that the sum set is unbounded below too.

1. Check nonemptiness and the possible extended values

Choose $a_0 \in A$ and $b_0 \in B$, which is possible because both sets are nonempty. The real number $a_0 + b_0$ belongs to $A + B$, so the sum set is nonempty. Put

$$\alpha = \inf A, \qquad \beta = \inf B.$$

Each of α, β is either a real number or $-\infty$. Neither can be $+\infty$ under the conventions for nonempty subsets of $\mathbb{R}$. Thus the expression $\alpha + \beta$ never involves the undefined combination $+\infty + (-\infty)$.

2. Suppose both A and B are bounded below

In this case $\alpha, \beta \in \mathbb{R}$. For every $a \in A$ and $b \in B$, the lower-bound properties give $a \geq \alpha$ and $b \geq \beta$. Adding yields

$$a + b \geq \alpha + \beta.$$

Hence $\alpha + \beta$ is a real lower bound for the nonempty set $A + B$. Corollary 4.5 supplies its real infimum $\gamma = \inf(A + B)$. Because γ is the greatest lower bound,

$$\gamma \geq \alpha + \beta.$$

To prove the reverse inequality, fix any $\varepsilon > 0$. The real number $\alpha + \varepsilon/2$ is larger than the greatest lower bound α, so it cannot be a lower bound for A. Consequently some $a \in A$ satisfies

$$\alpha \leq a < \alpha + \frac{\varepsilon}{2}.$$

The first inequality holds because α is a lower bound. The same reasoning for B supplies $b \in B$ with

$$\beta \leq b < \beta + \frac{\varepsilon}{2}.$$

Adding the two strict upper estimates gives

$$a + b < \alpha + \beta + \varepsilon.$$

Since $a + b \in A + B$ and γ is a lower bound for that set,

$$\gamma \leq a + b < \alpha + \beta + \varepsilon.$$

This holds for every $\varepsilon > 0$. If $\gamma > \alpha + \beta$, choose the positive real number $\varepsilon = \gamma - (\alpha + \beta)$. The displayed inequality would then give

$$\gamma < \alpha + \beta + \big(\gamma - (\alpha + \beta)\big) = \gamma,$$

a contradiction. Therefore $\gamma \leq \alpha + \beta$. Combining both comparisons proves

$$\inf(A + B) = \alpha + \beta$$

when both infima are finite. No upper bound on A or B was used. Thus this case also covers sets that are unbounded above but bounded below.

3. Suppose A is not bounded below

Here $\alpha = -\infty$. Keep the fixed real member $b_0 \in B$ from Step 1. Let D be any proposed real lower bound for $A + B$. Because A is not bounded below and $D - b_0$ is real, there is an $a \in A$ satisfying

$$a < D - b_0.$$

Adding the real number b_0 gives $a + b_0 < D$. The number $a + b_0$ belongs to $A + B$, so D is not a lower bound. As this works for every real D, the sum set is unbounded below, and $\inf(A + B) = -\infty$.

If B is bounded below, then β is finite and $\alpha + \beta = (-\infty) + \beta = -\infty$. If B is also unbounded below, then $\beta = -\infty$ and $\alpha + \beta = (-\infty) + (-\infty) = -\infty$. Thus the equality holds in both possibilities for B.

4. Suppose A is bounded below and B is not

The case with both sets unbounded below was already covered in Step 3. It remains to handle a finite α and $\beta = -\infty$. Fix the real member $a_0 \in A$. Given any real D, the absence of a lower bound for B supplies $b \in B$ with $b < D - a_0$. Then $a_0 + b < D$ and $a_0 + b \in A + B$. Hence $A + B$ is unbounded below and

$$\inf(A + B) = -\infty = \alpha + (-\infty) = \inf A + \inf B.$$

5. Account for all cases

Either both sets are bounded below, or A is unbounded below, or A is bounded below and B is unbounded below. Steps 2–4 prove the equality in these exhaustive cases. Every sum used to produce a member of $A + B$ was a sum of two real numbers; the infinite symbols were used only to record the resulting extended bounds.

Conclusion. For all nonempty $A, B \subseteq \mathbb{R}$, $\inf(A+B) = \inf A + \inf B$, with the stated extended-bound conventions.

6 A development of the real numbers

This section changes the point of view. A real number is to be constructed from rational numbers, rather than assumed in advance. Accordingly, *all elements of cuts and all auxiliary numbers in the construction below are rational.* We use the ordered-field rules for $\mathbb{Q}$, the Archimedean property for $\mathbb{Q}$, and elementary properties of integers. We do not assume that an irrational square root or cube root already exists in order to prove that a set is a cut.

The definition and two preliminary facts

A subset α of $\mathbb{Q}$ is a *Dedekind cut* when:

(i) α is nonempty and is not all of $\mathbb{Q}$;

(ii) it is downward closed: if $a \in \alpha$, $q \in \mathbb{Q}$, and $q < a$, then $q \in \alpha$;

(iii) it has no greatest element: for each $a \in \alpha$, there is $a' \in \alpha$ with $a < a'$.

An *outside element* is a rational number not belonging to the cut. The cut associated with a rational s is

$$s^* = \{q \in \mathbb{Q} : q < s\}.$$

For example, 0^* consists of the negative rational numbers. Its elements are numbers, whereas 0^* itself is a set of numbers.

Preliminary fact 1: an inside element is below every outside element.

Let $a \in \alpha$ and $b \in \mathbb{Q} \setminus \alpha$. Equality $a = b$ is impossible because one number cannot both belong and not belong to the same set. If $b < a$, downward closure would put b in α, also a contradiction. The order of $\mathbb{Q}$ gives

$$a < b.$$

Consequently, if $b \notin \alpha$ and $b \le c$ with $c \in \mathbb{Q}$, then $c \notin \alpha$: otherwise either $c = b$ contradicts the choice of b, or $b < c$ and downward closure puts b in α.

Preliminary fact 2: rational cuts satisfy the definition.

Fix $s \in \mathbb{Q}$. The number $s - 1$ belongs to s^*, so s^* is nonempty. The rational number s does not belong to s^*, so $s^* \neq \mathbb{Q}$. If $q < r$ and $r < s$, transitivity gives $q < s$, proving downward closure. Finally, if $q < s$, the rational midpoint

$$q' = \frac{q + s}{2}$$

satisfies

$$q' - q = \frac{s - q}{2} > 0, \qquad s - q' = \frac{s - q}{2} > 0.$$

Thus $q < q' < s$, and s^* has no greatest element.

6.1 Exercise 6.1: Order, equality, and addition for rational cuts

What we must show. For rational s, t, prove

$$s \le t \iff s^* \subseteq t^*, \qquad s = t \iff s^* = t^*, \qquad (s + t)^* = s^* + t^*.$$

Here the addition on the right of the last equality means

$$s^* + t^* = \{r + u : r \in s^*,\ u \in t^*\},$$

an operation on sets.

Part (a), Step 1: turn an inequality into an inclusion.

Assume $s \leq t$. Take any $q \in s^*$. The definition gives $q < s$. Combining $q < s$ with $s \leq t$ gives $q < t$. Therefore $q \in t^*$. This works for every $q \in s^*$, so $s^* \subseteq t^*$.

Part (a), Step 2: prove the converse by testing a midpoint.

Assume $s^* \subseteq t^*$. Suppose, for a contradiction, that $s > t$. Set $q = (s + t)/2$. This is rational, and

$$q - t = \frac{s - t}{2} > 0, \qquad s - q = \frac{s - t}{2} > 0.$$

Hence $t < q < s$. The inequality $q < s$ puts q in s^*, but $q > t$ prevents q from belonging to t^*. This contradicts the inclusion. Thus $s > t$ is impossible, and the total order gives $s \leq t$.

Conclusion. Both directions have been proved:

$$\boxed{s \leq t \iff s^* \subseteq t^*.}$$

Part (b), Step 1: equal rational numbers give the same set.

If $s = t$, the conditions $q < s$ and $q < t$ are identical. They select the same rational numbers, so $s^* = t^*$.

Part (b), Step 2: the same set determines only one rational endpoint.

If $s^* = t^*$, both $s^* \subseteq t^*$ and $t^* \subseteq s^*$ hold. Part (a), applied in both orders, gives $s \leq t$ and $t \leq s$. Antisymmetry gives $s = t$.

Conclusion. The rational-to-cut correspondence is injective:

$$\boxed{s = t \iff s^* = t^*.}$$

Part (c), Step 1: prove the sum is contained in the target cut.

Let $q \in s^* + t^*$. There are rational $r < s$ and $u < t$ with $q = r + u$. Adding u to the first inequality and s to the second gives

$$q = r + u < s + u < s + t.$$

Consequently $q \in (s + t)^*$.

Part (c), Step 2: split a number below the target endpoint.

Conversely, take $q \in (s + t)^*$. Then $q < s + t$, so

$$\delta = \frac{s + t - q}{2}$$

is a positive rational number. Put $r = s - \delta$ and $u = t - \delta$. Since $\delta > 0$, we have $r < s$ and $u < t$, hence $r \in s^*$ and $u \in t^*$. Moreover,

$$r + u = s + t - 2\delta = s + t - (s + t - q) = q.$$

Thus $q \in s^* + t^*$.

Conclusion. The two inclusions establish

$$\boxed{(s+t)^* = s^* + t^*.}$$

Ordinary rational addition agrees with addition of rational cuts.

6.2 Exercise 6.2: Why the sum of two cuts is another cut

What we must show. Given Dedekind cuts α, β, verify every defining condition for

$$\gamma = \alpha + \beta = \{a + b : a \in \alpha,\ b \in \beta\}.$$

All such sums are rational because $\mathbb{Q}$ is closed under addition.

Step 1: verify that the sum is nonempty.

Choose $a_0 \in \alpha$ and $b_0 \in \beta$, possible because both cuts are nonempty. Then $a_0 + b_0 \in \gamma$, so γ is nonempty.

Step 2: exhibit a rational missing from the sum.

Choose rational outside elements $A \notin \alpha$ and $B \notin \beta$. Preliminary fact 1 gives

$$a < A \quad (a \in \alpha), \qquad b < B \quad (b \in \beta).$$

Adding gives $a + b < A + B$ for every member $a + b$ of γ. In particular, the rational $A + B$ cannot belong to γ. Therefore $\gamma \neq \mathbb{Q}$.

Step 3: prove downward closure by changing one summand.

Let $x \in \gamma$ and let $y \in \mathbb{Q}$ satisfy $y < x$. Write $x = a + b$ with $a \in \alpha$, $b \in \beta$, and define

$$a' = a + (y - x).$$

Since $y - x < 0$, the rational number a' is less than a. Downward closure of α gives $a' \in \alpha$. Also

$$a' + b = a + (y - x) + b = (a + b) + (y - x) = x + y - x = y.$$

Thus $y \in \gamma$.

Step 4: prove that no member is greatest.

Take $x = a + b \in \gamma$. Choose $a' \in \alpha$ with $a' > a$, using the no-greatest-element property of α. Then

$$x' = a' + b \in \gamma, \qquad x' - x = a' - a > 0.$$

Every member x therefore has a larger member x'.

Conclusion. The set $\alpha + \beta$ is nonempty, proper, downward closed, and has no greatest element. Hence

$$\boxed{\alpha + \beta \text{ is a Dedekind cut.}}$$

6.3 Exercise 6.3: The zero cut and a correctly defined negative cut

What we must show. Prove (a) $\alpha + 0^* = \alpha$, and (b) define a cut $-\alpha$ and prove $\alpha + (-\alpha) = 0^*$.

Part (a), Step 1: prove the forward inclusion.

If $x \in \alpha + 0^*$, write $x = a + z$ with $a \in \alpha$ and $z < 0$. Then $x < a$, and downward closure gives $x \in \alpha$. Thus $\alpha + 0^* \subseteq \alpha$.

Part (a), Step 2: represent every original member as a sum.

Take $x \in \alpha$. Choose $a \in \alpha$ with $x < a$ and put $z = x - a$. Then $z \in \mathbb{Q}$, $z < 0$, and $a + z = x$. Hence $z \in 0^*$ and $x \in \alpha + 0^*$.

Conclusion. The two inclusions give

$$\boxed{\alpha + 0^* = \alpha.}$$

Part (b), Step 1: define the negative with a strict endpoint.

Set

$$\boxed{-\alpha = \{q \in \mathbb{Q} : \text{there exists } r \in \mathbb{Q} \setminus \alpha \text{ with } q < -r\}.} \tag{6.3.1}$$

This includes rational numbers strictly below negatives of outside elements.

The strict inequality is necessary. If $\alpha = 0^*$, simply negating every member of its complement would give $\{q \in \mathbb{Q} : q \leq 0\}$. That set has the greatest member 0, and hence is not a cut. Definition (6.3.1) avoids including such a boundary point.

Part (b), Step 2: prove nonemptiness.

Choose $r_0 \notin \alpha$. The rational $q_0 = -r_0 - 1$ satisfies $q_0 < -r_0$, so $q_0 \in -\alpha$.

Part (b), Step 3: prove properness.

Choose $a_0 \in \alpha$. For every rational $r \notin \alpha$, preliminary fact 1 gives $a_0 < r$, hence $-r < -a_0$. Therefore $-a_0 < -r$ cannot hold for any outside element r. Definition (6.3.1) shows $-a_0 \notin -\alpha$, so $-\alpha \neq \mathbb{Q}$.

Part (b), Step 4: prove downward closure.

If $q \in -\alpha$, choose $r \notin \alpha$ with $q < -r$. For any rational $p < q$, we have $p < q < -r$, so the same r witnesses $p \in -\alpha$.

Part (b), Step 5: prove there is no greatest member.

Let $q \in -\alpha$, with outside witness r such that $q < -r$. The rational midpoint $q' = (q - r)/2$ satisfies

$$q' - q = \frac{-r - q}{2} > 0, \qquad -r - q' = \frac{-r - q}{2} > 0.$$

Thus $q < q' < -r$, and $q' \in -\alpha$. Steps 2–5 prove that the proposed negative is a cut.

Part (b), Step 6: establish a rational small-gap lemma.

We need to find an inside point close to an outside point without assuming that a real endpoint already exists.

Claim. For every positive rational h, there is $a \in \alpha$ such that $a + h \notin \alpha$.

Choose $a_0 \in \alpha$, $b_0 \notin \alpha$. Preliminary fact 1 gives $b_0 - a_0 > 0$. The rational Archimedean property supplies $N \in \mathbb{N}$ with

$$Nh > b_0 - a_0.$$

Then $a_0 + Nh > b_0$, so $a_0 + Nh \notin \alpha$ by preliminary fact 1. Consider the finite set

$$J = \{j \in \{1, \dots, N\} : a_0 + jh \notin \alpha\}.$$

It contains N and is nonempty. Let j be its smallest member. The preceding point $a_0 + (j-1)h$ lies inside α: it is a_0 if $j = 1$, and otherwise its index is below the first outside index. Taking

$$a = a_0 + (j-1)h$$

gives $a \in \alpha$ and $a + h = a_0 + jh \notin \alpha$, as claimed. Only the rational Archimedean property and a finite least-index choice were used.

Part (b), Step 7: show that every sum lies in the zero cut.

Take $x = a + q \in \alpha + (-\alpha)$ with $a \in \alpha$, $q \in -\alpha$. Choose $r \notin \alpha$ with $q < -r$. Since $a < r$,

$$x = a + q < a - r < 0.$$

Hence $x \in 0^*$, proving $\alpha + (-\alpha) \subseteq 0^*$.

Part (b), Step 8: represent every negative rational as a sum.

Take $z \in 0^*$, so $z < 0$, and set $h = -z/2 > 0$. The small-gap lemma supplies $a \in \alpha$ with $r := a + h \notin \alpha$. Put $q = z - a$. Then

$$q + r = (z - a) + (a + h) = z + h = z - \frac{z}{2} = \frac{z}{2} < 0.$$

Thus $q < -r$. The outside witness r proves $q \in -\alpha$. Finally, $a + q = a + (z - a) = z$, so $z \in \alpha + (-\alpha)$.

Conclusion. Definition (6.3.1) gives a Dedekind cut satisfying

$$\boxed{\alpha + (-\alpha) = 0^*.}$$

It constructs the additive inverse required by A4 using rational numbers alone.

As a check, when $\alpha = s^*$, every outside element satisfies $r \geq s$. If $q < -r$, then $q < -s$, so $-\alpha \subseteq (-s)^*$. Conversely, if $q < -s$, the outside element $r = s$ witnesses $q \in -\alpha$. Hence $-(s^*) = (-s)^*$.

6.4 Exercise 6.4: Why the proposed product of cuts fails

What we must show. Examine the proposed rule

$$\alpha \cdot_{\text{naive}} \beta = \{ab : a \in \alpha,\ b \in \beta\}$$

using 0^*, 1^*, and $(-1)^*$, and explain why it cannot define multiplication of real numbers.

Part (a), Step 1: identify the three input sets.

Write

$$A = 0^* = \{q \in \mathbb{Q} : q < 0\}, \quad B = 1^* = \{q \in \mathbb{Q} : q < 1\}, \quad C = (-1)^* = \{q \in \mathbb{Q} : q < -1\}.$$

In this exercise, a product of letters means the proposed set of pairwise products. Rational multiplication is commutative, so $AB = BA$, $AC = CA$, and $BC = CB$. Six calculations suffice.

Part (a), Step 2: calculate AA.

Two negative rationals have a positive product, so $AA \subseteq \{q \in \mathbb{Q} : q > 0\}$. Conversely, for any rational $q > 0$,

$$q = (-1)(-q),$$

and both factors belong to A. Therefore

$$AA = \{q \in \mathbb{Q} : q > 0\}.$$

Part (a), Step 3: calculate AC.

A member of A and a member of C are both negative, so their product is positive. For a rational $q > 0$, write

$$q = \left(-\frac{q}{2}\right)(-2),$$

where $-q/2 \in A$ and $-2 \in C$. Consequently

$$AC = \{q \in \mathbb{Q} : q > 0\}.$$

Part (a), Step 4: calculate CC.

Members of C have the forms $-u, -v$ with $u, v > 1$. Their product is $uv > 1$. Hence $CC \subseteq \{q \in \mathbb{Q} : q > 1\}$.

For the reverse inclusion, let $q > 1$ be rational and put

$$r = -\frac{q+1}{2}, \qquad t = -\frac{2q}{q+1}.$$

The denominator $q + 1 > 2$ is nonzero, and $q + 1 > 2$ gives $r < -1$. Since the denominator is positive,

$$\frac{2q}{q+1} > 1 \iff 2q > q + 1 \iff q > 1.$$

Thus $t < -1$. Both factors belong to C, and

$$rt = \frac{q+1}{2}\,\frac{2q}{q+1} = q.$$

We conclude

$$CC = \{q \in \mathbb{Q} : q > 1\}.$$

Part (a), Step 5: calculate AB.

Every product is rational. Conversely, each rational q has a representation with one factor in A and one in B:

- If $q > 0$, use $q = (-q)(-1)$.
- If $q = 0$, use $q = (-1) \cdot 0$; the second factor satisfies $0 < 1$.

- If $q < 0$, use $q = (2q)(1/2)$; the first factor is negative and the second is less than 1.

Therefore $AB = \mathbb{Q}$.

Part (a), Step 6: calculate BB.

Because $q < 0$ implies $q < 1$, we have $A \subseteq B$. Hence

$$\mathbb{Q} = AB \subseteq BB \subseteq \mathbb{Q}.$$

The final inclusion is closure of rational multiplication. Thus $BB = \mathbb{Q}$.

Part (a), Step 7: calculate BC.

Every product again is rational. To represent any rational q:

- For $q > 0$, use $q = (-q/2)(-2)$.
- For $q = 0$, use $q = 0 \cdot (-2)$.
- For $q < 0$, put $u = -q > 0$ and use

$$q = \frac{u}{2(u+1)}\big(-2(u+1)\big).$$

  The second factor is less than -2, so it belongs to C. The first is positive and less than 1 because $2(u+1) - u = u + 2 > 0$. It belongs to B.

Thus $BC = \mathbb{Q}$.

Part (a), Step 8: collect the answers.

Let $\mathbb{Q}_{>0} = \{q \in \mathbb{Q} : q > 0\}$ and $\mathbb{Q}_{>1} = \{q \in \mathbb{Q} : q > 1\}$. All six answers are in the table:

$\cdot_{\text{naive}}$	0^*	1^*	$(-1)^*$
0^*	$\mathbb{Q}_{>0}$	$\mathbb{Q}$	$\mathbb{Q}_{>0}$
1^*	$\mathbb{Q}$	$\mathbb{Q}$	$\mathbb{Q}$
$(-1)^*$	$\mathbb{Q}_{>0}$	$\mathbb{Q}$	$\mathbb{Q}_{>1}$

Part (b), Step 1: none of these outputs is a cut.

The set $\mathbb{Q}$ is not a cut because it is not a proper subset of itself. The set $\mathbb{Q}_{>0}$ is not downward closed: it contains 1 but not the smaller rational 0. The set $\mathbb{Q}_{>1}$ is not downward closed either: it contains 2 but not the smaller rational 0.

Part (b), Step 2: rational multiplication would give different outputs.

A multiplication consistent with the rational correspondence would require

$$0^* \cdot 0^* = (0 \cdot 0)^* = 0^*.$$

Instead, the proposed rule gives the positive rationals, which are disjoint from 0^*. Likewise, it should give

$$(-1)^* \cdot (-1)^* = 1^*,$$

but instead gives $\mathbb{Q}_{>1}$. For example, $0 \in 1^*$ but $0 \notin \mathbb{Q}_{>1}$.

Conclusion. The proposed pairwise-product rule is not a binary operation on the Dedekind cuts and does not reproduce rational multiplication. A valid definition has to account for signs and downward closure.

6.5 Exercise 6.5: Constructing irrational cuts using rational inequalities

What we must show. Prove all three claims:

(a) $\alpha = \{q \in \mathbb{Q} : q^3 < 2\}$ is a cut, but $\{q \in \mathbb{Q} : q^2 < 2\}$ is not;

(b) α is not the cut of any rational number;

(c)
$$\beta = 0^* \cup \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}$$
is a cut and is not the cut of any rational number.

Plan. Rational polynomial inequalities will verify the cut conditions. Explicit positive rational increments will rule out a greatest element and will rule out rational endpoints. No pre-existing irrational root is needed.

Preliminary calculation 1: cubes preserve strict order.

For rational $u < v$,
$$v^3 - u^3 = (v - u)(v^2 + uv + u^2).$$
The first factor is positive. The other factor satisfies
$$v^2 + uv + u^2 = \left(v + \frac{u}{2}\right)^2 + \frac{3u^2}{4}.$$
If $u \neq 0$, the last term is positive and the square is nonnegative. If $u = 0$, then $v > 0$ and the factor equals $v^2 > 0$. Thus $v^3 - u^3 > 0$ for all rational signs, proving $u^3 < v^3$.

Preliminary calculation 2: bound explicit small increments.

Fix $s \in \mathbb{Q}$, let h be rational with $0 < h \leq 1$, and put
$$C_3(s) = 3s^2 + 3\,|s| + 1 > 0.$$
The absolute value is rational because it is either s or $-s$. Multiplying $h \leq 1$ by $h > 0$ gives $h^2 \leq h$; multiplying again gives $h^3 \leq h^2 \leq h$. Expansion yields
$$\begin{aligned}(s+h)^3 - s^3 &= 3s^2h + 3sh^2 + h^3 \\ &\leq 3s^2h + 3\,|s|\,h^2 + h^3 \\ &\leq (3s^2 + 3\,|s| + 1)h = C_3(s)h.\end{aligned}$$
Using $-s \leq |s|$ gives the corresponding decrement bound:
$$\begin{aligned}s^3 - (s-h)^3 &= 3s^2h - 3sh^2 + h^3 \\ &\leq 3s^2h + 3\,|s|\,h^2 + h^3 \\ &\leq C_3(s)h.\end{aligned}$$
The bounds hold even when s is negative or zero.

For squares, define the positive rational $C_2(s) = 2\,|s| + 1$. Expansion gives
$$(s+h)^2 - s^2 = 2sh + h^2 \leq C_2(s)h,$$

$$s^2 - (s-h)^2 = 2sh - h^2 \leq 2|s|\,h \leq C_2(s)h.$$

Every denominator $2C_3(s)$ and $2C_2(s)$ below is positive.

Part (a), Step 1: prove that the cubic set is nonempty and proper.

The rational 0 lies in α, since $0^3 = 0 < 2$. The rational 2 does not lie in α, since $2^3 = 8 > 2$. Hence α is a nonempty proper subset of $\mathbb{Q}$.

Part (a), Step 2: prove downward closure of the cubic set.

Let $q \in \alpha$ and let rational $p < q$. Strict increase of cubes gives

$$p^3 < q^3 < 2.$$

Thus $p \in \alpha$. This argument includes negative p and q.

Part (a), Step 3: construct a larger cubic-set member.

Let $q \in \alpha$, so the rational $\delta = 2 - q^3$ is positive. Choose

$$h = \min\left\{\frac{1}{2}, \frac{\delta}{2C_3(q)}\right\}.$$

Both entries of the minimum are positive rational numbers. Therefore h is rational, $0 < h \leq 1/2 < 1$, and $C_3(q)h \leq \delta/2$. The increment bound gives

$$(q+h)^3 \leq q^3 + C_3(q)h \leq q^3 + \frac{\delta}{2} < q^3 + \delta = 2.$$

Hence $q + h \in \alpha$ and $q + h > q$. Thus α has no greatest member.

Conclusion. The cubic set $\{q \in \mathbb{Q} : q^3 < 2\}$ satisfies all defining conditions and is a Dedekind cut.

Part (a), Step 4: test the unmodified square condition.

Let $E = \{q \in \mathbb{Q} : q^2 < 2\}$. It contains 0, since $0^2 < 2$. But $-2 < 0$ and $(-2)^2 = 4 > 2$, so $-2 \notin E$. This contradicts the downward closure that any cut containing 0 would have to satisfy.

Conclusion. The set $\{q \in \mathbb{Q} : q^2 < 2\}$ is not a cut. Its exclusion of sufficiently negative rationals prevents downward closure.

Part (b), Step 1: suppose the cubic cut has a rational endpoint.

Suppose $\alpha = s^*$ for some $s \in \mathbb{Q}$. If $s^3 < 2$, then $s \in \alpha$. But $s \notin s^*$ because $s < s$ is false. This contradicts equality of the two sets. Thus $s^3 < 2$ is impossible.

Part (b), Step 2: exclude an endpoint whose cube is too large.

If $s^3 > 2$, put $\delta = s^3 - 2 > 0$ and choose

$$h = \min\left\{\frac{1}{2}, \frac{\delta}{2C_3(s)}\right\} > 0.$$

The rational $r = s - h$ satisfies $r < s$, so $r \in s^*$. However,

$$s^3 - r^3 \leq C_3(s)h \leq \frac{\delta}{2}.$$

Rearranging gives

$$r^3 \geq s^3 - \frac{\delta}{2} > s^3 - \delta = 2.$$

Therefore $r \notin \alpha$, another contradiction. The total order excludes both strict alternatives and forces $s^3 = 2$.

Part (b), Step 3: prove that a rational cube cannot equal 2.

Write $s = m/n$ in lowest terms, with $m, n \in \mathbb{Z}$ and $n > 0$. The equation $s^3 = 2$ gives $m^3 = 2n^3$, so m^3 is even. An odd integer has the form $2k + 1$ and has odd cube:

$$(2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1.$$

Hence m must be even; put $m = 2u$ with $u \in \mathbb{Z}$. Substitution gives

$$8u^3 = 2n^3, \qquad n^3 = 4u^3.$$

Now n^3 is even, so the same parity argument makes n even. Then m and n have the common positive factor 2, contradicting lowest terms.

Conclusion. The cubic cut is not s^* for any rational s. It represents an irrational element of the cut construction, later identified with the positive cube root of 2 after the real-number construction is completed.

Part (c), Step 1: prove nonemptiness and properness.

The corrected set is

$$\beta = \{q \in \mathbb{Q} : q < 0\} \ \cup \ \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}.$$

The rational 0 belongs to its second part, since $0 \geq 0$ and $0^2 < 2$. Thus β is nonempty. The rational 2 is not negative and its square 4 is not less than 2. It lies in neither part, so $\beta \neq \mathbb{Q}$.

Part (c), Step 2: verify downward closure by separating signs.

Let $q \in \beta$ and let $p \in \mathbb{Q}$ satisfy $p < q$. If $p < 0$, it belongs to $0^* \subseteq \beta$. If $p \geq 0$, then $q > p \geq 0$. Thus q cannot belong to the negative part of β, and its membership must give $q^2 < 2$. Moreover,

$$q^2 - p^2 = (q - p)(q + p) > 0,$$

since $q - p > 0$ and $q + p > 0$. Hence $p^2 < q^2 < 2$. Together with $p \geq 0$, this puts p in the second part of β. Both signs of p therefore give membership.

Part (c), Step 3: find a larger member when the chosen member is negative.

If $q \in \beta$ and $q < 0$, put $q' = q/2$. Then $q' - q = -q/2 > 0$ and $q' < 0$. So $q' > q$ and $q' \in 0^* \subseteq \beta$.

Part (c), Step 4: find a larger member in the nonnegative case.

If $q \in \beta$ and $q \geq 0$, then $\delta = 2 - q^2 > 0$. Choose the rational

$$h = \min\left\{\frac{1}{2}, \frac{\delta}{2C_2(q)}\right\} > 0.$$

It satisfies $h \leq 1$ and $C_2(q)h \leq \delta/2$, so

$$(q + h)^2 \leq q^2 + C_2(q)h \leq q^2 + \frac{\delta}{2} < q^2 + \delta = 2.$$

Because $q + h \geq 0$, it lies in β, and $q + h > q$. This case includes $q = 0$.

Conclusion. The corrected square set β is nonempty, proper, downward closed, and has no greatest rational. It is a Dedekind cut. Including all negative rationals repairs the failure in part (a).

Part (c), Step 5: suppose the corrected square cut is rational.

Assume $\beta = s^*$ for some rational s. The rational 1 belongs to β, because $1 \geq 0$ and $1^2 < 2$. Therefore $1 \in s^*$, which means $1 < s$. In particular, $s > 0$.

If $s^2 < 2$, then $s \geq 0$ and $s^2 < 2$ put s in β. But $s \notin s^*$, contradicting equality. Thus $s^2 < 2$ is impossible.

Part (c), Step 6: exclude an endpoint whose square is too large.

If $s^2 > 2$, let $\delta = s^2 - 2 > 0$ and choose

$$h = \min\left\{\frac{1}{2}, \frac{s}{2}, \frac{\delta}{2C_2(s)}\right\}.$$

All entries are positive rationals. Thus $0 < h \leq 1$, $h \leq s/2$, and $C_2(s)h \leq \delta/2$. The rational $r = s - h$ satisfies

$$r \geq \frac{s}{2} > 0, \qquad r < s.$$

The latter inequality puts r in s^*, but

$$r^2 \geq s^2 - C_2(s)h \geq s^2 - \frac{\delta}{2} > s^2 - \delta = 2.$$

Since $r > 0$ and $r^2 > 2$, it is not in either part of β. This contradicts $\beta = s^*$. Both strict alternatives have been excluded, so equality of these cuts would force $s^2 = 2$.

Part (c), Step 7: exclude a rational square equal to 2.

Write $s = m/n$ in lowest terms, with $m, n \in \mathbb{Z}$ and $n > 0$. Then $s^2 = 2$ implies $m^2 = 2n^2$, so m^2 is even. An odd integer $2k + 1$ has odd square:

$$(2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Hence m is even; write $m = 2u$. Substitution gives

$$4u^2 = 2n^2, \qquad n^2 = 2u^2.$$

Now n^2 is even, so the same argument makes n even. Both numerator and denominator would be divisible by 2, contrary to lowest terms.

Conclusion. The corrected square cut is not the cut of a rational number. It represents the irrational element later identified with the positive square root of 2. Its cut conditions and irrationality were established using rational arithmetic, explicit inequalities, and the cut axioms, without assuming that real square root in advance.